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SCHOOL OF CIVIL, ENVIRONMENTAL AND LAND MANAGEMENT
ENGINEERING



POLITECNICO
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ASSIGNMENT 1: Preliminary Design of a Reinforced Concrete Medium-Rise Residential Building in Medium Seismicity Area

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ATTACHMENT TO THE REPORT

- DRAWINGS
- SUMMARY OF THE REPORT

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CHAPTER 1

INTRODUCTION

The assignment is aimed to perform a preliminary design of a structural resisting system for a 6-story residential building located in a moderate seismicity area. In order to develop the structural system, a base architectural layout of the building was provided.

1.1 Geometry

The proposed residential building is composed of 6 floors and a roof, where the first floor is aimed to host commercial activity. Hence, the ground floor is characterized by wider spaces in contrast to the upper floors, which are conceived for residential purposes. The upper floors can be accessed by stairs or an elevator. Both are located at the center of the structure. The overall building dimensions are represented in Figure 1.1 having a floor height of about 3 meters. The most suitable structural scheme to be used is the one of a *reinforced concrete dual frame structure*, which is composed of *beam* and *columns* frame structure carrying the gravity loads and *shear walls* that constitute the lateral load resisting system aimed to withstand the forces generated by seismic action. Therefore, from the seismic point of view, the building is designed as a *wall system*. This statement allows to design the walls as the primary seismic resisting elements while the secondary elements (e.g., beams and columns) can be designed to sustain only gravity loads.



Figure 1.1: Dimensions of the building

1.2 Materials

1.2.1 Concrete

According to Eurocode 2, 4.1 (4): “Corrosion protection of steel reinforcement depends on density, quality, and thickness of concrete cover and cracking. The cover density and quality are achieved by controlling the maximum water/cement ratio and minimum cement content and may be related to a minimum strength class of concrete.”.

Therefore, considering a mild exposure class of XC1, the concrete of class C30/37 is adopted using Table 4.3N of EN 1992-1-1:2004. The corresponding mechanical properties are given in the below table adopted from Table 3.1 of EN 1992-1-1:2004:

Table 1.1: Mechanical properties of concrete

Strength Class	f_{ck}	$f_{ck,cube}$	f_{ctm}	f_{cd}	E_{cm}
C30/37	30	37	2.9	17	33
	MPa	MPa	MPa	MPa	GPa

The concrete cover has been selected referring to Sec 4.4.1 of EN 1992-1-1:2004 (the bar size referred to for the calculations is 16mm). The details are given in the below table.

Table 1.2: Nominal Concrete Cover

Nominal Concrete Cover		26	mm	A+B
A	c_{min}	16	mm	max(i, ii, iii)
i	$c_{min,b}$	16	mm	
ii	$c_{min,dur} + c_{dur} - c_{dur,st} - c_{dur,add}$	15	mm	
iii		10	mm	
B	c_{dev}	10	mm	(4.4.1.3 (1), Eurocode 2)

Considering the output value of the minimum nominal cover, the selected cover value is 30 mm.

1.2.2 Steel Reinforcement

The reinforcing steel bars of class B450C. All relevant parameters are summarized in the following table.

Table 1.3: Mechanical Properties of Steel Reinforcement

Grade	B450C	
Characteristic Yield Strength (f_{yk})	450	MPa
Design Yield Strength (f_{yd})	391.3	MPa
Modulus of Elasticity (E)	210	GPa

CHAPTER 2

LOAD ANALYSIS

2.1 Variable Loads

The analysis of acting loads starts with classification of building. According to Section 3.1.4 of *Norme tecniche per le costruzioni* 2018 (NTC 2018), the building is designed for Category A and has importance class II. Hence, variable loads are $q = 2.00 \text{ kN/m}^2$ for internal environment and $q = 4.00 \text{ kN/m}^2$ for stairs.

The snow load is taken assuming that the building is in Zone I of the map defined in section 3.4.2 of NTC2018 at a height lower than 200m. Assuming to have a single flap roof and that the location is not exposed to wind flows, the resulting snow load is $q_{\text{snow}} = 1.32 \text{ kN/m}^2$.

2.2 Permanent Loads

The permanent loads comprise of *Structural Permanent Loads* (G1) and *Non – structural Permanent Loads* (G2). The structural permanent loads consist of dead loads caused by slab elements. The floor slab is realized in the form of a one-way slab through the combined use of precast concrete joists and lightweight blocks, as shown in Figure 2.1. The non-structural permanent loads comprise the internal partitions, the concrete cover, the spread and the floor tiles. The detailed geometry and load analysis of the permanent and non-permanent loads are given in Table 2.1.

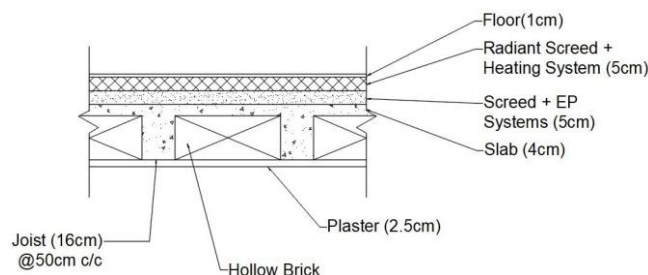


Figure 2.1: Slab Cross Section

Table 2.1: Permanent Load Details

Load Description	Item Description	Dimension		Unit Weight (kN/m ³)	No. of Members	Sub-Total (kN/m ²)
		Width (m)	Depth (m)			
Structural Dead Load (G1)	Joist	0.12	0.16	25	2	0.96
	Hollow Core	0.38	0.2	5	2	0.76
	Slab	1	0.04	25	1	1
	Total (G1)					2.72
Non-Structural (Superimposed) Dead Load (G2)	Concrete Cover	1	0.025	16.5	1	0.41
	Spread (Tech syst.)	1	0.05	10	1	0.50
	Spread (Sand+Cement)	1	0.05	20	1	1.00
	Floor Tiles	-	-	-	-	0.2
	Internal Partitions	-	-	-	-	1.2
	Total (G2)					3.31

2.3 Seismic Load

The seismic load is obtained by the generation of a response spectrum that is dependent on structural characteristics and the locality of the building being studied. In particular, the site is characterized by a soil type C, a peak ground acceleration (PGA) of 0.25g (reference PGA, of value 0.25g, factored with $\gamma = 1$, based on the importance class of I as per EN 1998-1/4.2.5(5)P) and structural damping of 5%.

The structure is characterized by a behavior factor ' q ' = 3.6 that must not be reduced owing to the building's regularity in height. Regularity in the plan will be analyzed after the definition of the structural scheme.

From the available data, both elastic and design response spectra can be computed using an online tool given in the footnote¹. The response spectra, both elastic and design, are shown in Figure 2.2. Their effect on the structure consists of an acceleration applied to the latter due to the dynamic interaction. To value this, the structural first natural period must be estimated to enter it as an input of the response spectrum function. The significant value of spectra is known as the time corresponding to the end of the plateau, resulting that $T_c = 0.61$.

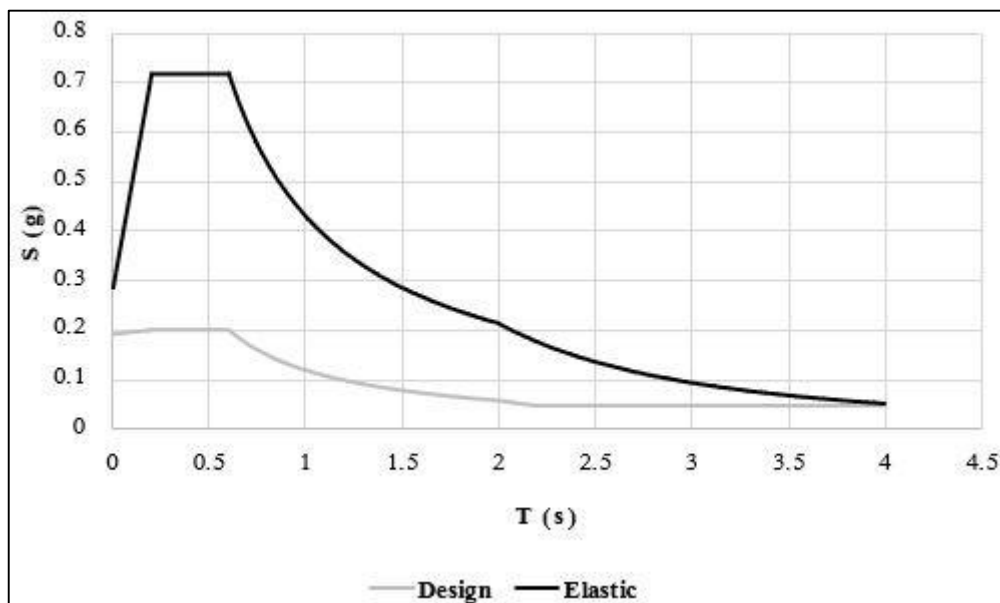


Figure 2.2: Elastic and Design Response Spectra

¹<https://eurocodeapplied.com/design/en1998/elastic-response-spectrum>

CHAPTER 3

STRUCTURAL SCHEME

The structural layout has been preliminarily developed based on the two constraints given below:

- Architectural layout
- Span-to-depth ratio
- Foundation

The chosen slab type is *One-Way Slab* as per the load distribution scheme shown in the below layout. The depth of the slab is taken considering the limit ratios. The ratios are computed below using the maximum spans L_b (Beam Span) and L_s (Slab Span). Based on the layout shown below the values of L_s and L_b are:

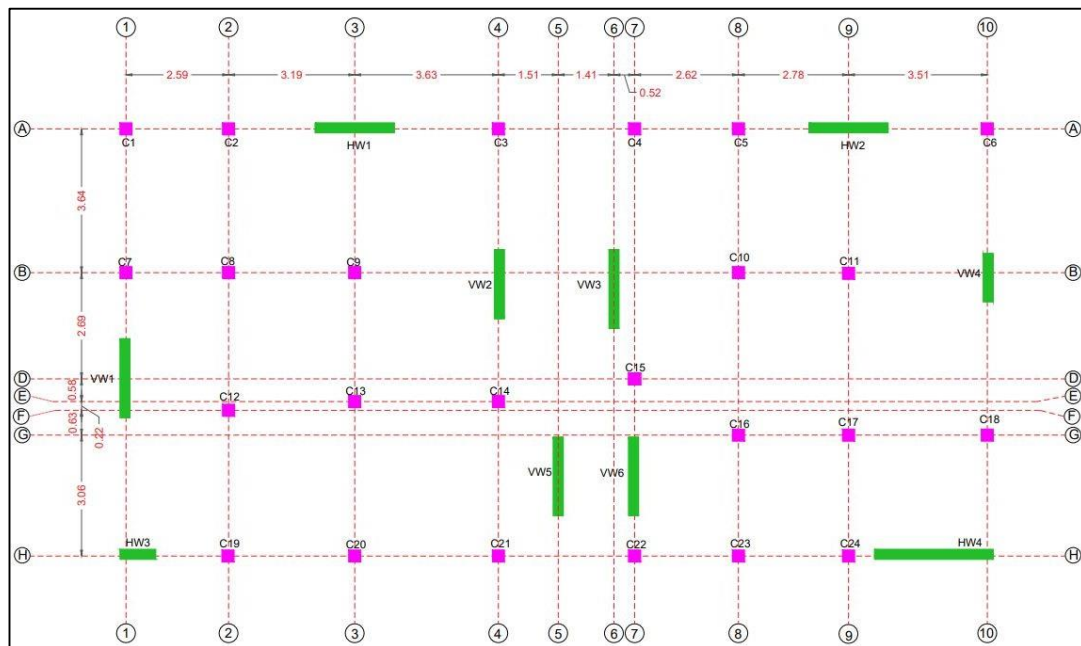


Figure 3.1: Center Line

$$L_s = 4.12 \text{ m}$$

$$L_b = 3.63 \text{ m}$$

Computing the limit slab depth using the ratio of $L_s/25$:

$$L_s/25 = 4.12/25 = 16.5 \text{ cm}$$

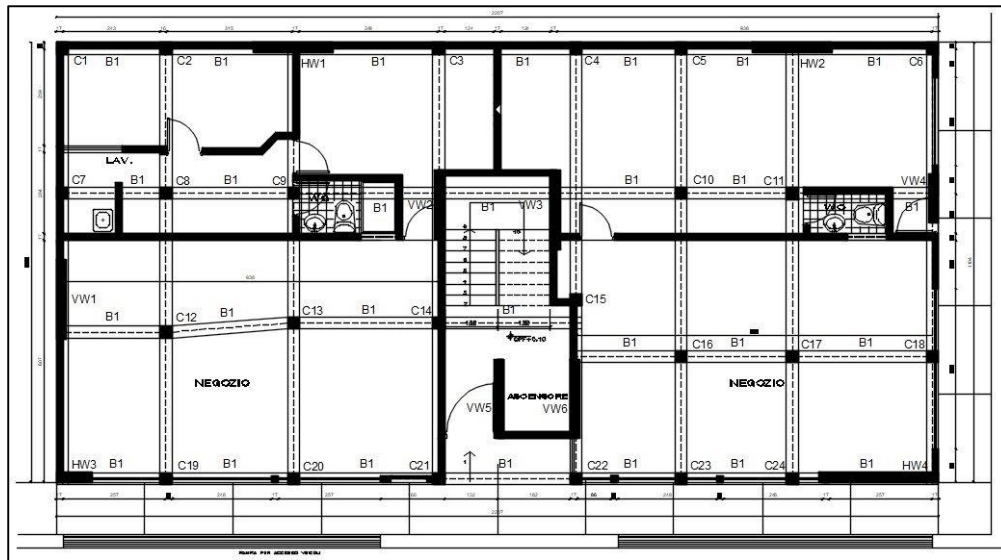


Figure 3.2: Structural Layout of Ground Floor (1/2)

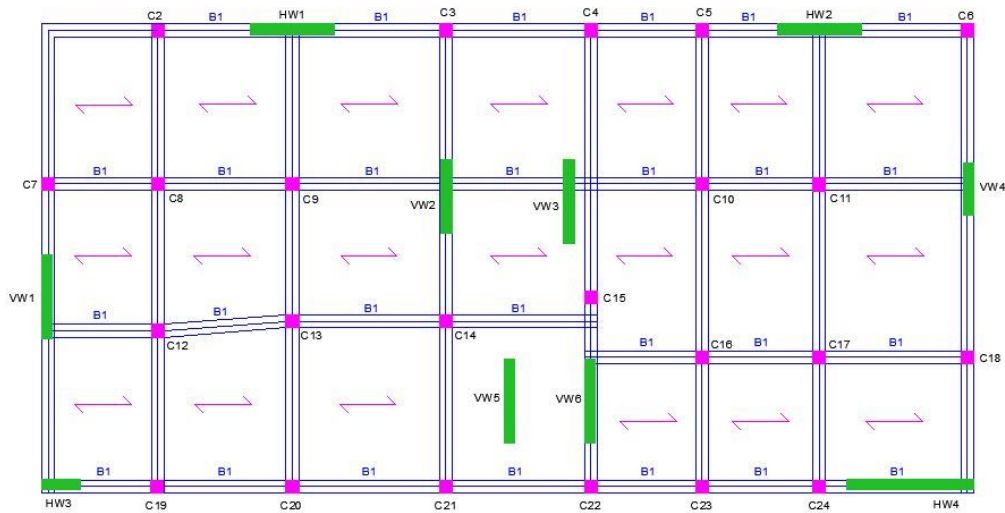


Figure 3.3: Structural Layout of Ground Floor (2/2)

Based on this, the depth of the slab has been taken equal to 20 cm with the scheme of 16+4 cm (i.e., Joist depth of 16cm and Slab Depth of 4cm). The length L_b is needed to be checked based on the selected slab depth as the beam depth will be taken equal to that of the slab. The calculation of allowable length is given below:

$$20 \times 20 = 400 \text{ cm} = 4 \text{ m} > L_b = 3.63 \text{ m}$$

The premises of the given architectural layout are respected with the exception of a few columns which are in conflict with the architectural layout which is to be negotiated with the architect for possible updating of the layout. Nevertheless, the proposed layout is respected on each floor to a greater extent. A typical structural layout is shown in Figures 3.2 and 3.3.

The detailed drawings of the structural layouts are given in the drawings annexure.

CHAPTER 4

Foundation

4.1 Deep Foundation

Deep foundations are the type of foundations that transmit the load of the structure further down the earth in comparison to shallow foundations. This foundation's depth-to-width ratio is usually more than 4 to 5, meaning the foundation is submerged deeper than its horizontal dimensions. This design factor is frequently used by deep foundations to improve their overall performance and load-bearing capacity. These foundations are used when the project requires support at significant depths or when shallow foundations are not feasible because of insufficient soil strength.

4.1.1 Types of Shallow foundation

Figure 4-1(a) shows the most common case in which the upper soil strata are too compressible or too weak to support heavy vertical loads. In this case, deep foundations transfer loads to a deeper dense stratum and act as toe-bearing foundations. In the absence of a dense stratum within a reasonable depth, the loads must be gradually transferred, mainly through soil resistance along the shaft, Figure 4-1(b). An important point to remember is that deep foundations transfer load from unsuitable layers to suitable layers. The foundation designer must define at what depth suitable soil layers begin in the soil profile.

Deep foundations are frequently needed because of the relative inability of shallow footings to resist inclined, lateral, or uplift loads and overturning moments. Deep foundations resist uplift loads by shaft resistance, Figure 4-1(c). Lateral loads are resisted either by vertical deep foundations in bending, Figure 4-1(d), or by groups of vertical and battered foundations, which combine the axial and lateral resistances of all deep foundations in the group, Figure 4-1(e). Lateral loads from overhead highway signs and noise walls may also be resisted by groups of deep foundations, Figure 4-1(f).

Deep foundations are often required when scour around footings could cause loss of bearing capacity at shallow depths, Figure 4-1(g). In this case the deep foundations must extend below the depth of scour and develop the full capacity in the support zone below the level of expected scour. FHWA (2001c) scour guidelines require the geotechnical analysis of bridge foundations to be performed on the basis that all stream bed materials in the scour prism have been removed and are not available for bearing or lateral support. Costly damage and the need for future underpinning can be avoided by properly designing for scour conditions.

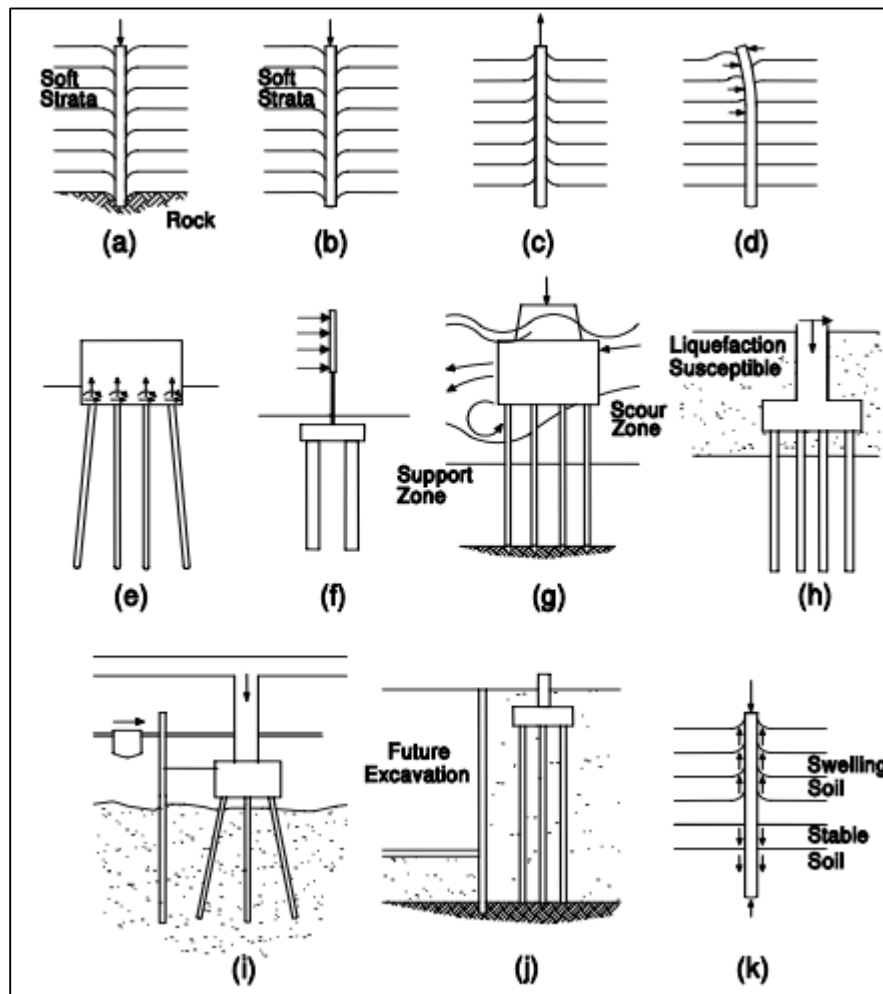


Table 4.1. Situations in which deep foundations may be needed (Vesic, 1977; FHWA, 2006a).

4.2 Bearing Capacity of Deep Foundation

The static capacity of a pile can be defined as the sum of soil/rock resistances along the pile shaft and at the pile toe available to support the imposed loads on the pile. Static analyses are performed to determine the ultimate capacity of an individual pile and of a pile group as well as the deformation response of a pile group to the applied loads. The ultimate capacity of an individual pile and of a pile group is defined as the smaller of:

- (1) The capacity of the surrounding soil/rock medium to support the loads transferred from the pile(s) or,
- (2) The structural capacity of the pile(s).

Soil-structure interaction analysis methods are used to determine the deformation response of piles and pile groups to lateral loads; such methods can also be used for deformation evaluation under vertical loads. The results from these analyses as well as the results of static analysis of pile group settlement are compared to the performance criteria established for the structure.

The ultimate geotechnical pile capacity, Q_u , of a pile in homogeneous soil may be expressed as follows in terms of the shaft (commonly known as “skin”) resistance, R_s , toe resistance, R_t , and the weight, W , of the pile:

$$Q_u = R_s + R_t - W$$

In most cases, such as H-Piles and open ended pipe piles, the weight W is small compared to the shaft and toe resistance and is neglected. However, the weight of pipe piles, particularly large diameter pipes, filled with concrete may be significant and may be included in the analysis. In this chapter, the W term is neglected. Equation 9-1, without the W term, may also be expressed in the form

$$Q_u = f_s A_s + q_t$$

Where f_s is the unit shaft resistance over the shaft surface area, A_s , and q_t is the unit toe resistance over the pile toe area, A_t . The above equations for pile bearing capacity assume that both the pile toe and the pile shaft have moved sufficiently concerning the adjacent soil to simultaneously develop the ultimate shaft and toe resistances. Generally, the displacement needed to mobilize the shaft resistance is smaller than that required to mobilize the toe resistance. This simple rational approach has been commonly used for all piles except very large diameter piles where such an approach may not be valid.

The figure illustrates typical load transfer profiles for a single pile. The load transfer distribution can be obtained from a static load test where strain gages or telltale rods are attached to a pile at different depths along the pile shaft. Figure 9-4 shows the measured ultimate axial load, Q_u , in the pile plotted against depth. The shaft resistance transferred to the soil is represented by R_s , and R_t represents the resistance at the pile toe. In Figure, the load transfer distribution for a pile with no shaft resistance is illustrated. In this case, the full axial load at the pile head is transferred to the pile toe. In Figure, the axial load versus depth for a uniform shaft resistance distribution typical of a cohesive soil is illustrated. The figure presents the axial load in the pile versus depth for a triangular shaft resistance distribution typical of cohesionless soils.

CHAPTER 5

DESIGN OF GRAVITY

5.1 Column Design

The columns are designed for ULS as reinforced concrete elements. The columns are designed considering the material details given in the first chapter. In order to develop the design, the applied load (the demand) is needed to be determined. Considering the conceptual nature of the design, only the most loaded column has been designed and the corresponding design is replicated for the entire building. In order to determine the most loaded column, the tributary area scheme is adopted. The tributary area (in other words, the influence area) for each of the columns was determined considering the structural layout of a typical floor. The tributary area of each column is shown in the figure below:

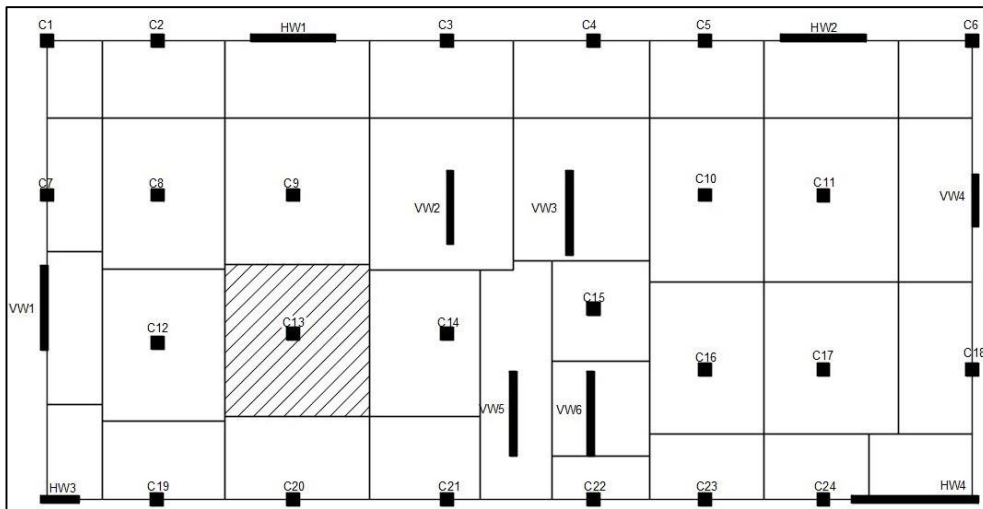


Figure 5.1: Tributary Area

Based on this scheme and considering a uniformly distributed load for the entire floor, the 'C13' was determined to be the most loaded column with tributary area of 12.18 m^2 .

For the sake of calculation of the required column dimension ('B' - square column being considered), the steel area ratio was considered to be 1.5%. The formulation

Adopted for the calculation is given below:

$$B = \sqrt{\frac{N_{base}}{0.8 \times f_{cd} \times A_s \times f_{yd}}}$$

Where:

N_{base} = Tributary area (m²) $\times$ ($\gamma_f * G_k + \gamma_f * Q_k$ in kN/m²)

f_{cd} = Design concrete strength

A_s = Area of steel (based on the 1.5% steel ratio)

f_{sd} = Design steel strength

Table 5.1: Column Dimensioning

$\gamma_f (G1, G2)$	1.35, 1.5	
$\gamma_f (Q)$	1.50	
G	6.03	kN/m ²
Q	2	kN/m ²
Tributary Area	12.18	m ²
N_{ed}	840.77	kN
f_{cd}	17	MPa
A_s	1.5%	
f_{yd}	391.3	MPa
B	24.4	cm

Based on this, the column size of 30cm x 30cm was chosen. Furthermore, the column was provided with 10x16mm steel bars. Further discussion on the provision of steel bars is done in Chapter 5 where column design is checked for ULS limit state. The structural drawing of the column section is given in the attached annex. The cross-section is also depicted in the following figure:

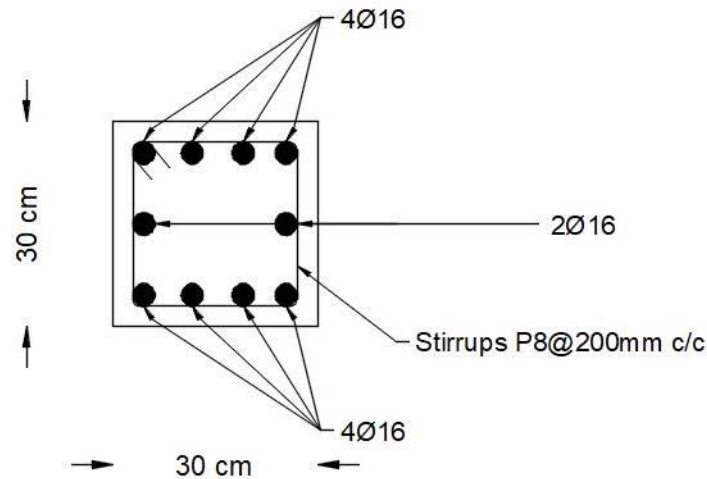


Figure 5.2: Column Cross-Section

5.2 Beam Design

For the purpose of design of the primary beam, a multi-bay beam was selected as shown below. The total load being applied to the beam was computed considering the influence strip of the subject multi bay-beam. The selected multi bay-beam is shown in the below figure. The strip width was multiplied by the applied ULS load to compute the line load on the beam. Since the load being applied on the bay- beam consists of both permanent and live load, the bending moment and the shear force calculations were made using the concept of the envelope. In order to compute the envelope, 'TRAVE' tool by *Prof. Gelfi* was used. The resulting envelopes are shown below. In order to account for the fixity condition on the end of the beams, the envelope was computed for both the cases, pinned and fixed ends. The envelopes for both of them are shown.

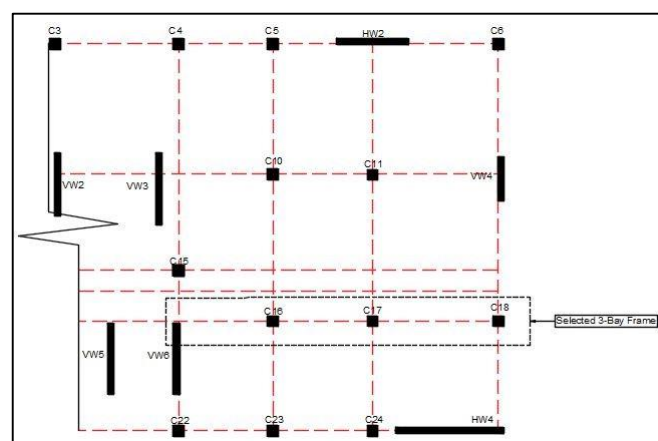


Figure 5.3: Selected Multi Bay Beam

The reinforcement design was developed for the sections depicted in Figure 4.8.

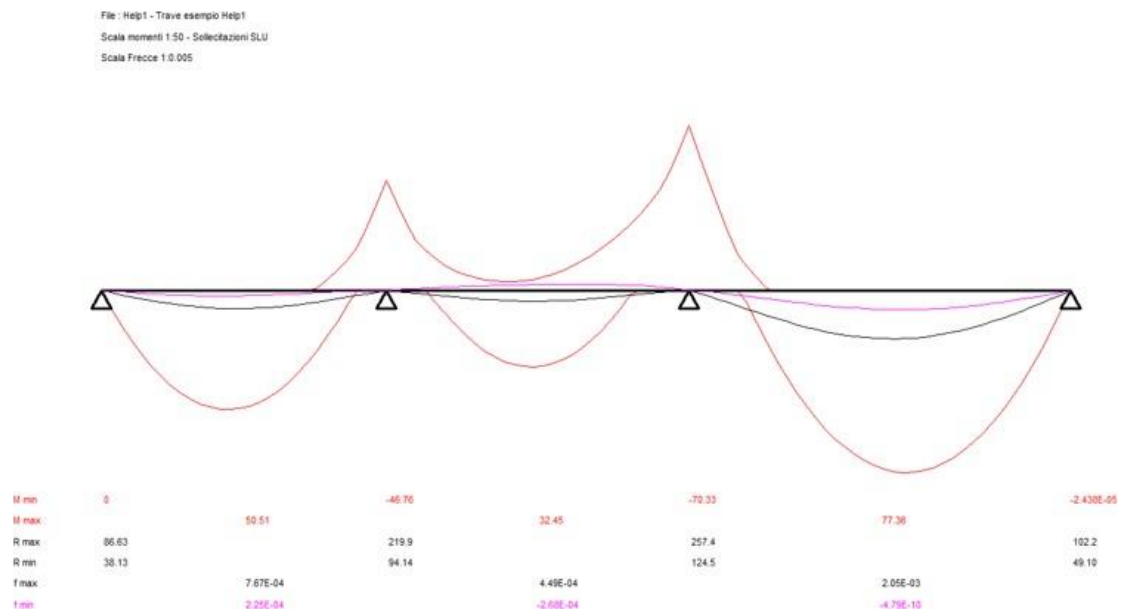


Figure 5.4: Bending Moment Envelope - Pinned Ends

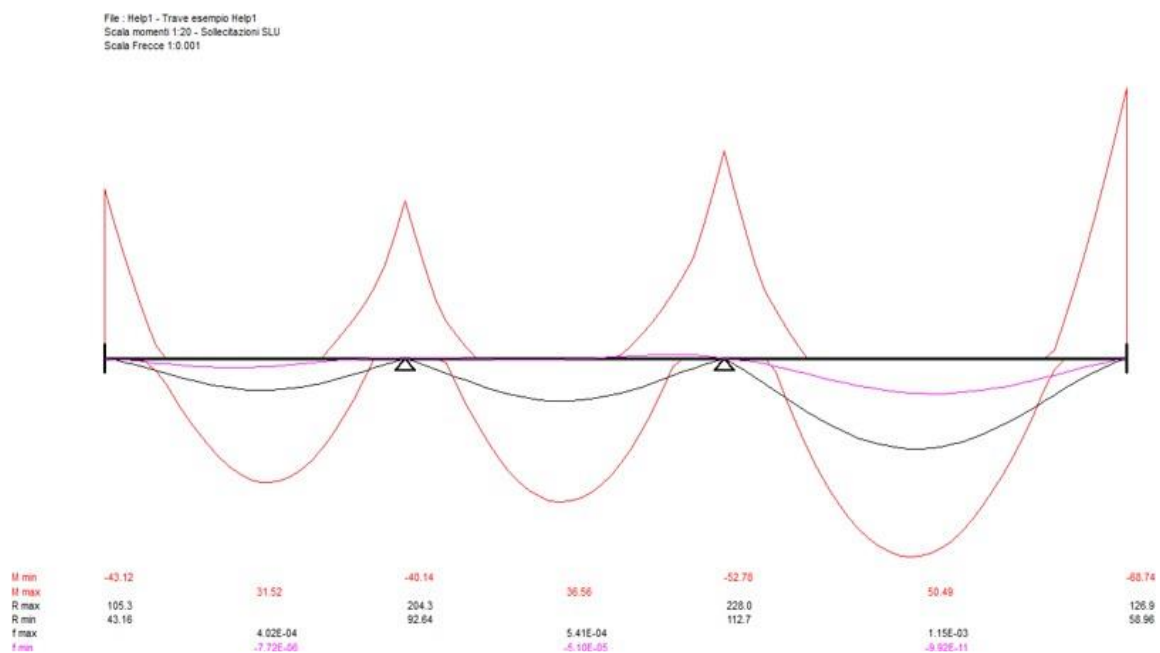


Figure 5.5: Bending Moment Envelope - Fixed Ends

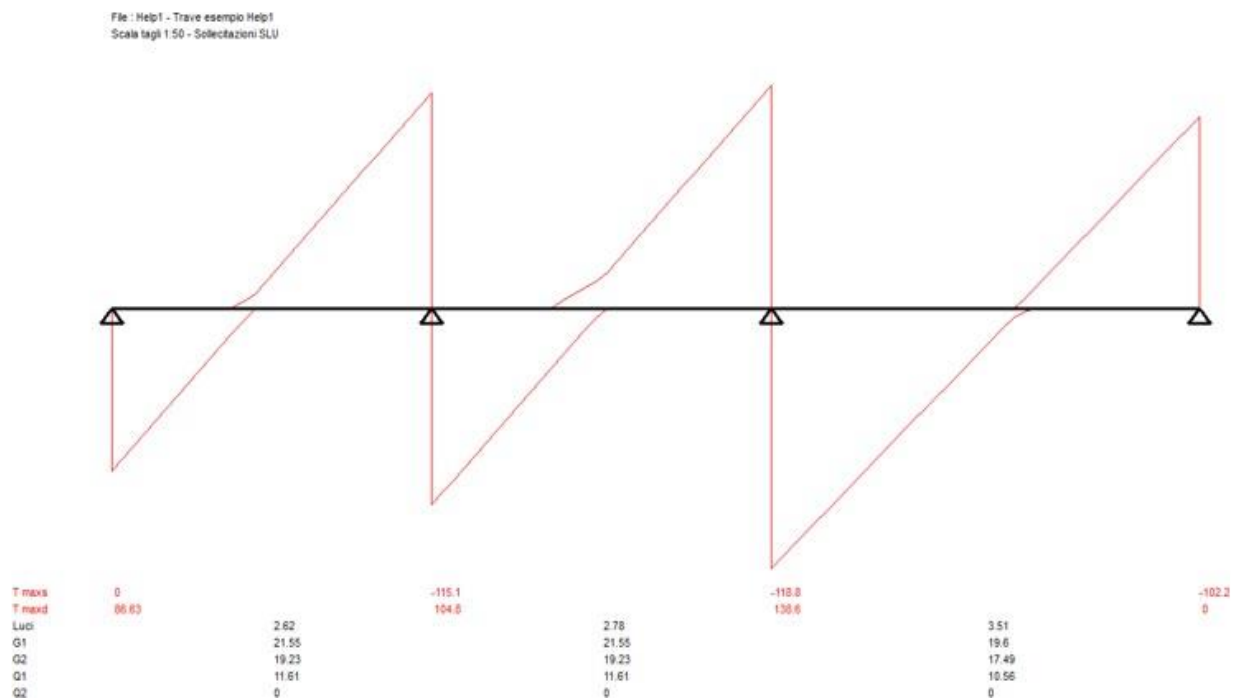


Figure 5.6: Shear Force Envelope - Pinned Ends

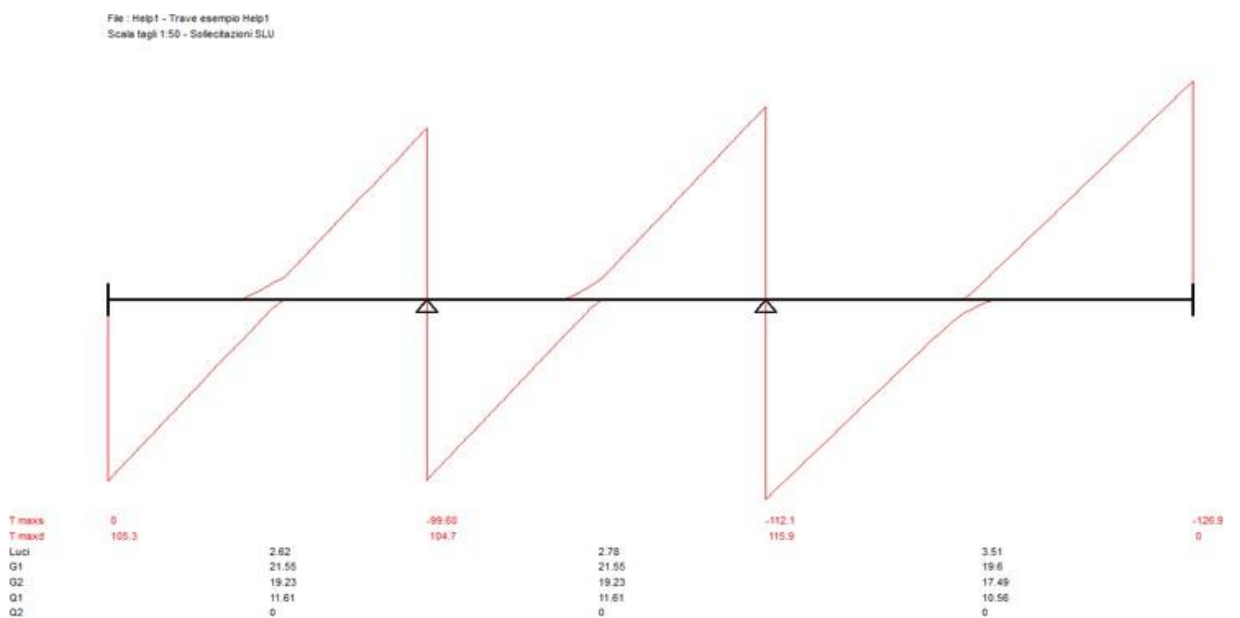


Figure 5.7: Shear Force Envelope - Fixed Ends

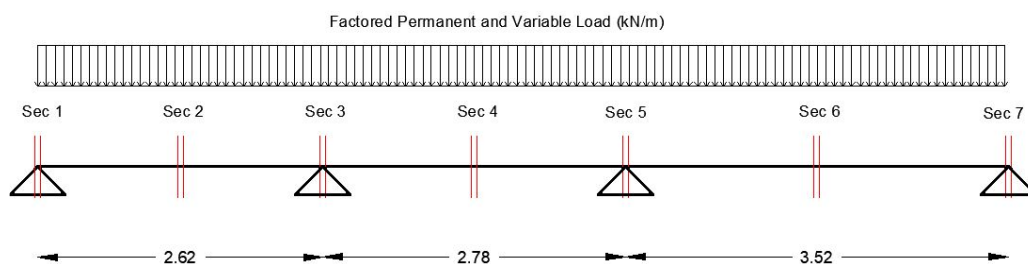


Figure 5.8: Multi Bay Beam and Selected Design Cross-Sections

For each of the sections, the positive (bottom) and negative (top) moment was selected from the absolute maximum of the two cases (i.e., fixed and pinned ends). The design cross-section of the beams was selected as 40cmx20cm. Based on the applied moments, the required minimum steel area was computed and resultantly steel bars were provided considering both the requirement and the ease of the construction. The requirement and the selected steel area along with number of the bars are tabulated below.

Table 5.2: Steel Area Requirement

		Sec 1	Sec 2	Sec 3	Sec 4	Sec 5	Sec 6	Sec 7
Applied Moment	kNm	-43.12	50.51	-46.76	36.56	-70.33	77.36	-68.74
b	mm	450	450	450	450	450	450	450
d	mm	170	170	170	170	170	170	170
Min Req. Area	mm ²	720.2	843.7	781.0	610.7	1174.7	1292.1	1148.2
Provided (No.16 Bar)	Top	6	6	6	6	8	8	8
	Bott	6	6	6	6	8	8	8
Provided Steel Area (mm²)	Top	1206	1206	1206	1206	1608	1608	1608
	Bott	1206	1206	1206	1206	1608	1608	1608

The structural drawings of the cross-sections are given in the annex. The cross sections are also illustrated in the following figures:

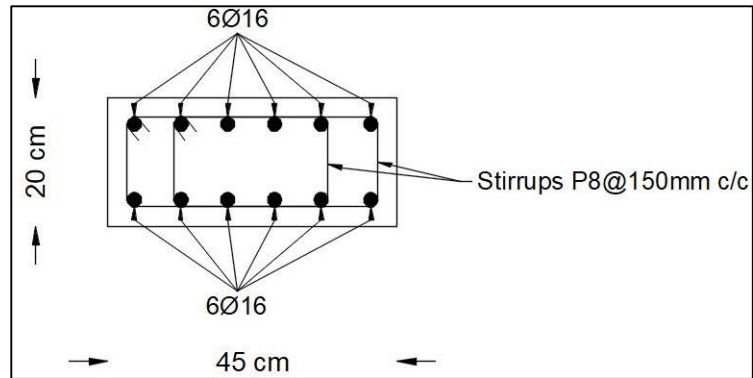


Figure 5.9: Beam Cross-Section - Sec 1,2,3,4

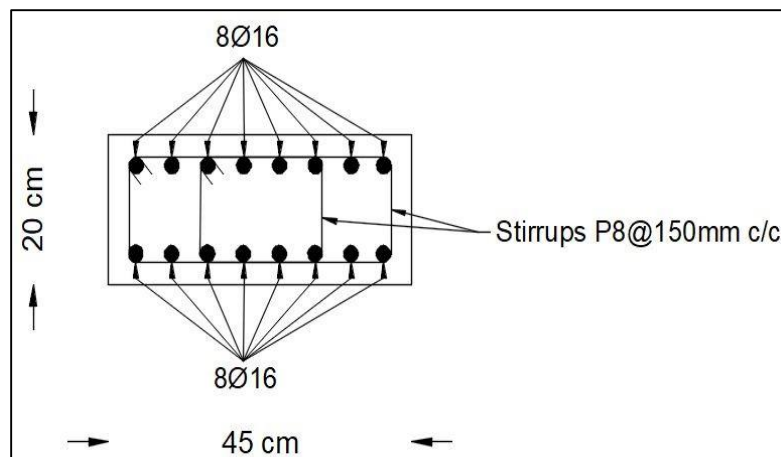


Figure 5.10: Beam Cross-Section - Sec 5,6,7

CHAPTER 6

CHECKS FOR GRAVITY DESIGN

The beam and column sections, designed in the previous chapter, are checked for ULS and SLS limits in the chapter.

6.1 ULS Check on Column

6.1.1 Axial and Flexure

The ULS check on the designed column was carried out considering the applied ULS axial load discussed in the previous chapter. Furthermore, the effect of Accidental Eccentricity is required to be considered. Based on the applied ULS axial load, the moment due to eccentricity is calculated as follows:

$$\text{Moment} = \text{Axial force} * \text{Eccentricity} = 842.84\text{kN} * 0.005\text{m} = 4.56 \text{ kNm}$$

The tool by Prof. Gelfi titled 'Verifica CA SLU' was used to calculate the capacity of the column. In addition to the above, the moment from the beam ends was also added which has a value of 79.8 kNm. Therefore, a total of bending moment equal to 83.49 kNm was considered for ULS Check.

Note: The authors are aware of the more robust method for the calculation of the bending moment to be taken by the subject column considering the ratio of column stiffness with beams at the joint. However, a conservative approach is taken here considering the conceptual nature of the design, and the complete bending moment is taken by the column.

The resulting M-N diagram is shown in Figure 6.4.1. It can be observed that the demand is well within the capacity.

The following checks are made for minimum and maximum steel area. The provided steel area is:

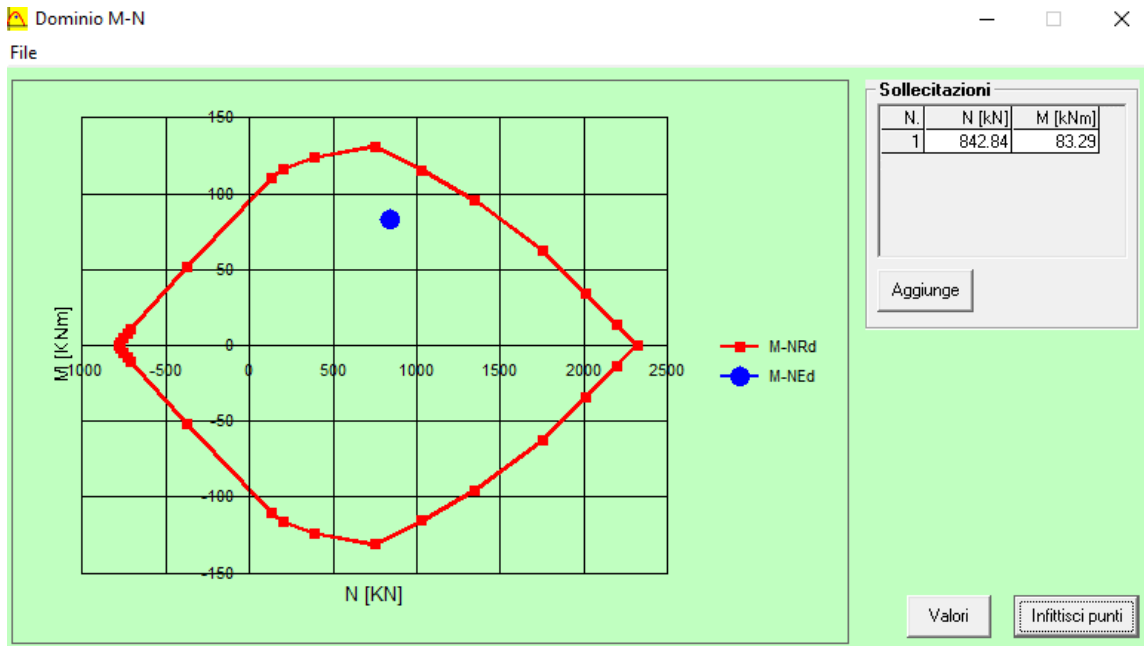


Figure 6.1: MN Interaction Diagram - Column

$$A_{s,provided} = 20.1 \text{ cm}^2 \text{ (10 - No.16 bars)}$$

$$A_{s,min} = 0.1 * \frac{N_{ed}}{f_{yd}} = \frac{0.1 * 842840 \text{ N}}{391 \text{ MPa}} = 2.15 \text{ cm}^2$$

$$A_{s,max} = 0.04 * A_c = 0.04 * 30 * 30 = 36 \text{ cm}^2$$

The provided area is within the computed extremes.

6.1.2. Shear

Taking the moment coming from beams and dividing by the height of the column, it is possible to have the first estimate of the shear acting on the column. The expression is given below:

$$V_{ed} = \gamma_{rd} * \frac{2M_{rd}}{H}$$

Where:

M_{rd} = Moment from the beam

H = Height of the column

The values of the parameters and corresponding applied shear are given in the following table.

Table 6.1: Acting Shear on Column

Acting Shear		
γ_{rd}	1.3	
$M_{rd,beam}$	-79.08	kNm
H	3	m
V_{ed}	68.54	kN

Proceeding with the design, the shear reinforcement 2-legged $\phi 8@20\text{cm}$ was chosen, and checks of the resisting shear could be performed, according to the code as per the below equations.

$$V_{rd} = \min(V_{Rsd}, V_{Rcd})$$

$$V_{Rsd} = \frac{A_{sw}}{S} * 0.9 * d * f_{ydw} * (\cot(\alpha) + \cot(\theta)) * \sin(\alpha)$$

$$V_{Rcd} = \frac{f_{cd}}{\cot(\theta) + \tan(\theta)} * 0.9 * d * b_w * \alpha_{cw} * v_1$$

Results are reported in the below table:

Table 6.2: Column Shear Capacity

Shear Capacity		
d	270	mm
z	243	mm
n	2	-
Bar Size	8	mm
A_{sw}	101	mm ²
s	200	mm
f_{yd}	391.3	MPa
θ	30	Degree
cot(θ)	1.73	-
b_w	300	mm
f_{cd}	17	MPa
α_{cw}	1	-
v	0.6	-
V_{rds}	92.77	kN
V_{rdmax}	231.34	kN
V_{rd}	92.77	kN

The shear capacity is higher than the applied shear force.

6.2 ULS Check on Beam

6.2.1 Flexure

As mentioned, the beam is analyzed for the most stressed midspan and the supporting sections as discussed in the previous chapter. The obtained results with reinforcement layouts were analyzed for capacity using the Verifica CA SLU tool. The capacity was computed at each section being referred. The capacities at each section are tabulated against demand in the below table.

Table 6.3: Beam Capacity at ULS

Bending Moment		Sec 1	Sec 2	Sec 3	Sec 4	Sec 5	Sec 6	Sec 7
Demand	kNm	-43.12	50.51	-46.76	36.56	-70.33	77.36	-68.74
Capacity - Top	kNm	-69.36	-69.36	-69.36	-69.36	-91.28	-91.28	-91.28
Capacity - Bottom	kNm	69.36	69.36	69.36	69.36	91.28	91.28	91.28

The check for steel ratio is carried below:

$$A_{s,provided} = 16.08 \text{ cm}^2 \text{ (8 * 16mm bars)}$$

$$A_{s,min} = \max\left(0.26 * \frac{f_{ctm}}{f_{yd}} * b_t * d, 0.0013 * b_t * d\right) = 151 \text{ mm}^2$$

$$A_{s,max} = 0.04 * A_c = 0.04 * 30 * 30 = 36 \text{ cm}^2$$

The steel area check is satisfied.

The cross-section shall also be checked for minimum bar spacing. As per EC2, the clear bar spacing should not be less than the maximum of the bar diameter, maximum aggregate size (mm) + 2mm and 20mm. Considering the bar size of 16mm and maximum aggregate size of 37.5mm (1-1/2 in), the limit comes out to be 39.5 mm.

Based on the beam cross-section, considering the beam width of 450mm and 8x16 bars, the clear bar distance comes out to be 40 mm (considering the cover of 30mm and a total of 7 spaces in between). This satisfies the clear space requirement.

6.2.2 Shear

After the bending case, shear checks were performed for the beam. The applied shear load is $V_{ed} = 138.6 \text{ kN}$. According to the chosen stirrup's layout, the resisting shear

was computed as the minimum value between the resisting shear of concrete and the one of steel. Then, it was compared to the demanding shear. The computation of the resisting shear forces was done as follows. The shear reinforcement 4-legged $\phi 8@15\text{cm}$ was chosen:

$$V_{rd} = \min (V_{Rsd}, V_{Rcd})$$

$$V_{Rsd} = \frac{A_{sw}}{s} * 0.9 * d * f_{ydw} * (\cot(\alpha) + \cot(\theta)) * \sin(\alpha)$$

$$V_{Rcd} = \frac{f_{cd}}{\cot(\theta) + \tan(\theta)} * 0.9 * d * b_w * \alpha_{cw} * v_1$$

Table 6.4: Beam Shear Capacity

A_{sw}	201	mm ²
s	150	mm
d	175	mm
b	450	mm
f_{yd}	391.3	MPa
a_{cw}	1	-
θ	30	degree
f_{cd}	17	MPa
v₁	0.51	-
α	0.78540	rad
V_{Rsd}	159.54	kN
V_{Rcd}	224.92	kN
V_{rd}	159.54	kN

Thus the shear capacity is satisfied.

6.3 Strong Column-Weak Beam Check

Finally, considering the seismic design, the capacity rule was applied which assumes columns are stronger than beams, it follows:

$$\sum M_{Rdc} = 252 \text{ KN.m} > 1.3 * \sum M_{Rdb} = 237.3 \text{ KN.m}$$

The check is only applied to the reference column considering the beam section at the corresponding joint. In the case of a detailed design, the check should be repeated for each column.

6.4 SLS Check on Beam

Section 4.1.2.2.1 of NTC2018 prescribes the verification of the following limit states:

- Deformation limit state
- Cracking limit state
- Stress limitation limit state
- Vibration limit state

The first three checks were performed for the selected bay-beam. Due to the residential category and the lack of information on the surrounding environment, the vibration limit state check was not performed for any structural element.

6.4.1 Deformation limit state

According to section C4.1.2.2.2 of CM2019, the check can be implicitly considered respected for spans lower than 10m, if their span-to-depth ratio (L/h) is below a certain limit. Considering the multi-bay beam scheme, the beam cross-section, and the reinforcement from the ULS design, the check was referred to as the longest span (of length equal to 3.63m). The limit L/h ratio was computed using the below formula:

$$\frac{l}{h} \leq K \left[11 + \frac{0.015 f_{ck}}{\rho + \rho'} \right] \left[\frac{500 A_{s,eff}}{f_{yk} A_{s,calc}} \right]$$

The limit ratio comes out to be '18'. Considering the longest span of 3.63m and beam depth of 20cm, the actual L/h ratio is 18 which follows the limit value. Thus the check is satisfied.

6.4.2 Cracking limit state

Section 4.1.2.2.4 of NTC2018 states that the cracking check to be performed depends on several factors such as environmental exposure class, load combination, and reinforcement sensitivity to the corrosion. Assuming not to be in an aggressive environment and having ordinary rebars not susceptible to corrosion, the structure is in class A, for which the crack opening check must be performed. Moreover, the code also defines limit crack openings $w_3 = 0.4\text{mm}$ for the frequent load combination and $w_2 = 0.3\text{mm}$ for the quasi-permanent load combination.

Checks could be performed in the form of the stress limitation check for the two different cases depending on the minimum bar diameter present in the beam longitudinal reinforcement.

Following section C4.1.2.2.4.5 of CM2019. Tensile stress limits are prescribed by Table C4.1 II of CM2019 and chosen values are taken by the interpolation because parameters regarding the adopted bars diameters defined in Chapter 4 (Section 4.2) are not present.

Stresses in tensioned steel are computed using “Verifica CA SLU” software for sections of the maximum acting bending moment. Finally, values of stresses are reported in the following table for both frequent and quasi-permanent load combinations, performing comparison both at the midspan and at the support, due to the most severe moment applied. Hence, it results that all those checks were satisfied.

Table 6.5: Comparison of stresses at frequent and quasi-permanent load combination

Frequent	Applied Moment (kN.m)	Stress (MPa)
Max Allowable	-	280
At Support	-43.79	191
At Midspan	48.04	209
Quasi-Permanent	Applied Moment (kNm)	Stress (MPa)
Max Allowable	-	240
At Support	-41.49	181
At Midspan	45.47	198

6.4.3 Stress Limit State

According to section 4.1.2.2.5 of NTC2018, some stress limitations are imposed for different load combinations both for concrete and steel.

Steel is imposed at the characteristic load combination with $\sigma_s \leq 0.8f_{yk}$. Although this condition is considered as verified yet because the cracking limit state is already verified since it has a more restrictive condition.

The limit stress for concrete is imposed as $\sigma_c \leq 0.6f_{ck}$ at the characteristic load combination and as $\sigma_c \leq 0.45f_{ck}$ at the quasi-permanent load combination.

As done previously, moments and correspondent stresses are computed in both most loaded midspan by means of simple software calculation. The values are reported in the below table. The checks are observed to be satisfactory.

Table 6.6: Comparison of stresses at characteristic and quasi-permanent load combination

Characteristic	Stress (MPa)
Max Allowable	-18.0
At Support	-10.4
At Midspan	-11.4
Quasi-Permanent	Stress (MPa)
Max Allowable	-13.5
At Support	-9.3
At Midspan	-10.2

CHAPTER 7

PRELIMINARY SEISMIC DESIGN

7.1 Seismic Mass

Seismic mass was evaluated according to NTC18, 2.5.3 in seismic configuration.

$$M = G_1 + G_2 + \sum_j \psi_{2j} \cdot Q_{kj}$$

Where:

M : Seismic Mass;

G_1 : Structural Permanent Load;

G_2 : Non-structural Permanent Load;

ψ_{2j} : 0.3 – Coefficient for residential use (NTC18, Tab. 2.5.I);

Q_{kj} : Variable Loads, 2kN/m²

The calculations for the structural and non-structural permanent loads and variable loads were carried out earlier in the form of area load. The same have been recalled here to calculate the mass of each floor considering the floor area. The results are summarized in the table below for the floors and the roof.

Table 7.1: Seismic Mass - Ground Floor

Ground Floor		
G₁	2.72	kN/m ²
G₂	3.31	kN/m ²
Q	2	kN/m ²
S_i	0.3	-
S_i * Q	0.6	kN/m ²

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Table 7.1: Seismic Mass - Ground Floor
(Continued)

Total	6.63	kN/m ²
Floor Area	240.1	m ²
Total Load	1592.46	kN
	160	tons

Table 7.2: Seismic Mass - All floors except
ground and roof

All floors except ground and roof		
G₁	2.72	kN/m ²
G₂	3.31	kN/m ²
Q	2	kN/m ²
S_i	0.3	
S_i * Q	0.6	kN/m ²
Total	6.63	kN/m ²
Floor Area	228.29	m ²
Total Load	1514.13	kN
	152	tons

Table 7.3: Seismic Mass - Roof

Roof		
G₁	2.72	kN/m ²
G₁	3.31	kN/m ²
Q	2	kN/m ²
S_i	0.3	-
S_i * Q	0.6	kN/m ²
Total	6.63	kN/m ²
Floor Area	226.06	m ²
Total Load	1499.34	kN
	150	tons

In addition to the above loads, the mass of columns and shear walls is needed to be

considered for the sake of completeness. Therefore, considering the geometry and material of the columns and shear walls, the following tables summarize the mass of columns and shear walls to be added.

Table 7.4: Seismic Mass - Columns

Columns		
Area single column	0.09	m ²
Height	3	m
Unit Weight	25	kN/m ³
Volume	0.27	m ³
No: of columns	24	-
Total weight x floor	162	kN
Mass	16.26	ton

Table 7.5: Seismic Mass - Shear Walls

Shear Walls		
Thickness	0.25	m
Linear meter	18.85	m
Height	3	m
Unit weight	25	kN/m ³
Volume	14.14	m ³
Total weight x floor	353.4	kN
Mass	35.47	ton

Conclusively, the floor mass for each storey is summarized in the table below:

Table 7.6: Total Seismic Mass

Floor	Mass (Ton)
Ground	211.55
1	203.69
2	203.69
3	203.69
4	203.69
Roof	202.21

Table 7.6: Total Seismic Mass
(Continued)

Total	1228.5
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7.2 Seismic Shear

According to section 7.3.3.2 NTC2018, the linear static analysis could be applied if the structure is regular in elevation, and its natural period of vibrations satisfies the following inequality:

$$T_1 < 2.5T_c = 2.5 \cdot 0.61s = 1.525s$$

Simplified formula for computation of T_1 was applied, according to C7.3.3.2:

$$T_1 = C_t \times H^{3/4} = 0.075 \times (18)^{3/4} = 0.66s$$

The check for the time period was satisfied, and the structure was considered regular in the elevation, thus the linear static analysis could be applied in order to obtain the distribution of the seismic shear along the building. The seismic shear per floor is computed as:

$$F_i = V_b \frac{\sum z_i M_i}{z_j M_j}$$

$$V_b = S_d(T_1) \cdot M_{\text{tot}} \cdot \lambda = 0.182 \times 9.81 \times 1228.5 \times 0.85 = 1864.41 \text{ kN}$$

Where:

V_b – seismic base shear force;

z_i – height of i-floor from the ground level;

M_i – mass of i-floor;

$S_d(T_1)$ – ordinate of the design response spectrum at T_1 ; λ

– correction factor

Considering the equation for the distribution of shear force stated above the resulting calculation for shear force distribution at each floor is given below:

Table 7.7: Seismic Shear Distribution

Floor	$z_i(m)$	$m_i(t)$	$z_i \times m_i$	$F_i(kN)$
Ground	3	211.55	634.65	92.23
1	6	203.69	1222.14	177.61
2	9	203.69	1833.21	266.41

Continued on next page

Table 7.7: Seismic Shear Distribution (Continued)

3	12	203.69	2444.28	355.21
4	15	203.69	3055.35	444.01
Roof	18	202.21	3639.7	528.93
	Total	1228.52	12829.34	1864.41

7.3 Shear force distribution among lateral load resisting system

In order to determine the distribution of the unit seismic base shear in the horizontal plane (X and Y) of the structure, both wall and column properties were taken into account. Contributions from all columns were included in both directions in the calculation of the distribution of unit seismic base shear. The center of mass (CM) was found with respect to a reference point (QPF is shown in the plans given in the drawing). On the other hand, the center of stiffness (CR) was calculated with respect to the center of mass.

After finding the values of the moment of inertia for each structural element, the dimensionless parameters S'_{ix} , S'_{iy} , S''_{ix} and S''_{iy} were computed as follows:

$$S'_{ix} = I_{iy} / \sum I_{iy}$$

$$S'_{iy} = I_{ix} / \sum I_{ix}$$

$$S''_{ix} = e_y \times y_i \times I_{iy} / \sum (x_i^2 \times I_{ix} + y_i^2 \times I_{iy})$$

$$S''_{iy} = e_x \times x_i \times I_{ix} / \sum (x_i^2 \times I_{ix} + y_i^2 \times I_{iy})$$

Then, these contributions were summed up in each direction in order to find the distribution of the unit seismic base shear on the floor as:

$$S'_{ix} + S''_{ix}$$

$$S'_{iy} + S''_{iy}$$

The table below summarizes the aforementioned calculation of base shear redistribution for the X and Y directions. Moreover, it can be inferred from the tables that 97.60% of the seismic base shear is taken by the shear walls in the X direction and 97.29% by the shear walls in the Y direction. The contribution of each column is relatively small to the walls. Therefore, since walls play a more dominant role in stiffness distribution, taking much more than 65% of the seismic base shear, the structural system of the building is defined as a Wall System.

Table 7.8: Distribution of unit seismic base shear in the y-direction

i	I_{ix} (m ⁴)	S'_{iy} (m)	x'_i (m)	$x'_i S'_{iy}$ (m)	x_i	$x_i I_{ix}$ (m ⁵)	$x_i^2 I_{ix}$ (m ⁶)	$x_i^2 I_{ix} + y_i^2 I_{iy}$	$S_i''y$	S_{iy}
VW1	0.1000	0.1982	-11.39	-2.2572	-9.77	-0.9773	9.5506	9.5522	0.0000	0.1982
VW2	0.0670	0.1328	-1.85	-0.2453	-0.23	-0.0156	0.0036	0.0141	0.0000	0.1328
VW3	0.1000	0.1982	0.97	0.1928	2.59	0.2587	0.6694	0.6807	0.0000	0.1982
VW4	0.0216	0.0428	10.52	0.4505	12.14	0.2622	3.1820	3.1906	0.0000	0.0428
VW5	0.1000	0.1982	-0.36	-0.0708	1.26	0.1257	0.1581	0.1643	0.0000	0.1982
VW6	0.1000	0.1982	1.47	0.2919	3.09	0.3087	0.9531	0.9593	0.0000	0.1982
HW1	0.0016	0.0031	-5.54	-0.0172	-3.92	-0.0061	0.0240	4.7396	0.0000	0.0031
HW2	0.0016	0.0031	6.94	0.0215	8.56	0.0134	0.1144	4.8299	0.0000	0.0031
HW3	0.0007	0.0014	-11.02	-0.0154	-9.40	-0.0066	0.0622	0.2082	0.0000	0.0014
HW4	0.0023	0.0046	9.35	0.0435	10.97	0.0257	0.2819	5.6901	0.0000	0.0046
C1	0.0004	0.0008	-11.32	-0.0091	-9.70	-0.0039	0.0381	0.0563	0.0000	0.0008
C2	0.0004	0.0008	-8.47	-0.0068	-6.85	-0.0028	0.0190	0.0372	0.0000	0.0008
C3	0.0004	0.0008	-1.65	-0.0013	-0.03	0.0000	0.0000	0.0182	0.0000	0.0008
C4	0.0004	0.0008	1.79	0.0014	3.41	0.0014	0.0047	0.0229	0.0000	0.0008
C5	0.0004	0.0008	4.42	0.0036	6.04	0.0024	0.0148	0.0330	0.0000	0.0008
C6	0.0004	0.0008	10.71	0.0086	12.33	0.0050	0.0615	0.0798	0.0000	0.0008
C7	0.0004	0.0008	-11.32	-0.0091	-9.70	-0.0039	0.0381	0.0419	0.0000	0.0008

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Table 7.8: Distribution of unit seismic base shear in the y-direction (Continued)

C8	0.0004	0.0008	-8.47	-0.0068	-6.85	-0.0028	0.0190	0.0228	0.0000	0.0008
C9	0.0004	0.0008	-5.28	-0.0042	-3.66	-0.0015	0.0054	0.0092	0.0000	0.0008
C10	0.0004	0.0008	4.42	0.0036	6.04	0.0024	0.0148	0.0186	0.0000	0.0008
C11	0.0004	0.0008	7.20	0.0058	8.82	0.0036	0.0315	0.0353	0.0000	0.0008
C12	0.0004	0.0008	-8.47	-0.0068	-6.85	-0.0028	0.0190	0.0191	0.0000	0.0008
C13	0.0004	0.0008	-5.28	-0.0042	-3.66	-0.0015	0.0054	0.0054	0.0000	0.0008
C14	0.0004	0.0008	-1.65	-0.0013	-0.03	0.0000	0.0000	0.0000	0.0000	0.0008
C15	0.0004	0.0008	1.79	0.0014	3.41	0.0014	0.0047	0.0048	0.0000	0.0008
C16	0.0004	0.0008	4.42	0.0036	6.04	0.0024	0.0148	0.0152	0.0000	0.0008
C17	0.0004	0.0008	7.20	0.0058	8.82	0.0036	0.0315	0.0319	0.0000	0.0008
C18	0.0004	0.0008	10.71	0.0086	12.33	0.0050	0.0615	0.0620	0.0000	0.0008
C19	0.0004	0.0008	-8.47	-0.0068	-6.85	-0.0028	0.0190	0.0258	0.0000	0.0008
C20	0.0004	0.0008	-5.28	-0.0042	-3.66	-0.0015	0.0054	0.0123	0.0000	0.0008
C21	0.0004	0.0008	-1.65	-0.0013	-0.03	0.0000	0.0000	0.0068	0.0000	0.0008
C22	0.0004	0.0008	1.79	0.0014	3.41	0.0014	0.0047	0.0115	0.0000	0.0008
C23	0.0004	0.0008	4.42	0.0036	6.04	0.0024	0.0148	0.0216	0.0000	0.0008
C24	0.0004	0.0008	7.20	0.0058	8.82	0.0036	0.0315	0.0383	0.0000	0.0008
	0.5045	1.0		-1.62			15.4587	30.6591		1.0

Table 7.9: Distribution of unit seismic base shear in the x-direction

i	I_{iy} (m ⁴)	S'_{ix} (m)	y'_i (m)	$y'_i S'_{ix}$ (m)	y_i	$y_i I_{iy}$ (m ⁵)	$y_i^2 * I_{iy}$ (m ⁶)	$x_i^2 * I_{ix} + y_i^2 * I_{iy}$	$S_i''x$	S_{ix}
VW1	0.0016	0.0028	2.1429	0.0059	1.0230	0.0016	0.0016	9.5522	0.0000	0.0028
VW2	0.0014	0.0024	-1.6471	-0.0040	-2.7670	-0.0038	0.0105	0.0141	0.0000	0.0024
VW3	0.0016	0.0028	-1.5671	-0.0043	-2.6870	-0.0042	0.0113	0.6807	0.0000	0.0028
VW4	0.0009	0.0017	-1.9171	-0.0032	-3.0370	-0.0028	0.0086	3.1906	0.0000	0.0017
VW5	0.0016	0.0028	3.1129	0.0086	1.9930	0.0031	0.0062	0.1643	0.0000	0.0028
VW6	0.0016	0.0028	3.1129	0.0086	1.9930	0.0031	0.0062	0.9593	0.0000	0.0028
HW1	0.1000	0.1770	-5.7471	-1.0174	-6.8670	-0.6867	4.7155	4.7396	0.0000	0.1770
HW2	0.1000	0.1770	-5.7471	-1.0174	-6.8670	-0.6867	4.7155	4.8299	0.0000	0.1770
HW3	0.0091	0.0161	5.1229	0.0826	4.0030	0.0365	0.1460	0.2082	0.0000	0.0161
HW4	0.3375	0.5975	5.1229	3.0607	4.0030	1.3510	5.4082	5.6901	0.0000	0.5975
C1	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0563	0.0000	0.0007
C2	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0372	0.0000	0.0007
C3	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0182	0.0000	0.0007
C4	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0229	0.0000	0.0007
C5	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0330	0.0000	0.0007
C6	0.0004	0.0007	-5.5871	-0.0040	-6.7070	-0.0027	0.0182	0.0798	0.0000	0.0007
C7	0.0004	0.0007	-1.9471	-0.0014	-3.0670	-0.0012	0.0038	0.0419	0.0000	0.0007

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Table 7.9: Distribution of unit seismic base shear in the x-direction (Continued)

C8	0.0004	0.0007	-1.9471	-0.0014	-3.0670	-0.0012	0.0038	0.0228	0.0000	0.0007
C9	0.0004	0.0007	-1.9471	-0.0014	-3.0670	-0.0012	0.0038	0.0092	0.0000	0.0007
C10	0.0004	0.0007	-1.9471	-0.0014	-3.0670	-0.0012	0.0038	0.0186	0.0000	0.0007
C11	0.0004	0.0007	-1.9471	-0.0014	-3.0670	-0.0012	0.0038	0.0353	0.0000	0.0007
C12	0.0004	0.0007	1.5329	0.0011	0.4130	0.0002	0.0001	0.0191	0.0000	0.0007
C13	0.0004	0.0007	1.3129	0.0009	0.1930	0.0001	0.0000	0.0054	0.0000	0.0007
C14	0.0004	0.0007	1.3129	0.0009	0.1930	0.0001	0.0000	0.0000	0.0000	0.0007
C15	0.0004	0.0007	0.7329	0.0005	-0.3870	-0.0002	0.0001	0.0048	0.0000	0.0007
C16	0.0004	0.0007	2.1629	0.0016	1.0430	0.0004	0.0004	0.0152	0.0000	0.0007
C17	0.0004	0.0007	2.1629	0.0016	1.0430	0.0004	0.0004	0.0319	0.0000	0.0007
C18	0.0004	0.0007	2.1629	0.0016	1.0430	0.0004	0.0004	0.0620	0.0000	0.0007
C19	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0258	0.0000	0.0007
C20	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0123	0.0000	0.0007
C21	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0068	0.0000	0.0007
C22	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0115	0.0000	0.0007
C23	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0216	0.0000	0.0007
C24	0.0004	0.0007	5.2229	0.0037	4.1030	0.0017	0.0068	0.0383	0.0000	0.0007
	0.5649	1.0		1.12			15.2005	30.6591		1.0

7.4 Regularity Check

In general, the regularity in the plan can be checked when the structural model is defined. The criteria for regularity in the plan are described in EN 1998-1 (4.2.3.2):

- The slenderness of the building shall be not higher than 4 ($\lambda = \frac{L_{\max}}{L_{\min}}$)
- The structural eccentricity shall be smaller than 30% of the torsional radius ($e_{ox} \leq 0.30r_x$, $e_{oy} \leq 0.30r_y$).
- The torsional radius shall be larger than the radius of the gyration of the floor mass in plan ($r_x \geq l_s$, $r_y \geq l_s$).

Considering the first point, the check is satisfied since the $\lambda = L_{\max}/L_{\min} = 22/11 = 2$ is less than 4.

The eccentricity of the plan was computed by determining the center of stiffness with respect to the center of gravity. The resulting eccentricity is computed as:

$$e_x = 1.62 \text{ m}$$

$$e_y = 1.12 \text{ m}$$

In order to compute the admissibility of the eccentricity, the torsional radii are computed using the following formula.

$$x_{\text{CR}} = \frac{\sum (x E I_y)}{\sum (E I_y)}; \quad y_{\text{CR}} = \frac{\sum (y E I_x)}{\sum (E I_x)}$$

$$r_x = \sqrt{\frac{\sum [(x - x_{\text{CR}})^2 E I_y + (y - y_{\text{CR}})^2 E I_x]}{\sum (E I_y)}}$$

$$r_y = \sqrt{\frac{\sum [(x - x_{\text{CR}})^2 E I_y + (y - y_{\text{CR}})^2 E I_x]}{\sum (E I_x)}}$$

The torsional radii were determined as:

$$r_x = 7.37 \text{ m}$$

$$r_y = 7.79 \text{ m}$$

The check for eccentricity is:

$$e_x = 1.62 \text{ m} < 0.3r_x = 0.3 \times 7.37 = 2.21 \text{ m}$$

$$e_y = 1.12 \text{ m} < 0.3r_y = 0.3 \times 7.79 = 2.34 \text{ m}$$

The check is satisfied for both directions. Therefore, the second condition is satisfied. Considering the third condition, the radius of gyration of the floor mass is defined as:

$$A = 22 \text{ m}$$

$$B = 11 \text{ m}$$

$$I_s = \sqrt{\frac{A^2 + B^2}{12}} = 7.1 \text{ m}$$

Since, both r_x and r_y are greater than I_s , therefore, the third condition for regularity is satisfied.

The above-discussed geometrical parameters are shown for a typical plan in the below figure.

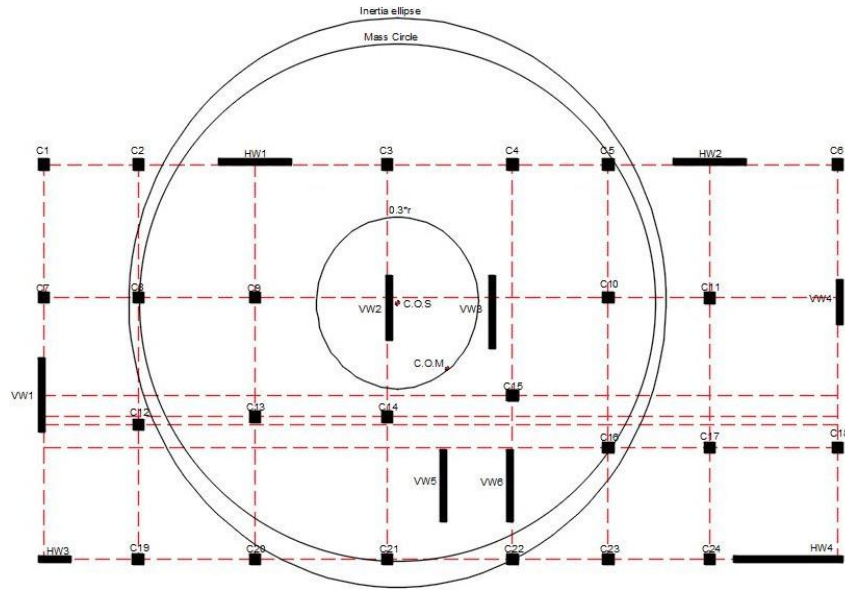


Figure 7.1: Plan Eccentricity

7.5 Seismic Vulnerability

The seismic vulnerability of the building is assessed based on two indices:

- Wall Area Index - I
- Wall Area Index - II

The first index is related to an outline of earthquake provisions from the Japanese Building Code (Aoyama H,1981) and the second one is a study by Hassan and Sozen based on the assessment of seismic vulnerability.

7.5.1 Wall Area Index - I

The seismic vulnerability is assessed by plotting the ratio of the total wall area to floor area (WI%) against the average shear stress computed by dividing weight by the cumulative cross-sectional area of walls and columns. The parameters are plotted by superimposing them over a literature graph from the Japanese Building Code. The values of the indices are given in the table below for each of the floors and cumulatively. In each of these cases, it is observed that:

WI% is greater than 0.3% and Stress is less than 1.2; therefore, the check is within the safe region of the literature graph (see picture below with superimposed points for both X and Y directions).

7.5.2 Wall Area Index - II

In order to assess the Wall Area Index II, the indices proposed by Hassan and Sozen are to be computed. The formulation of both indices is given below:

$$WI\% = \frac{A_{wt}}{A_{ft}} * 100 \quad (6.2)$$

$$CI\% = \frac{A_{ce}}{A_{ft}} * 100 \quad (6.3)$$

Where:

- WI and CI represent the wall index and column index, respectively;
- $A_{wt} = A_{cw} + A_{mw}/10$ = Effective cross-sectional area of walls in a given horizontal direction;
- A_{cw} is the total cross-sectional area of the reinforced concrete walls in one horizontal direction at the base;
- A_{mw} is the cross-sectional area of the non-reinforced masonry filler walls in one horizontal direction at the base;
- A_{ft} is the total floor area above the base;
- $A_{ce} = A_{col}/2$ = Effective cross-sectional area of columns at the base;
- A_{col} is the total cross-sectional area of columns above the base.

Based on the formulas above, the wall and column indexes were calculated for each floor and are reported in the following tables.

Table 7.10: Area Indices in X-Direction

Wall Area Index (X-Direction)	Wall Area Index 1							Wall Area Index 2		
Floor	A _w [m ²]	A _c [m ²]	A _f [m ²]	W _I [%]	A _w + A _c [m ²]	W [kN]	W/(A _w + A _c) [MPa]	C _I [%]	A _{mw} [m ²]	W _I [%]
0	1.975	2.16	240.1	0.82%	4.14	-	-	0.45%	3.24	0.96%
1	1.975	2.16	228.29	0.87%	4.14	-	-	0.47%	3.65	1.03%
2	1.975	2.07	228.29	0.87%	4.05	-	-	0.45%	3.21	1.01%
3	1.975	2.07	228.29	0.87%	4.05	-	-	0.45%	3.21	1.01%
4	1.975	2.07	228.29	0.87%	4.05	-	-	0.45%	3.21	1.01%
5	1.975	2.07	228.29	0.87%	4.05	-	-	0.45%	3.21	1.01%
6	1.00	1.66	226.06	0.44%	2.66	-	-	0.37%	3.13	0.58%
Total	12.85	14.26	1607.61	0.80%	27.11	12240.96	0.45	0.44%	22.86	0.94%

Table 7.11: Area Indices in Y-Direction

Wall Area Index (Y-Direction)	Wall Area Index 1							Wall Area Index 2		
Floor	A _w [m ²]	A _c [m ²]	A _f [m ²]	W _I [%]	A _w + A _c [m ²]	W [kN]	W/(A _w + A _c) [MPa]	C _I [%]	A _{mw} [m ²]	W _I [%]
0	2.74	2.16	240.1	1.14%	4.90	-	-	0.45%	1.98	1.22%
1	2.74	2.16	228.29	1.20%	4.90	-	-	0.47%	1.9	1.28%
2	2.74	2.07	228.29	1.20%	4.81	-	-	0.45%	1.54	1.27%
3	2.74	2.07	228.29	1.20%	4.81	-	-	0.45%	1.54	1.27%
4	2.74	2.07	228.29	1.20%	4.81	-	-	0.45%	1.54	1.27%
5	2.74	2.07	228.29	1.20%	4.81	-	-	0.45%	1.54	1.27%
6	2.74	1.66	226.06	1.21%	4.39	-	-	0.37%	0.5	1.23%
Total	19.16	14.26	1607.61	1.19%	33.42	12240.96	0.37	0.44%	10.54	1.26%

The values of WI and CI are superimposed on the limit graph by Hassan and Sozen for each of the directions. It can be observed that the indices are out of the vulnerable zone.

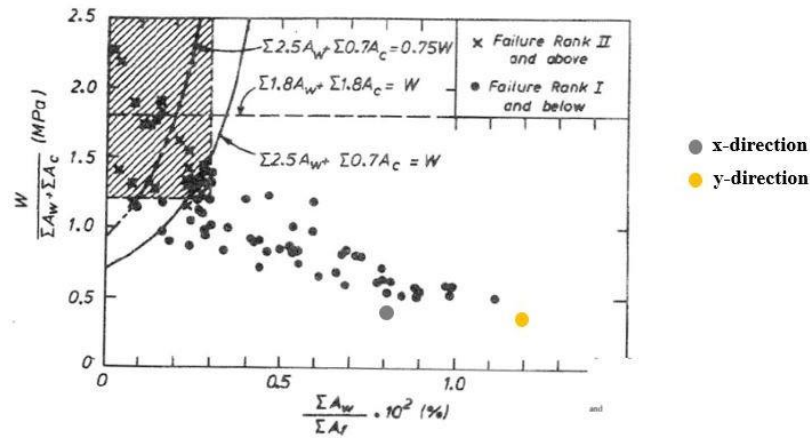


Figure 7.2: Area Index - 1

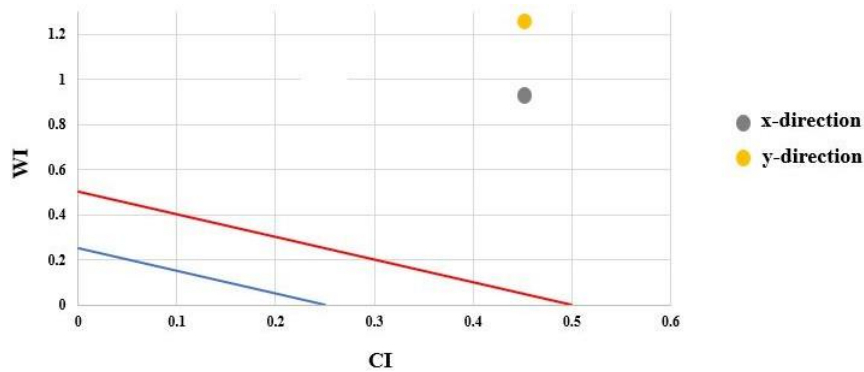


Figure 7.3: Area Index - 2

7.6 Deformability Checks

Checks for deformability were performed verifying the inter-story drift according to NTC 7.3.6. To compute it, the reference was to a simple cantilever model with lumped masses at floors. The loading were applied as obtained from analysis of base shear done earlier. The modelling of the deformable cantilever was performed considering the total inertia of columns and walls for each side. From displacement values shown in the model results of the following figure, it was evaluated the maximum inter story drift of 20.79 mm along Y direction and of 18.57 mm along X direction. Moreover, it is assumed that connections between structural and non-structural elements do not interfere with structural deformations. So, it results that:

$$d_{ry} = 20.79 \text{ m} < 0.01 \cdot h_i = 0.01 \cdot 3000 = 30 \text{ mm}$$

$$d_{rx} = 18.57 \text{ m} < 0.01 \cdot h_i = 0.01 \cdot 3000 = 30 \text{ mm}$$

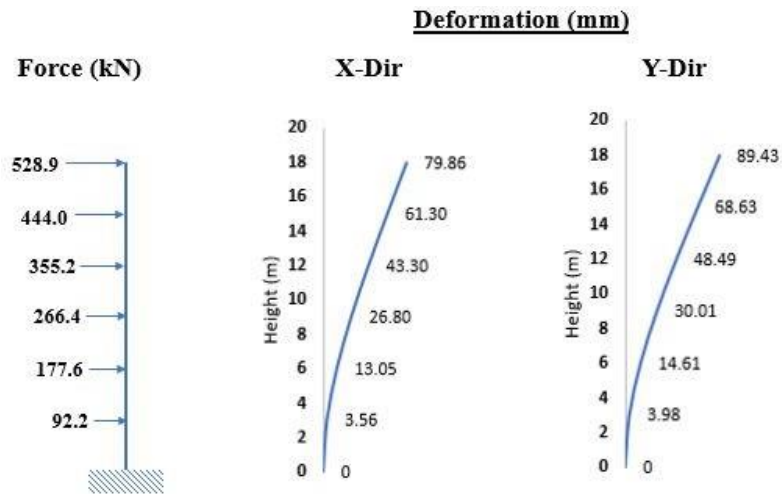


Figure 7.4: Deformability Checks

The checks are satisfied

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POLITECNICO DI MILANO

SCHOOL OF CIVIL, ENVIRONMENTAL AND LAND MANAGEMENT
ENGINEERING



POLITECNICO
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ASSIGNMENT 2: Preliminary Design of a
Medium-Rise Steel Residential Building in
Medium Seismicity Area

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ATTACHMENTS TO THIS REPORT

- DRAWINGS
- SUMMARY OF THE REPORT

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Chapter 1

INTRODUCTION

The assignment is aimed to perform a preliminary design of a structural resisting system, based on '*Structural Steel*' as construction material, for a 6-story residential building located in a moderate seismicity area. In order to develop the structural system, a base architectural layout of the building was provided.

1.1 Geometry

The proposed residential building is composed of 6 floors and a roof, where the first floor is aimed to host commercial activity. Hence, the ground floor is characterized by wider spaces in contrast to the upper floors, which are conceived for residential purposes. The upper floors can be accessed by stairs or an elevator. Both are located at the center of the structure. The overall building dimensions are represented in Figure 1.1 having a floor height of about 3 meters.

The selected structural scheme to be used is a *Steel-Reinforced Concrete Composite Structure* which is composed of *beams* and *columns* steel frame structure carrying the gravity loads and reinforced concrete *shear walls* that constitute the lateral load resisting system aimed to withstand the forces generated by seismic action. Therefore, from the seismic point of view, the building is designed as a *wall system*. This statement allows to design the walls as the primary seismic resisting elements while the secondary elements (e.g., beams and columns) can be designed to sustain only gravity loads.

1.2 Materials

1.2.1 Structural Steel

The proposed building construction material for the frame structure is '*Structural Steel*'. The selected steel grade is S355. S355 steel is a European standard structural steel that is commonly used in Europe. S355 grade steel is a medium tensile, low carbon, manganese metal with a high strength-to-weight ratio. The numerical indicator of "355" refers to the material's yield stress (when at a nominal thickness of 16mm), with the "S" indicating that it is indeed made from structural steel.

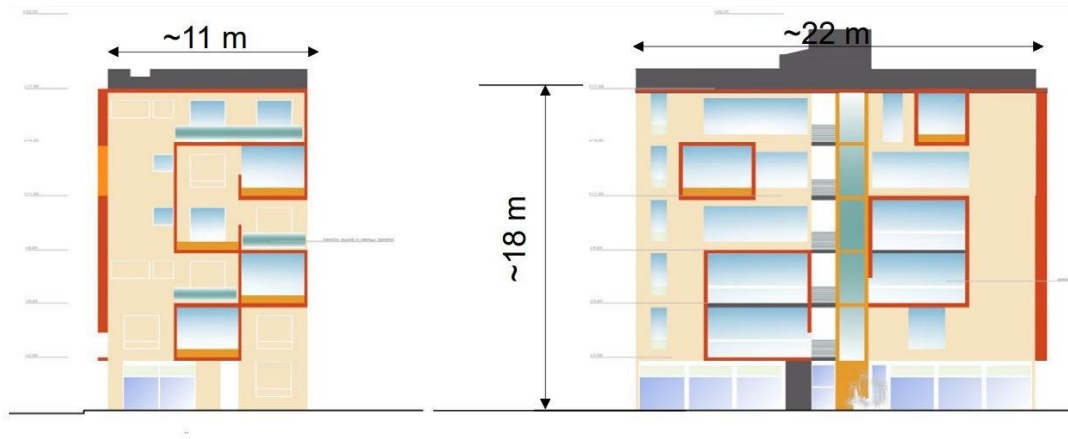


Figure 1.1: Dimensions of the building

The relevant properties of the selected steel grade, as per EN 10025-2 are tabulated in the following table:

Table 1.1: Structural Steel Properties

Yield Strength ($f_y - t \leq 16mm$)	355	MPa
Tensile Strength ($f_u - 3mm < t < 16mm$)	470-630	MPa
Unit Weight (γ)	78.5	kN/m ³
Modulus of Elasticity (E)	210000	MPa
Shear Modulus (G)	81000	MPa
Poisson's Ratio (in elastic range)	0.3	-
Coefficient of linear thermal expansion (α)	12×10^{-6}	K ⁻¹

1.2.2 Concrete

The shear walls are composed of Reinforced Concrete. According to Eurocode 2, 4.1 (4): “Corrosion protection of steel reinforcement depends on density, quality, and thickness of concrete cover and cracking. The cover density and quality are achieved by controlling the maximum water/cement ratio and minimum cement content and may be related to a minimum strength class of concrete.”. As a bracing system is needed so shear walls composed of concrete are chosen instead of a vertical bracing system made of steel members.

Therefore, considering a mild exposure class of XC1, the concrete of class C30/37 is adopted using Table 4.3N of EN 1992-1-1:2004. The corresponding mechanical properties are given in the following table adopted from Table 3.1 of EN 1992-1-1:2004:

Table 1.2: Mechanical properties of concrete

Strength Class	f_{ck}	$f_{ck,cube}$	f_{ctm}	f_{cd}	E_{cm}
C30/37	30	37	2.9	17	33
	MPa	MPa	MPa	MPa	GPa

The concrete cover has been selected referring to Sec 4.4.1 of EN 1992-1-1:2004 (the bar size referred to for the calculations is 16mm). The details are given in the below table.

Table 1.3: Nominal Concrete Cover

Nominal Concrete Cover		26	mm	A+B
A	c_{min}	16	mm	max(i, ii, iii)
i	$c_{min,b}$	16	mm	
ii	$c_{min,dur} + c_{dur} - c_{dur,st} - c_{dur,add}$	15	mm	
iii		10	mm	
B	c_{dev}	10	mm	(4.4.1.3 (1), Eurocode 2)

Considering the output value of the minimum nominal cover, the selected cover value is 30 mm.

1.2.3 Steel Reinforcement

The reinforcing steel bars of class B450C are used in the shear wall. All relevant parameters are summarized in the following table.

Table 1.4: Mechanical Properties of Steel Reinforcement

Grade	B450C	
Characteristic Yield Strength (f_{yk})	450	MPa
Design Yield Strength (f_{yd})	391.3	MPa
Modulus of Elasticity (E)	210	GPa

Chapter 2

LOAD ANALYSIS

2.1 Variable Loads

The analysis of acting loads starts with the classification of the building. According to the Section 3.1.4 of *Norme tecniche per le costruzioni* 2018 (NTC 2018), the building is designed for Category A and has an importance class II. Hence, variable loads are $q = 2.00$ kN/m² for the internal environment and $q = 4.00$ kN/m² for stairs.

The snow load is taken assuming that the building is in Zone I of the map defined in section 3.4.2 of NTC2018 at a height lower than 200m. Assuming to have a single flap roof and that the location is not exposed to wind flows, the resulting snow load is $q_{snow} = 1.32$ kN/m².

2.2 Permanent Loads

The permanent loads comprise of *Structural Permanent Loads* (G_1) and *Non-structural Permanent Loads* (G_2). The structural permanent loads consist of dead loads caused by slab elements. The floor slab is selected as a 'Traditional Composite Slab' for steel structure which is realized in the form of a one-way slab through the combined use of Steel joists, Metal corrugated deck, Lightweight concrete, Dry Screed, and other flooring materials as shown in Figure 2.1. The detailed geometry and load analysis of the permanent and non-permanent loads are given in Table 2.1.

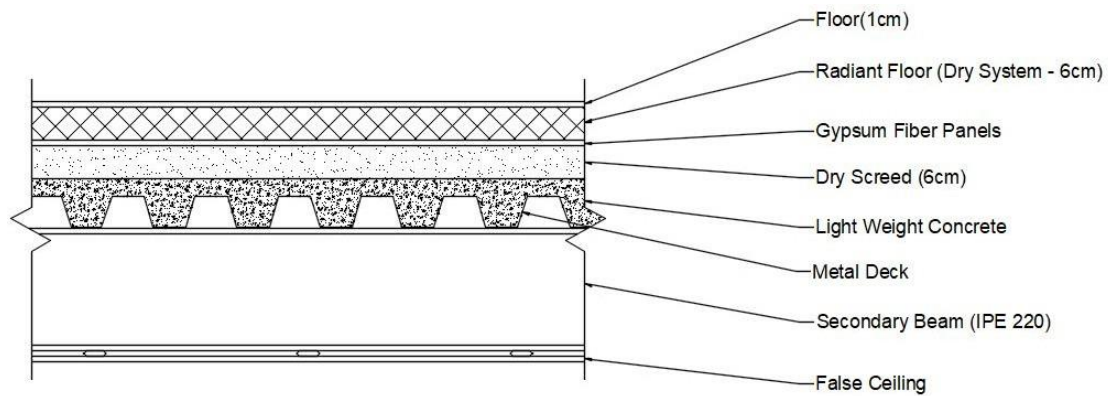


Figure 2.1: Slab Cross Section Table

2.1: Permanent Loading

Load Description	Item Description	Load (kN/m ²)
Structural Dead Load (G₁)	Joist (IPE 200, i = 1.5m)	0.17
	Light Weight Concrete	1.16
	Metal Deck (0.7mm)	0.09
	Total (G₁)	1.42
Non-Structural (Superimposed) Dead Load (G₂)	Dry Screed	0.24
	Gypsum Fiber Panels	0.29
	Radiant Floor (Dry 20+20+20 mm)	0.79
	Floor (Laminate Timber)	0.2
	Partition Walls (Dry System)	0.8
	Fall Ceiling	0.25
	Total (G₂)	2.57

2.3 Seismic Load

The seismic load is obtained by the generation of a response spectrum that is dependent on structural characteristics and the locality of the building being studied. In particular, the site is characterized by a soil type C, a peak ground acceleration (PGA) of 0.25g (reference PGA, of value 0.25g, factored with $\gamma = 1$, based on the importance class of I as per EN 1998-1/4.2.5(5)P) and structural damping of 5%.

The structure is characterized by a behavior factor ' q ' = 3.3 which has been obtained from Eurocode 8 Sec 5.2.2.2 assuming the structure of medium ductility (refer to Table 5.1 of EC8). The ' q ' value has been obtained for the Dual System. The value

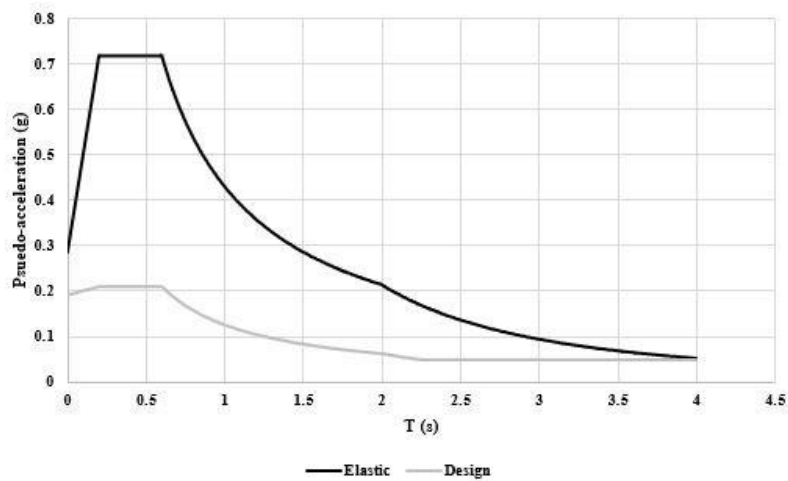


Figure 2.2: Elastic and Design Response Spectra of

k_w being used for the calculation of 'q' was obtained as '0.86'.

From the available data, both elastic and design response spectra can be computed using an online tool given in the footnote¹. The response spectra, both elastic and design, are shown in the above figure.

Their effect on the structure consists of an acceleration applied to the latter due to the dynamic interaction. To value this, the structural first natural period must be estimated to enter it as an input of the response spectrum function. The significant value of spectra is known as the time corresponding to the end of the plateau, resulting that $T_c = 0.61$.

¹<https://eurocodeapplied.com/design/en1998/elastic-response-spectrum>

Chapter 3

STRUCTURAL SCHEME

The gravity load-bearing system consists of steel columns and beams (primary and secondary). The beams are simply connected to the columns. The slab is considered to be rigid in plan. The structural layout has been preliminarily developed based on the three constraints given below:

- Architectural layout
- Span-to-depth ratio
- Regularity of running beams (due to steel sections)

The chosen slab type is *One-Way Slab* as per the load distribution scheme shown in the structural layout. The depth of the joist in the selected slab type is taken considering limit ratio. The initial depth of the joist was taken as per below-described scheme but later it was revised when secondary beam was designed as discussed in the next chapter. The limiting ratio is computed below using the maximum spans L_s (Slab Span). Based on the layout shown below the value of L_s is:

$$L_s = 4.22 \text{ m}$$

Computing the limit slab depth using the ratio of $L_s/25$:

$$L_s/25 = 4.22/25 = 16.8 \text{ cm}$$

Based on this and after the considerations given in the next chapter, the depth of the steel joist in the slab has been taken equal to 22 cm. The checks pertaining to the serviceability of the primary beam are performed in the next chapter since the depth of the primary beam is higher than 22cm (the depth of the primary beam is 30cm, which permits a length of $30\text{cm} * 20\text{cm}$ which is equal to 6m).

The premises of the given architectural layout are respected apart from a few columns which are in conflict with the architectural layout which is to be negotiated with the architect for possible editing of the layout. Nevertheless, the proposed layout is respected on each floor to a greater extent. As required, the structural layout of a typical floor is shown in the following figures. The detailed drawings of the structural layouts are given in the drawings annexure.

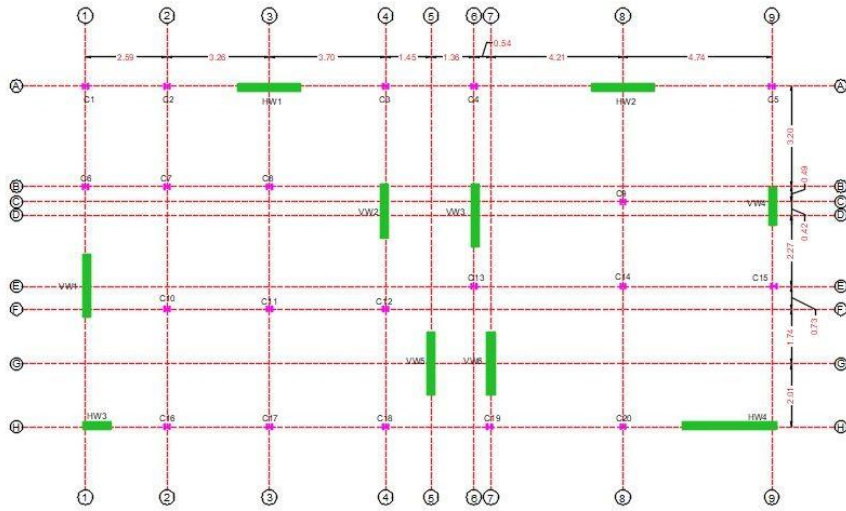


Figure 3.1: Center Line

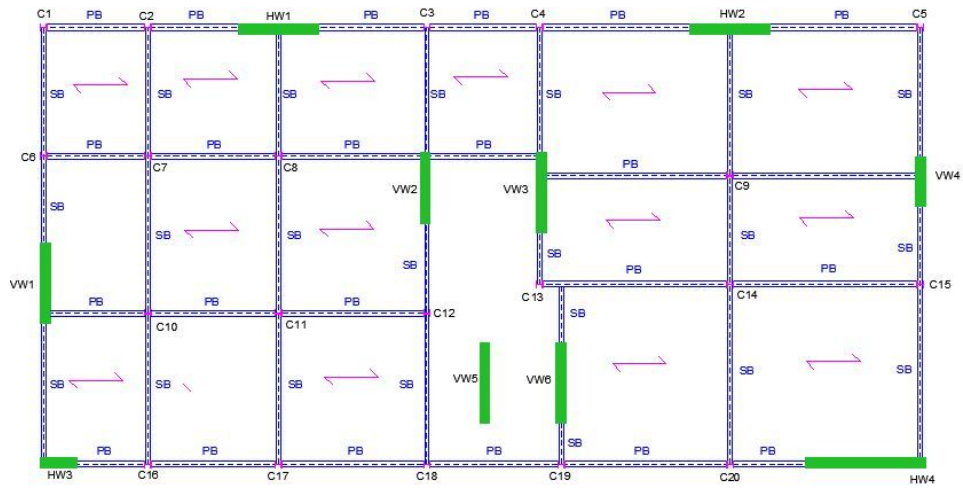


Figure 3.2: Ground and First Floor Structural Layout (1/2)



Figure 3.3: Ground and First Floor Structural Layout (2/2)

Chapter 4

DESIGN FOR GRAVITY AND SAFETY CHECKS

4.1 Column Design

The column design has been carried out based on EC3. In order to develop the design, the applied ULS loading on the column was determined using the scheme of the tributary area as shown in the figure below. The most loaded column (C14) was determined based on largest tributary area being supported (16.71 m^2) which leads to ULS axial load:

$$N_{ed} = 879 \text{ kN}$$

Since the beams are simply connected to the columns, the bending moment transfer is zero. However, due to the eccentricity of shear load transfer from the primary beam, there will be a small moment applied to the column about its stronger axis (H-Section considered).

$$M_{yed} = 1.17 \text{ kNm}$$

The moment due to shear transfer from the secondary beam is negligible due to the

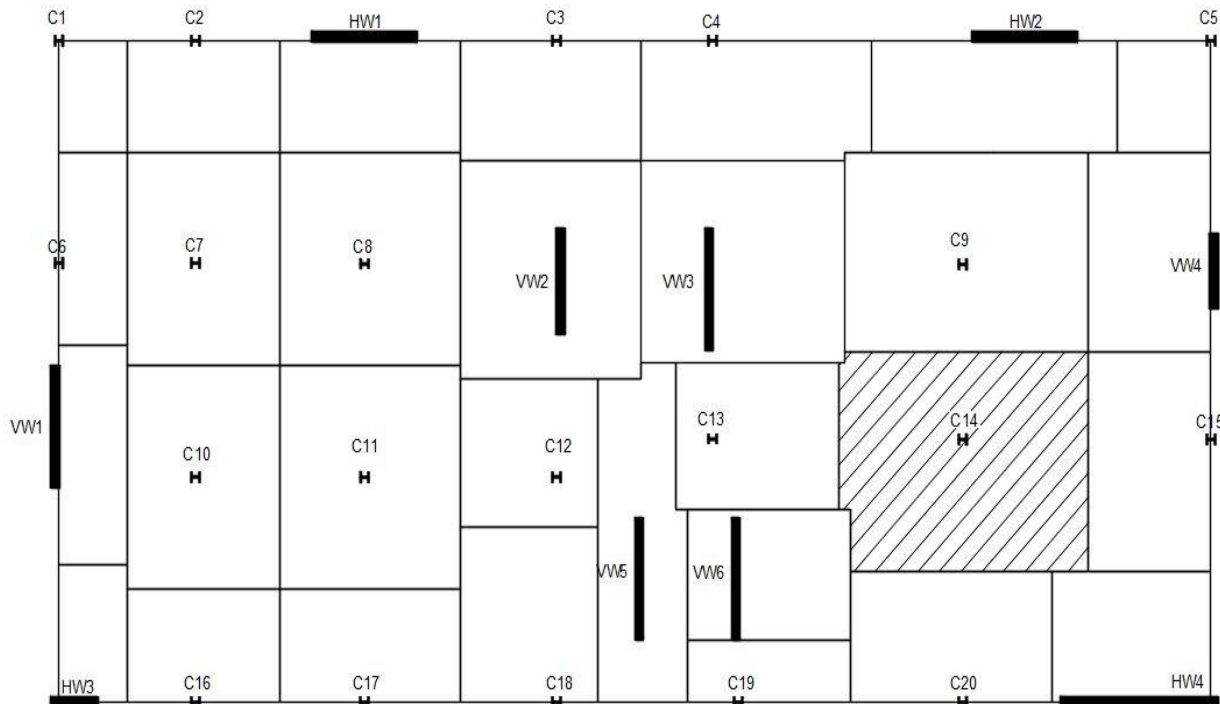


Figure 4.1: Tributary Area

very small eccentricity.

Note: Since the steel compression member is susceptible to failure in buckling, the relevant design checks have been carried. The checks can be followed in the following design procedure.

4.1.1 Design and Checks

In order to develop the design different trial sections were tested and finally 'HE160A' (European Wide Flange Section) was selected which satisfied the required checks. The geometrical properties of the selected section are given in the following table:

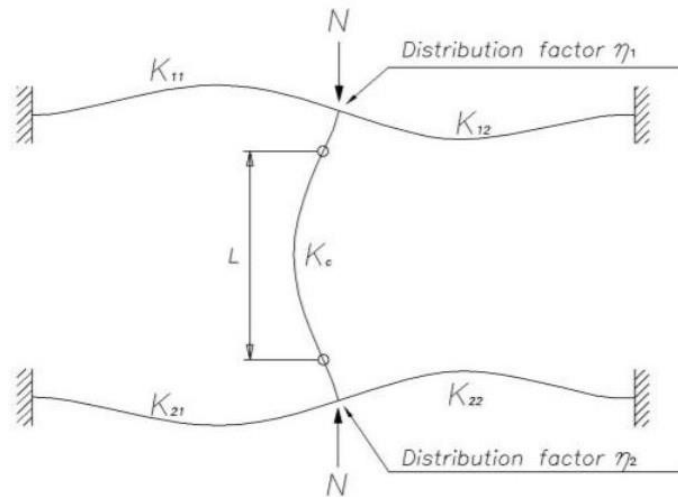
Table 4.1: Section Details - HE160A

Section	HEA 160	
Geometrical Data in	h	152
	b	160
	t _w	6
	t _f	9
	r	15
	A(cm ²)	38.8
	p(kg/m)	30.4
Strong Axis	J _y (cm ⁴)	1673
	W _y (cm ³)	220
	i _y (cm)	6.57
Weak Axis	J _z (cm ⁴)	616
	W _z (cm ³)	76.9
	i _z (cm)	3.98
	I _T (cm ⁴)	88704
	I _W (cm ⁶)	31491460000

The suitability of the selected section is checked as follows.

At the start of the checks, the support conditions of the column were determined. The supported conditions are determined in the form of the 'K' factor which is obtained using the below formula.

$$\eta = \frac{K_c}{K_c + K_{11} + K_{12}} \quad (4.1)$$



Consider the designed secondary beam of IPE220 (about the weaker axis of the column), the value of K_c , K_{11} and K_{12} are computed as:

$$K = \frac{I}{h} \quad (4.2)$$

Where:

- I = Moment of Inertia of the element
- h = length of the element

Table 4.2: Value of η

	$I \text{ (mm}^4\text{)}$	$h \text{ (m)}$	$K \text{ (m}^3\text{)}$	η
Column	6146500	3	2.04883E-06	0.11
Beam (11)	27720000	4.22	6.56872E-06	
Beam (12)	27720000	2.87	9.65854E-06	

Considering the value of η same for the top and the bottom, the resulting value of 'K', as per Eurocode, is obtained to be 0.525 by using the formula below:

$$K = 0.5 + (0.14 * (\eta_1 + \eta_2)) + (0.055 * (\eta_1 + \eta_2)^2) = 0.525$$

In the next step, the class of the section is to be determined as per Table 5.2 of EC3.

Flange Check:

$$\frac{c}{t} \leq 14\epsilon$$

$$\frac{c}{t} \leq 14 * \sqrt{\frac{235}{f_y}}$$

$$\frac{75}{9} \leq 14 * \sqrt{\frac{235}{355}}$$

$$8.33 \leq 11.39$$

Flange is in Class 3
Web Check:

$$\frac{c}{t} \leq 72\varepsilon$$

$$\frac{c}{t} \leq 72 * \sqrt{\frac{235}{f_y}}$$

$$\frac{152}{6} \leq 72 * \sqrt{\frac{235}{355}}$$

$$25.33 \leq 58.58$$

Web is in Class 2

Overall, the section is in Class 3 as per Table 5.2 of EC3.

The next step includes the determination of non-dimensional slenderness as per the below formula (Sec 6.3.1.3(1) of EC3):

$$\begin{aligned}\bar{\lambda} &= \frac{L}{i} \times \frac{1}{\pi} \left(\sqrt{\frac{f_y}{E}} \right) \\ &= \frac{1.575 * 1000}{40.8} \times \frac{1}{\pi} \times \left(\sqrt{\frac{f_y}{E}} \right) \\ &= 0.51\end{aligned}$$

L above was the free length that was computed by $K * L$ where L is the length of the column.

The imperfection factor is determined based on the value of α obtained considering the buckling curve 'c' as per Table 6.2 of EC3. The value of α as per Table 6.1 is 0.49. The value of α is taken from table 6.1 as $h/b \leq 1.2$ and $t_f \leq 100\text{mm}$. The formula for the computation of the imperfection factor is:

$$\varphi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$$

$$\alpha = 0.49$$

$$\varphi = 0.5[1 + 0.49(0.51 - 0.2) + 0.51^2]$$

$$\varphi = 0.702$$

The reduction factor χ is computed as per the below formula as per Sec 6.3.1.2(1) of EC3:

$$\chi = \frac{1}{\varphi + \sqrt{\varphi^2 - \lambda^2}}$$

$$\chi = 0.84$$

Based on these parameters, the resisting buckling load is checked as per the following formula (Sec 6.3.1.1(3) of EC3):

$$N_{b,Rd} = \frac{\chi A f_y}{\gamma_{M1}} = \frac{0.84 * 38.8 * 100 * 355}{1} = 1157kN$$

$$\frac{N_{ed}}{N_{b,Rd}} = \frac{879.3}{1157} = 0.76 < 1$$

These are the only checks needed to check the suitability of the column cross-section.

The check is satisfied for the selected section HE160A. Considering the value to be closer to 1, the section is also an economical choice. The drawing of the section is given in the attached drawing annexure.

4.2 Beam Design

4.2.1 Primary Beam

As per the selected scheme, the beams are designed to be simply supported. For the purpose of primary beam design, the longest beam in the structural layout was selected and designed. The selected beam in the structural layout is shown in the following figure.

The influence area of the beam was determined. Since the width of the influence area is different across, the loading is obtained for them separately. The loading is given in the following table:

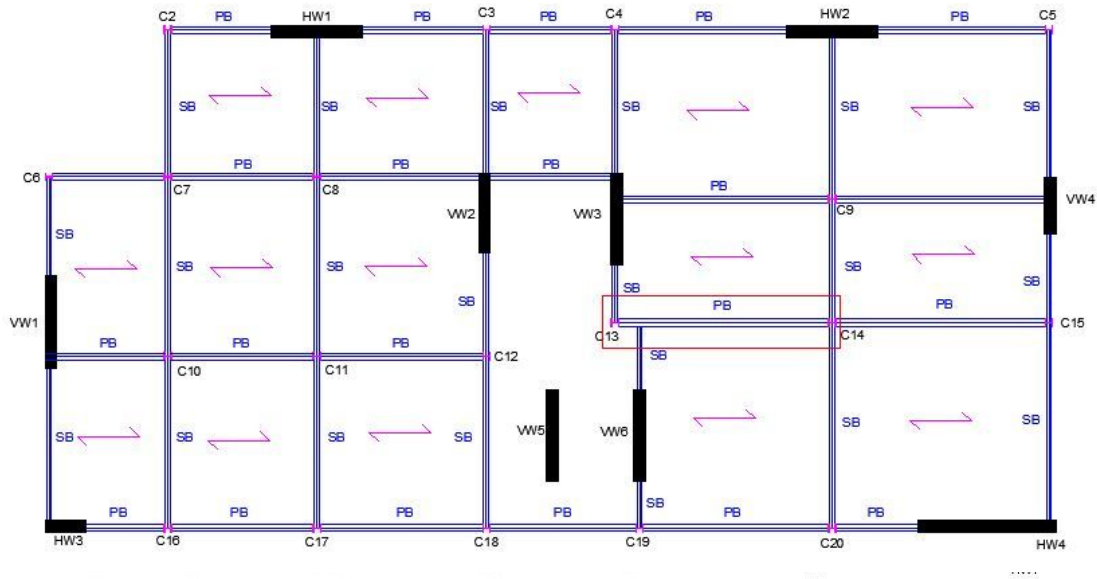


Figure 4.2: Design Primary Beam

Table 4.9: Beam Applied load

	Width (m)	Length (m)	Load (G1) kN/m	Load (G2) kN/m	Load (Q) kN/m
Strip 1	2.38	2.61	4.57	9.17	7.14
Strip 2	3.73	2.12	7.17	14.38	11.19
	Total Length	4.73			

The design procedure for the beam is explained in the following paragraphs. The beam has been designed as a laterally restrained beam (considering the slab to be rigid) as per the procedure given in the EC3. For the purpose of design, multiple trial sections were checked and finally, the IPE300 section was selected which satisfied all of the ULS and SLS checks. The checks are reproduced below. The following table summarizes the material properties.

Before the application of checks, the value of loading is to be updated for the self-weight. Considering the section, the additional load due to self-weight is 0.41 kNm. Considering the final loading, the ULS moment and shear come out to be:

$$M_{ed} = 104.67 \text{ kNm}$$

$$V_{ed} = 88.3 \text{ kN}$$

The class of the section is determined as per the geometrical properties of the section (as done in the case of column). With reference to Table 5.2 of EC3 and considering the geometry of the section the class checks are as follows.

$$\frac{c}{t} \leq 9\epsilon$$

$$\frac{c}{t} \leq 9 * \sqrt{\frac{235}{f_y}}$$

$$\frac{56.45}{10.7} \leq 9 * \sqrt{\frac{235}{355}}$$

$$5.28 \leq 7.32$$

Web is in Class 1

$$\frac{c}{t} \leq 72\epsilon$$

$$\frac{c}{t} \leq 72 * \sqrt{\frac{235}{f_y}}$$

$$\frac{278.6}{7.1} \leq 72 * \sqrt{\frac{235}{355}}$$

$$30.24 \leq 58.32$$

Flange is in Class 1

Based on this, the section is classified as 'Class 1'.

Table 4.10: Section Details - IPE300

Section		IPE 300
Geometrical Data	h	300
	b	150
	t _w	7.1
	t _f	10.7
	r	15
	A(cm ²)	53.8
	p(kg/m)	42.2
Strong Axis	J _y (cm ⁴)	8356
	W _y (cm ³)	557
	i _y (cm)	12.5
Weak Axis	J _z (cm ⁴)	604
	W _z (cm ³)	80.5
	i _z (cm)	3.35
	I _T (cm ⁴)	158295.4
	I _W (cm ⁶)	1.26379E+11

The shear area and subsequently shear resistance are to be determined as per the below formula (Sec 6.2.6(3) of EC3). For I beam with load applied on flange, consider the case of rolled I section with load parallel to web.

$$A_v = A - 2b \cdot t_f + (t_w + 2r) \cdot t_f$$

Based on the geometrical properties of the selected beam, value of A_v comes out to:

$$A_v = 2566.97 \text{ mm}^2$$

Subsequently, value of shear resistance is obtained from formula (Sec 6.2.6(2)):

$$V_{pl,Rd} = \frac{A_v \cdot f_y}{\sqrt{3} \cdot \gamma_{M0}} = \frac{2566.97 \cdot 355}{\sqrt{3} \cdot 1.0} \cdot 10^{-3} = 526.12 \text{ kN}$$

Considering the shear area, yield strength, and γ_{M0} , the value of shear resistance is obtained as 526.12 kN. Checking the ratio:

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{88.3}{526.12} = 0.17 < 1$$

This check is satisfied.

Subsequently, the check for combined shear and bending failure is applied. If the below ratio is < 0.5 , the reduction in bending resistance is not required (Sec 6.2.8(2) of EC3).

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{88.3}{526.12} = 0.17 < 0.5$$

Therefore, the reduction in bending resistance is not required.

Subsequently, the check for shear buckling failure is applied and if satisfied then the shear buckling check is not required (Sec 6.2.6(6) of EC3).

$$\frac{h_w}{t_w} < 72 \frac{\varepsilon}{\eta}; \frac{300 - (2 \cdot 10.7) - (2 \cdot 15)}{17.1} < 72 \cdot \frac{\sqrt{(235/355)}}{1.0}$$

$$14.54 < 58.58; \text{SATISFIED}$$

Therefore, the shear buckling check is not required.

The next step is to determine the bending moment resistance. The formula for the calculation of the resistance bending moment is (Sec 6.2.5(2) of EC3) (α is the shape factor that is taken as 1.1):

$$M_{b,Rd} = \frac{W_{pl,y} f_y}{\gamma_{M0}} = \frac{W_{el,y} \alpha f_y}{\gamma_{M0}} = \frac{557000 \cdot 1.1 \cdot 355}{1} = 217.51 \text{ kNm}$$

The adequacy of the bending resistance is checked against the applied moment as follows:

$$\frac{M_{ed}}{M_{b,Rd}} = \frac{104.67}{217.51} = 0.48 < 1$$

Therefore, the check is satisfied.

Subsequently, SLS checks are applied for the deflection limit check. The deflection due to loading is given by below formula:

$$\delta = \frac{5 \cdot w \cdot L^4}{384 \cdot E \cdot I}$$

Where 'w' is the load at SLS state. The value of w is 26.61 kN/m here. Based on material and geometrical properties, the deflection comes out to be:

$$\delta = 9.97 \text{ mm}$$

The allowable deflection, as per ENV 1993-1, is given by:

$$\delta_{allowable} = \frac{L}{250}$$

The value of allowable deflection comes out to be:

$$\delta_{allowable} = \frac{L}{250} = \frac{4730}{250} = 18.92 \text{ mm}$$

The adequacy of the deflection is assessed as per below ratio:

$$\frac{\delta}{\delta_{allowable}} = \frac{9.97}{18.92} = 0.53 < 1$$

The check is satisfied.

The second deflection check as per ENV 1993-1 is given by:

$$\delta_{2,allowable} = \frac{L}{300}$$

The value of allowable deflection of live loads come out to be:

$$\delta_{2,allowable} = \frac{L}{300} = \frac{4730}{300} = 15.77 \text{ mm}$$

The deflection of the live load where 'w' is 8.96 Kn/m comes out to be:

$$\delta_2 = 3.35 \text{ mm}$$

The adequacy of the deflection is assessed as per below ratio:

$$\frac{\delta_2}{\delta_{2,allowable}} = \frac{3.35}{15.77} = 0.21 < 1$$

This check is satisfied.

It is required that for building structures for floors regularly loaded by people, the lowest natural frequency of the structure is not less than 3 Hz

$$\text{Natural Frequency, } f_0 = \frac{18}{\sqrt{\delta_m}} = \frac{18}{\sqrt{9.97}} = 5.7 \text{ Hz}$$

Next for the primary beam, it is susceptible to Lateral Torsional Buckling, so the Buckling Resistance of the primary beam also has to be computed:

$$\lambda_{LT} = \sqrt{\frac{W_{pl,y} f_y}{M_{cr}}}$$

$$M_{cr} = C_1 \frac{\pi^2 EI_z}{(k_z L)^2} \left[\sqrt{\left(\frac{k_z}{k_w} \right)^2 \frac{I_W}{I_z} + \frac{(k_z L)^2 GJ_T}{\pi^2 EI_z} + (C_2 z_g)^2} - C_2 z_g \right]$$

z_g is taken as 0 as it is assumed that Load is applied on the shear centre. C_1 depends on the moment distributions and is taken as 1.879.

Condizioni di carico e di vincolo	Diagramma del momento flesso	Valori di k_z	Valori dei coefficienti		
			C_1	C_2	C_3
	$\psi = +1$	1.0	1.000	—	1.000
		0.7	1.000	—	1.113
		0.5	1.000	—	1.144
	$\psi = +3/4$	1.0	1.141	—	0.998
		0.7	1.270	—	1.165
		0.5	1.305	—	2.283
	$\psi = +1/2$	1.0	1.523	—	0.992
		0.7	1.473	—	1.556
		0.5	1.514	—	2.271
	$\psi = +1/4$	1.0	1.563	—	0.977
		0.7	1.739	—	1.531
		0.5	1.788	—	2.245
	$\psi = 0$	1.0	1.879	—	0.939
		0.7	2.092	—	1.473
		0.5	2.150	—	2.150
	$\psi = -1/4$	1.0	2.281	—	0.855
		0.7	2.538	—	1.340
		0.5	2.609	—	1.957
	$\psi = -1/2$	1.0	2.704	—	0.676
		0.7	3.009	—	1.059
		0.5	3.093	—	1.546
	$\psi = -3/4$	1.0	2.927	—	0.866
		0.7	3.009	—	0.575
		0.5	3.093	—	0.837
	$\psi = -1$	1.0	2.752	—	0.000
		0.7	3.063	—	0.000
		0.5	3.149	—	0.000

K_z and k_w are 1.0.

$$M_{cr} = \frac{1.879 * \pi^2 * 210000 * 6040000}{(1 * 1500)^2} \left[\sqrt{\left(\frac{1}{1} \right)^2 * \frac{1.26379E + 11}{6040000} + \frac{(1 * 1500)^2 * 81000 * 158295.4}{\pi^2 * 210000 * 6040000}} \right]$$

$$M_{cr} = 1593.34 \text{ kNm}$$

$$\lambda_{LT} = 0.37$$

$$\varphi_{LT} = 0.5[1 + \alpha_{LT}(\lambda_{LT} - 0.2) + \lambda_{LT}^2]$$

$$\varphi_{LT} = 0.5[1 + 0.34(0.37 - 0.2) + 0.37^2]$$

$$\varphi_{LT} = 0.597$$

$$\chi_{LT} = \frac{1}{\varphi_{LT} + \sqrt{\varphi_{LT}^2 - \lambda_{LT}^2}}$$

$$\chi = 0.939$$

$$\text{Buckling Resistance, } M_{b,pl,Rd} = \frac{X_{LT} * W_{pl} * f_y}{\gamma_{m0}} = \frac{0.939 * 557000 * 1.1 * 355}{1} = 204.24 \text{ kNm}$$

$$\frac{M_{ed}}{M_{b,pl,Rd}} = \frac{104.67}{204.24} = 0.51 < 1$$

The check for the lateral torsional buckling is also satisfied.

The four checks are reproduced below:

$$\frac{M_{ed}}{M_{b,Rd}} = \frac{104.67}{222.94} = 0.4 < 1$$

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{88.3}{526.12} = 0.17 < 1$$

$$\frac{\delta}{\delta_{allowable}} = \frac{9.97}{18.92} = 0.53 < 1$$

$$\frac{M_{ed}}{M_{b,pl,Rd}} = \frac{104.67}{204.24} = 0.51 < 1$$

Since all of the ratios are satisfied, the selected section is safe. Since one of the ratios is close to the value of 1, therefore, the section is economical. The section drawing is given in the attached drawing annexure.

4.2.2 Secondary Beam

The secondary beams (slab joists) support the slab structure. The loading on the secondary is a function of joist spacing. Given the joist spacing of 1.5m, the width of the influence area is 1.5m. Based on this, the loading analysis is:

Table 4.11: Secondary Beam - Applied Load

	Width (m)	Length (m)	G1 (kN/m)	G2 (kN/m)	Q (kN/m)
Strip	1.5	4.22	2.9	5.8	4.5

The design procedure for the beam is similar to the primary beam explained in the previous sections. The beam has been designed as a laterally restrained beam (considering the slab to be rigid) as per the procedure given in the EC3. For the purpose of design, multiple trial sections were checked and finally, the IPE220 section was selected which satisfied all of the ULS and SLS checks. The checks are reproduced below. The following table summarizes the material properties.

Before the application of checks, the value of loading is to be updated for the self- weight. However, the self-weight is already accounted for in the G1 loading. Considering the final loading, the ULS moment and shear come out to be:

$$M_{ed} = 41.06 \text{ kNm}$$

$$V_{ed} = 38.9 \text{ kN}$$

Table 4.12: Section Details - IPE220

Section		IPE 220
Mass per meter, W (kg/m)		26.2
Mass per meter, W (kN/m)		0.26
Geometrical Data	h	220
	b	110
	t _w	5.9
	t _f	9.2
	r	12
	A(cm ²)	33.4
	p(kg/m)	26.2
Strong Axis	J _y (cm ⁴)	2772
	W _y (cm ³)	252
	i _y (cm)	9.11
Weak Axis	J _z (cm ⁴)	205
	W _z (cm ³)	37.3
	i _z (cm)	2.48
	I _T (cm ⁴)	72164.91333
	I _W (cm ⁶)	22773778000

The class of the section is determined as per the geometrical properties of the section (as done in the case of primary beam). With reference to Table 5.2 of EC3 and considering the geometry of the section the class checks are as follows. The values of ϵ and η are 0.81 and 1 as per EC3.

$$\frac{c}{t} \leq 9\epsilon$$

$$\frac{c}{t} \leq 9 * \sqrt{\frac{235}{f_y}}$$

$$\frac{52.05}{9.2} \leq 9 * \sqrt{\frac{235}{355}}$$

$$5.66 \leq 7.32$$

Flange is in Class 1

$$\frac{c}{t} \leq 72\epsilon$$

$$\frac{c}{t} \leq 72 * \sqrt{\frac{235}{f_y}}$$

$$\frac{208.2}{5.9} \leq 72 * \sqrt{\frac{235}{355}}$$

$$35.29 \leq 58.58$$

Web is in Class 1

Based on this, the section is classified as 'Class 1'.

The shear area and subsequently shear resistance are to be determined as per the below formula (Sec 6.2.6(3) of EC3). For I beam with load applied on flange, consider the case of rolled I section with load parallel to web.

$$A_v = A - 2b \cdot t_f + (t_w + 2r) \cdot t_f$$

Based on the geometrical properties of the selected beam, the value of A_v comes out to:

$$A_v = 1591.08 \text{ mm}^2$$

Subsequently, the value of shear resistance is obtained from the following formula (Sec 6.2.6(2) of EC3):

$$V_{pl,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}}$$

Considering the shear area, yield strength, and γ_{M0} , the value of shear resistance is obtained as 326.11 kN. Checking the ratio

$$V_{pl,Rd} = \frac{A_v * f_y}{\sqrt{3} * \gamma_{M0}} = \frac{1591.08 * 355}{\sqrt{3} * 1.0} * 10^{-3} = 326.11 \text{ kN}$$

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{38.9}{326.11} = 0.12 < 1$$

The check is satisfied.

Subsequently, the check for combined shear and bending failure is applied (Sec 6.2.8(2) of EC3). If the below ratio is <0.5, the reduction in bending resistance is not required.

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{38.9}{326.11} = 0.12 < 0.5$$

Therefore, no reduction in bending moment is required.

Subsequently, the check for shear buckling failure is applied and if satisfied then the shear buckling check is not required (Sec 6.2.6(6) of EC3).

$$\frac{h_w}{t_w} < 72 \frac{\varepsilon}{\eta}; \frac{220 - (2 * 9.2) - (2 * 12)}{5.9} < 72 * \frac{\sqrt{(235/355)}}{1.0}$$

$$30.10 < 58.58; \text{SATISFIED}$$

Therefore, the shear buckling check is not required.

The next step is to determine the bending moment resistance. The formula for the calculation of the resistance bending moment is (Sec 6.2.5(2) of EC3):

$$M_{b,Rd} = \frac{W_{pl,y} f_y}{\gamma_{M0}}$$

Based on this, the value comes out to be (α is the shape factor that is taken as 1.1):

$$M_{b,Rd} = \frac{W_{pl,y} f_y}{\gamma_{M0}} = \frac{W_{el,y} \alpha f_y}{\gamma_{M0}} = \frac{252000 * 1.1 * 355}{1} = 98.41 \text{ kNm}$$

The adequacy of the bending resistance is checked against the applied moment as follows:

$$\frac{M_{ed}}{M_{b,Rd}} = \frac{41.06}{98.41} = 0.42 < 1$$

Therefore, the check is satisfied.

There is no need for ULS-Lateral Torsional Buckling Check for Secondary Beams as the primary beams restrain the secondary beams and prevent them from Laterally Buckling.

Subsequently, SLS checks are applied for the deflection limit check. The deflection due to loading is given by below formula:

$$\delta = \frac{5 \cdot w \cdot L^4}{384 \cdot E \cdot I}$$

Where 'w' is the load at SLS state. The value of w is 13.16 kN/m here. Based on

material and geometrical properties, the deflection comes out to be:

$$\delta = 9.34 \text{ mm}$$

The allowable deflection, as per ENV 1993-1, is given by:

$$\delta_{allowable} = \frac{L}{250}$$

The value of allowable deflection comes out to be:

$$\delta_{allowable} = \frac{L}{250} = \frac{4220}{250} = 16.88 \text{ mm}$$

The adequacy of the deflection is assessed as per below ratio

$$\frac{\delta}{\delta_{allowable}} = \frac{9.34}{16.88} = 0.55 < 1$$

This check is satisfied.

The second deflection check as per ENV 1993-1 is given by:

$$\delta_{2,allowable} = \frac{L}{300}$$

The value of allowable deflection of live loads come out to be:

$$\delta_{2,allowable} = \frac{L}{300} = \frac{4220}{300} = 14.07 \text{ mm}$$

The deflection of the live load where 'w' is 4.5 kN/m comes out to be:

$$\delta_2 = 3.19 \text{ mm}$$

The adequacy of the deflection is assessed as per below ratio:

$$\frac{\delta_2}{\delta_{2,allowable}} = \frac{3.19}{14.07} = 0.23 < 1$$

This check is satisfied.

It is required that for building structures for floors regularly loaded by people, the lowest natural frequency of the structure is not less than 3 Hz

$$\text{Natural Frequency, } f_0 = \frac{18}{\sqrt{\delta_m}} = \frac{18}{\sqrt{9.34}} = 5.9 \text{ Hz}$$

The three checks are reproduced below:

$$\frac{V_{ed}}{V_{pl,Rd}} = \frac{38.9}{326.11} = 0.12 < 1$$

$$\frac{M_{ed}}{M_{b,Rd}} = \frac{41.06}{98.41} = 0.42 < 1$$

$$\frac{\delta}{\delta_{allowable}} = \frac{9.34}{16.88} = 0.55 < 1$$

Since all of the ratios are satisfied, the selected section is safe. Since one of the ratios is close to the value of 1, therefore, the section is economical. The section details are given in the drawings annexure.

4.3 Comments on Connection Design

Considering the conceptual nature of the design, the design pertaining to the connections was not carried out. However, a conceptual connection of Column-Primary Beam (simple connection) and Wall-Primary Beam (simple connection) was developed and illustrated below. The detailed design of the connections should be carried out to comprehensively consider the design parameters.

4.3.1 Column-Primary Beam Connection

The connection between column and primary beam is realized through 'simple' bolted connection achieved through double angle section on the both sides of web of primary beam. The illustration of connection is shown in Figure 4.3.

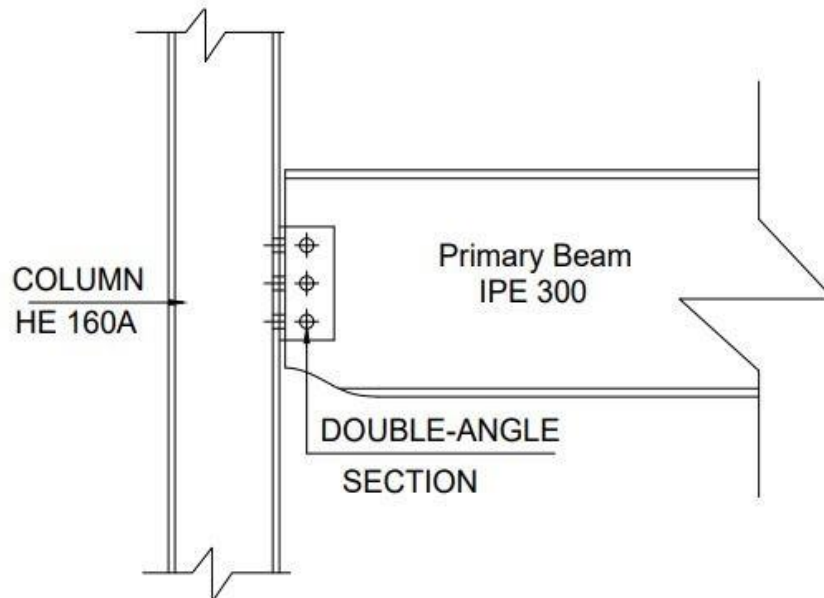


Figure 4.3: Column-Primary Beam Connection

4.3.2 Wall-Primary Beam Connection

The connection between wall and primary beam is realized through simple double angle connection in which the angle plate is anchored to the RCC wall by anchor bolts and the beam by bolts in beam. The illustration of connection is shown in Figure 4.4. For such a connection, the design rules are available in EC:2-4: Design of Fasteners for use in Concrete. Since this is a wall-primary beam connection and since the beams have been assumed to be simply supported so only shear checks will be needed at these connections. The important checks will be the pryout failure of anchor rods, concrete cone failure and concrete edge failure. Also it should be observed that while drilling and installation of the anchor rods, the anchor rods or the drill do not collide with the reinforcement in the shear wall. These would be the checks for the anchor bolts. For the bolts connected to the beam, the checks in EC 3:1-8 Design of Joints will be used.

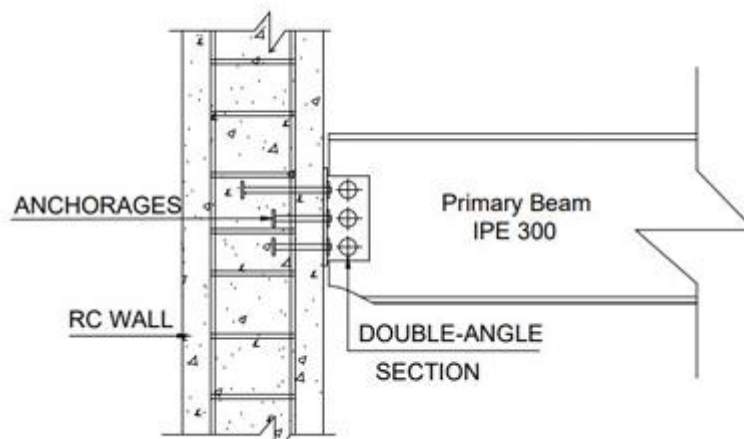
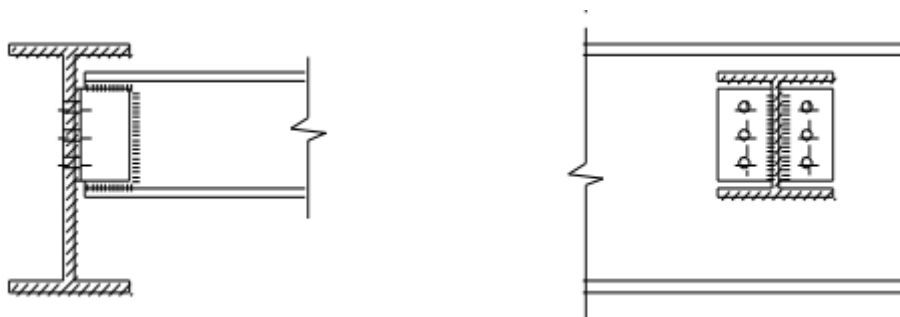


Figure 4.4: Wall-Primary Beam Connection

4.3.3 Primary Beam – Secondary Beam Connection

For the connection of the primary beam to the secondary beam, a simple connection for the simply supported ends will be provided. A web bolted-cleated connection will be provided in which a double angle section can be used which is fillet welded with a 10mm depth to the secondary beam and bolted to the primary beam.



The secondary beam has to be connected to the primary beam at such a height that the metal corrugated sheet and the lightweight concrete after being installed and poured come onto the same level as the primary beam.

4.3.4 Comments on Metal Corrugated Sheets

The metal corrugated sheet chosen was of nominal thickness 0.7mm. As the sheet is to be placed over the secondary beams so the sheet has to be chosen that has deflection not greater than the limits and can take on the load being applied that is 4.57KN/m^2 . The length of one sheet is 1.50m.

Sp. Th. mm	Larghezza appoggio: 100 mm			
	CAMPATA SINGOLA			
L=m	1,50	1,75	2,00	2,25
0,50	831	611	462	363
0,60	1018	743	567	446
0,70	1205	886	677	534
0,80	1397	1023	781	616
1,00	1782	1309	1001	787
1,20	2211	1623	1238	979
1,50	2882	2118	1617	1276

4.3.5 Comments on Composite Concrete-Metal Deck Slab Behaviour

Some comments will be also made on the composite slab and what provisions to be provided according to Eurocode 4: Design of Composite Steel and Concrete Structures Part 1.1: General Rules and Rule for Buildings. Very basic details will be discussed as this project is on the conceptual design of the building. As per EC4-1.1: Sec 9.2.1, the overall depth of the height of the concrete above the ribs of the metal corrugated sheet to be a minimum of 40 mm and the overall depth of the composite slab to be a minimum of 80 mm. Transverse and Longitudinal Reinforcement is to be provided in the height of concrete above the ribs and it is to be a minimum of $80\text{mm}^2/\text{m}$. For the longitudinal reinforcement, another requirement to be fulfilled from Sec 7.4.1(4) should not be less than 0.4% for propped construction and 0.2% for unpropped construction.

The nominal size of the aggregate will not exceed the minimum of $0.40 \cdot h_c$ (where h_c is the height of concrete above the ribs), $b_0/3$ (where b_0 is the mean width of the ribs) and 31.5 mm (Sieve C 31.5) as per 9.2.2.

Chapter 5

FIRE RESISTANCE CHECK

5.1 Primary Beam

The fire safety check has been applied for the primary beam based on *Eurocode 3: Design of steel structures - Part 1-2: General rules - Structural fire design*. The beam has been considered as laterally restrained. For the fire resistance, the beam is protected for the period of R90. The safety check calculations have been made as below.

The section classification has already been obtained as Class 1. The next step is to calculate the fire resistance of the unprotected beam (since the beam is considered unprotected from fire initially). The design value of the resistance moment at time $t=0$, $M_{fi,0,Rd}$, according to following formula:

$$M_{fi,0,Rd} = \frac{W_{pl,y} \cdot f_y}{k_1 \cdot k_2 \cdot \gamma_{M,fi}}$$

Where:

- $k_1 = 0.7$ for an unprotected beam exposed on three sides, with a concrete slab on the fourth side;
- $k_2 = 1.0$ for sections not at the supports

Referring to the section properties of selected primary beam section, the value of $M_{fi,0,Rd}$ is computed as:

$$M_{fi,0,Rd} = 310.7 \text{ kNm}$$

The degree of utilisation takes the value:

$$\mu_o = \frac{M_{fi,Ed}}{M_{fi,0,Rd}} = \frac{104.67}{310.73} = 0.33$$

The critical temperature comes out to be:

$$\theta_{a,cr} = 654^{\circ}\text{C}$$

Subsequently, the box value of the section factor was calculated:

$$[A_m/V]_b = \frac{2 \cdot h + b}{A} = 139 \text{ m}^{-1}$$

The correction factor for shadow effect was applied to the box section factor. The resultant value is:

$$k_{sh}[A_m/V] = 0.9 \cdot 139 = 125 \text{ m}^{-1}$$

Interpolation of the values given in Table A.4 of reference book, *FIRE DESIGN OF STEEL STRUCTURES, Franssen and Real*, yields a time of 18 min. to reach the critical temperature, $\theta_{a,cr} = 654^{\circ}\text{C}$. This is less than the required 90 min. and therefore fire protection is necessary to achieve the required fire resistance.

As the value of k_1 depends on whether or not the profile is protected, a new critical temperature must be obtained for the protected section. Considering $k_1 = 0.85$, the design value of the resistance moment at time $t=0$, $M_{fi,0,Rd}$, comes out to be:

$$M_{fi,0,Rd} = 255.9 \text{ kNm}$$

The revised degree of utilisation takes the value:

$$\mu_o = \frac{104.67}{255.9} = 0.409$$

The critical temperature comes out to be:

$$\theta_{a,cr} = 624^{\circ}\text{C}$$

For the purpose of required fire protection, Fiber-cement boards are selected. The thickness of contour encasement protection with fiber-cement boards necessary to prevent the steel temperature exceeding the critical temperature before the required time of 90 minutes is required to be computed.

The following thermal properties of the fiber-cement are defined:

$$\lambda_p = 0.15 \frac{W}{m \cdot K}$$

$$c_p = 1200 \frac{J}{kg \cdot K}$$

$$\rho_p = 800 \frac{kg}{m^3}$$

The massivity factor for the IPE 300 with hollow encasement heated on three sides is:

$$[A_p/V] = 139 \text{ m}^{-1}$$

By interpolation in Table A.5 from Annex A, for a temperature of 654°C, at 90 minutes of standard fire exposure, the modified section factor is:

$$\frac{A_p \cdot \lambda_p}{V \cdot d_p} \leq 1210 \frac{W}{m^3 \cdot K}$$

The thickness is:

$$d_p \geq \frac{A_p \cdot \lambda_p}{V \cdot 1210} = \frac{139 \cdot 0.15}{1210} = 17 \text{ mm}$$

This thickness can be corrected if the amount of heat stored in the protection, Φ , is taken into account to obtain the corrected thickness.

$$\frac{A_p \cdot \lambda_p}{V \cdot d_p \cdot (1 + \phi)/2} \leq 1210 \frac{W}{m^3 \cdot K}$$

The iterative procedure is performed and the revised value of thickness is 14.3 mm.

5.2 Column

Similar to the primary beam, the fire safety check has been applied on the column based on Sec 4.2.3.5 of EC3 Structural Fire Design. The detailed calculations are given below:

The cross-section was classified as Class 3. The reduction factor ($k_{y\theta}$) for the yield strength is obtained:

$$k_{y\theta} = \frac{N_{fi,Ed}}{A * f_y / \gamma_{M,fi}} + \frac{M_{fi,Ed}}{W_{el,y} * f_y / \gamma_{M,fi}}$$

Steel Temperature θ_a	Reduction factors at temperature θ_a relative to the value of f_y or E_a at 20°C			
	Reduction factor (relative to f_y) for effective yield strength $k_{y,\theta} = f_{y,\theta}/f_y$	Reduction factor (relative to f_y) for proportional limit $k_{p,\theta} = f_{p,\theta}/f_y$	Reduction factor (relative to E_a) for the slope of the linear elastic range $k_{E,\theta} = E_{a,\theta}/E_a$	Reduction factor (relative to f_y) for the design strength of hot rolled and welded thin walled sections (Class 4) $k_{0.2p,\theta} = f_{0.2p,\theta}/f_y$
20 °C	1.000	1.000	1.000	1.000
100 °C	1.000	1.000	1.000	1.000
200 °C	1.000	0.807	0.900	0.890
300 °C	1.000	0.613	0.800	0.780
400 °C	1.000	0.420	0.700	0.650
500 °C	0.780	0.360	0.600	0.530
600 °C	0.470	0.180	0.310	0.300
700 °C	0.230	0.075	0.130	0.130
800 °C	0.110	0.050	0.090	0.070
900 °C	0.060	0.0375	0.0675	0.050
1000 °C	0.040	0.0250	0.0450	0.030
1100 °C	0.020	0.0125	0.0225	0.020
1200 °C	0.000	0.0000	0.0000	0.000
NOTE: For intermediate values of the steel temperature, linear interpolation may be used.				

Figure 5.1: Reduction Factors

$$N_{fi,Ed} = 879.9 \text{ kN}$$

$$M_{fi,Ed} = 1.18 \text{ kNm}$$

The values of $k_{y,\theta}$ comes out to be:

$$k_{y,\theta} = 0.69$$

Interpolating in the above table, gives a critical temperature of:

$$\theta_{a,cr} = 530^\circ\text{C}$$

The uniform temperature of the profile after 90 minutes of standard fire exposure was evaluated and the value is given below:

$$\theta_a = 1003^\circ\text{C}$$

Because the temperature in the member is higher than the critical temperature calculated above, it can be concluded that the fire resistance of 90 minutes is not achieved.

The expression of EC3 is applied with revised value of $k_{y,\theta}$ at this temperature. The revised value is:

$$k_{y\theta} = 0.04$$

The following check is applied:

$$\frac{N_{fi,Ed}}{A * f_y / \gamma_{M,fi}} + \frac{M_{fi,Ed}}{W_{el,y} * f_y / \gamma_{M,fi}} \leq 1.0$$

Inserting the parameters, we get:

$$17.2 > 1.0$$

Thus, the member has no fire resistance after 90 minutes of standard fire exposure. The adequate fire protection should be provided.

Chapter 6

PRELIMINARY SEISMIC DESIGN

6.1 Seismic Mass

Seismic mass was evaluated according to NTC18, 2.5.3 in seismic configuration.

$$M = G_1 + G_2 + \sum_j \psi_{2j} \cdot Q_{kj}$$

where:

M : Seismic Mass;

G_1 : Structural Permanent Load;

G_2 : Non-structural Permanent Load;

ψ_{2j} : 0.3 – Coefficient for residential use (NTC18, Tab. 2.5.I);

Q_{kj} : Variable Loads, 2kN/m²

The calculations for the structural and non-structural permanent loads and variable loads were carried out earlier in the form of area load. The same have been recalled here to calculate the mass of each floor considering the floor area. The results are summarized in the table below for the floors and the roof since the floor area of the roof is significantly lower than area of floors.

Table 6.1: Seismic Mass - Ground Floor

Floor (Ground)		
G1	1.42	kN/m ²
G2	2.57	kN/m ²
Q	2.00	kN/m ²
Si	0.30	

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Table 6.1: Seismic Mass - Ground Floor (Continued)

Si*Q	0.60	kN/m ²
Total	4.59	kN/m ²
Floor Area	240.1	m ²
Total Load	1102.86	kN
	111	tons

Table 6.2: Seismic Mass - All Floors (ex. Ground/Roof)

All Floors (ex. Ground/Roof)		
G1	1.42	kN/m ²
G2	2.57	kN/m ²
Q	2	kN/m ²
Si	0.3	
Si*Q	0.6	kN/m ²
Total	4.59	kN/m ²
Floor Area	228.29	m ²
Total Load	1048.61	kN
	105	tons

Table 6.3: Seismic Mass - Roof

Roof		
G1	1.42	kN/m ²
G2	2.57	kN/m ²
Q	2	kN/m ²
Si	0.3	-
Si*Q	0.6	kN/m ²
Total	4.59	kN/m ²
Floor Area	226.06	m ²
Total Load	1038.37	kN

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Table 6.3: Seismic Mass - Roof
(Continued)

	104	tons
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In addition to the above loads, the mass of columns and shear walls is needed to be considered for the sake of completeness. Therefore, considering the geometry and material of the columns and shear walls, the following tables summarize the mass of columns and shear walls to be added.

Table 6.4: Seismic Mass - Column

Columns		
Area single column	0.00369	m ²
Height	3	m
Unit Weight	78.5	kN/m ³
Volume	0.011061	m ³
no: of columns	20	-
Total weight x floor	17.4	kN
Mass	1.74	ton

Table 6.5: Seismic Mass - Shear Walls

Shear Walls		
Thickness	0.25	m
Linear meter	18.85	m
Height	3	m
Unit weight	25	kN/m ³
Volume	14.14	m ³
Total weight x floor	353.4	kN
Mass	35.47	ton

Conclusively, the floor mass for each storey is summarized in the table below:

Table 6.6: Total Seismic Mass

Floor	Mass (Ton)
Ground	147.99

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Table 6.6: Total Seismic Mass
(Continued)

1	142.54
2	142.54
3	142.54
4	142.54
Roof	141.51
Total	859.7

6.2 Seismic Shear

According to section 7.3.3.2 NTC2018, the linear static analysis could be applied if the structure is regular in elevation, and its natural period of vibrations satisfies the following inequality:

$$T_1 < 2.5T_c = 2.5 \cdot 0.61s = 1.525s$$

Simplified formula for computation of T_1 was applied, according to C7.3.3.2:

$$T_1 = C_t \times H^{3/4} = 0.085 \times (18)^{3/4} = 0.743s$$

The check for the time period was satisfied, and the structure was considered regular in the elevation, thus the linear static analysis could be applied in order to obtain the distribution of the seismic shear along the building. The seismic shear per floor is computed as:

$$F_i = V_b \frac{\sum z_i M_i}{z_j M_j}$$

$$V_b = S_d(T_1) \cdot M_{\text{tot}} \cdot \lambda = 0.171 \times 9.81 \times 859.7 \times 0.85 = 1225.78 \text{ kN}$$

Where:

V_b – seismic base shear force;

z_i – height of i-floor from the ground level;

M_i – mass of i-floor;

$S_d(T_1)$ – ordinate of the design response spectrum at T_1 ; λ – correction factor

Considering the equation for the distribution of shear force stated above the resulting calculation for shear force distribution at each floor is given below:

Table 6.7: Seismic Shear Distribution

Floor	$z_i(m)$	$m_i(t)$	$z_i \times m_i$	$F_i(kN)$
Ground	3	147.99	443.96	60.58
1	6	142.54	855.25	116.70
2	9	142.54	1282.87	175.05

3	12	142.54	1710.50	233.40
4	15	142.54	2138.12	291.75
Roof	18	141.51	2547.24	347.57
	Total	859.66	8977.93	1225.04

6.3 Shear force distribution among lateral load resisting system

In order to determine the distribution of seismic base shear in horizontal plane (X Y) of structure, both wall and column properties were taken into account. Contributions from all columns were included in both directions in calculation of distribution of base shear. Center of mass (CM) was found with respect to reference point (QPF point given in the plan drawing). The center of stiffness (CR) was calculated with respect to the center of mass.

After finding the values of the moment of inertia for each structural element, the dimensionless parameters S'_{ix} , S'_{iy} , S''_{ix} and S''_{iy} were computed as follows:

$$S'_{ix} = I_{iy} / \sum I_{iy}$$

$$S'_{iy} = I_{ix} / \sum I_{ix}$$

$$S''_{ix} = e_y \times y_i \times I_{iy} / \sum (x_i^2 \times I_{ix} + y_i^2 \times I_{iy})$$

$$S''_{iy} = e_x \times x_i \times I_{iy} / \sum (x_i^2 \times I_{ix} + y_i^2 \times I_{iy})$$

Then, these contributions were summed up in each direction in order to find the distribution of the unit seismic base shear on the floor as:

$$\frac{S'_{ix} + S''_{ix}}{S'_{iy} + S''_{iy}}$$

The following tables summarize the aforementioned calculation of base shear redistribution for the X and Y directions. Moreover, it can be inferred from the tables that 99% of the seismic base shear is taken by the shear walls in the X direction and 99% by the shear walls in the Y direction. The contribution of each column is relatively very small to the walls. Therefore, since walls play a more dominant role in stiffness distribution, taking much more than 65% of the seismic base shear, the structural system of the building is defined as a Wall System.

Table 6.8: Distribution of unit seismic base shear in the y-direction

i	I _{iy} (m ⁴)	S' _{ix} (m)	y' _i (m)	y' _i S' _{ix} (m)	y _i	y _i I _{iy} (m ⁵)	y _i ² *I _{iy} (m ⁶)	x _i ² *I _{ix} + y _i ² *I _{iy}	S _i ''x	S _{ix}
VW1	0.1000	0.2020	-11.3872	-2.3007	-9.7461	-0.9746	9.4987	9.5003	0.0000	0.2020
VW2	0.0670	0.1353	-1.8472	-0.2500	-0.2061	-0.0138	0.0028	0.0135	0.0000	0.1353
VW3	0.1000	0.2020	0.9728	0.1965	2.6139	0.2614	0.6832	0.6947	0.0000	0.2020
VW4	0.0216	0.0436	10.5228	0.4592	12.1639	0.2627	3.1959	3.2047	0.0000	0.0436
VW5	0.1000	0.2020	-0.3572	-0.0722	1.2839	0.1284	0.1648	0.1709	0.0000	0.2020
VW6	0.1000	0.2020	1.4728	0.2976	3.1139	0.3114	0.9696	0.9757	0.0000	0.2020
HW1	0.0016	0.0032	-5.5372	-0.0175	-3.8961	-0.0061	0.0237	4.7665	0.0000	0.0032
HW2	0.0016	0.0032	5.7128	0.0180	7.3539	0.0115	0.0845	4.8273	0.0000	0.0032
HW3	0.0007	0.0014	-11.0172	-0.0157	-9.3761	-0.0066	0.0618	0.2064	0.0000	0.0014
HW4	0.0023	0.0047	9.3528	0.0443	10.9939	0.0258	0.2833	5.6381	0.0000	0.0047
C1	0.000010	0.0000	-11.3872	-0.0002	-9.7461	-0.0001	0.0009	0.0011	0.0000	0.0000
C2	0.0000	0.0000	-8.7972	-0.0002	-7.1561	-0.0001	0.0005	0.0007	0.0000	0.0000
C3	0.0000	0.0000	-1.8372	0.0000	-0.1961	0.0000	0.0000	0.0002	0.0000	0.0000
C4	0.0000	0.0000	0.9728	0.0000	2.6139	0.0000	0.0001	0.0002	0.0000	0.0000
C5	0.0000	0.0000	10.4528	0.0002	12.0939	0.0001	0.0014	0.0016	0.0000	0.0000
C6	0.0000	0.0000	-11.3872	-0.0002	-9.7461	-0.0001	0.0009	0.0010	0.0000	0.0000

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Table 6.8: Distribution of unit seismic base shear in the y-direction (Continued)

C7	0.0000	0.0000	-8.7972	-0.0002	-7.1561	-0.0001	0.0005	0.0005	0.0000	0.0000
C8	0.0000	0.0000	-5.5372	-0.0001	-3.8961	0.0000	0.0001	0.0002	0.0000	0.0000
C9	0.0000	0.0000	5.7128	0.0001	7.3539	0.0001	0.0005	0.0006	0.0000	0.0000
C10	0.0000	0.0000	-8.7972	-0.0002	-7.1561	-0.0001	0.0005	0.0005	0.0000	0.0000
C11	0.0000	0.0000	-5.5372	-0.0001	-3.8961	0.0000	0.0001	0.0001	0.0000	0.0000
C12	0.0000	0.0000	-1.8372	0.0000	-0.1961	0.0000	0.0000	0.0000	0.0000	0.0000
C13	0.0000	0.0000	0.9728	0.0000	2.6139	0.0000	0.0001	0.0001	0.0000	0.0000
C14	0.0000	0.0000	5.7128	0.0001	7.3539	0.0001	0.0005	0.0005	0.0000	0.0000
C15	0.0000	0.0000	10.4528	0.0002	12.0939	0.0001	0.0014	0.0014	0.0000	0.0000
C16	0.0000	0.0000	-8.7972	-0.0002	-7.1561	-0.0001	0.0005	0.0005	0.0000	0.0000
C17	0.0000	0.0000	-5.5372	-0.0001	-3.8961	0.0000	0.0001	0.0002	0.0000	0.0000
C18	0.0000	0.0000	-1.8372	0.0000	-0.1961	0.0000	0.0000	0.0001	0.0000	0.0000
C19	0.0000	0.0000	1.5128	0.0000	3.1539	0.0000	0.0001	0.0002	0.0000	0.0000
C20	0.0000	0.0000	5.7128	0.0001	7.3539	0.0001	0.0005	0.0006	0.0000	0.0000
	0.4950	1.0		-1.64			14.9773	30.0081		1.00

Table 6.9: Distribution of unit seismic base shear in the x-direction

i	I_{iy} (m ⁴)	S'_{ix} (m)	y'_i (m)	y'_i	S'_{ix} (m)	y_i	y_i	I_{iy} (m ⁵)	$y_i^2 * I_{iy}$ (m ⁶)	$x_i^2 * I_{ix} + y_i^2 * I_{iy}$	$S_i''x$	S_{ix}
VW1	0.0016	0.0028	2.1429		0.0060	1.0032		0.0016	0.0016	9.5003	0.0000	0.0028
VW2	0.0014	0.0025	-1.6471		-0.0041	-2.7868		-0.0038	0.0106	0.0135	0.0000	0.0025
VW3	0.0016	0.0028	-1.5671		-0.0044	-2.7068		-0.0042	0.0114	0.6947	0.0000	0.0028
VW4	0.0009	0.0017	-1.9171		-0.0032	-3.0568		-0.0029	0.0088	3.2047	0.0000	0.0017
VW5	0.0016	0.0028	3.1129		0.0088	1.9732		0.0031	0.0061	0.1709	0.0000	0.0028
VW6	0.0016	0.0028	3.1129		0.0088	1.9732		0.0031	0.0061	0.9757	0.0000	0.0028
HW1	0.1000	0.1801	-5.7471		-1.0351	-6.8868		-0.6887	4.7428	4.7665	0.0000	0.1801
HW2	0.1000	0.1801	-5.7471		-1.0351	-6.8868		-0.6887	4.7428	4.8273	0.0000	0.1801
HW3	0.0091	0.0164	5.1229		0.0841	3.9832		0.0363	0.1446	0.2064	0.0000	0.0164
HW4	0.3375	0.6078	5.1229		3.1139	3.9832		1.3443	5.3548	5.6381	0.0000	0.6078
C1	0.000004	0.0000	-5.7071		0.0000	-6.8468		0.0000	0.0002	0.0011	0.0000	0.0000
C2	0.0000	0.0000	-5.7071		0.0000	-6.8468		0.0000	0.0002	0.0007	0.0000	0.0000
C3	0.0000	0.0000	-5.7071		0.0000	-6.8468		0.0000	0.0002	0.0002	0.0000	0.0000
C4	0.0000	0.0000	-5.7071		0.0000	-6.8468		0.0000	0.0002	0.0002	0.0000	0.0000
C5	0.0000	0.0000	-5.7071		0.0000	-6.8468		0.0000	0.0002	0.0016	0.0000	0.0000
C6	0.0000	0.0000	-2.2471		0.0000	-3.3868		0.0000	0.0000	0.0010	0.0000	0.0000

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Table 6.9: Distribution of unit seismic base shear in the x-direction (Continued)

C7	0.0000	0.0000	-2.2471	0.0000	-3.3868	0.0000	0.0000	0.0005	0.0000	0.0000
C8	0.0000	0.0000	-2.2471	0.0000	-3.3868	0.0000	0.0000	0.0002	0.0000	0.0000
C9	0.0000	0.0000	-2.2471	0.0000	-3.3868	0.0000	0.0000	0.0006	0.0000	0.0000
C10	0.0000	0.0000	1.3729	0.0000	0.2332	0.0000	0.0000	0.0005	0.0000	0.0000
C11	0.0000	0.0000	1.3729	0.0000	0.2332	0.0000	0.0000	0.0001	0.0000	0.0000
C12	0.0000	0.0000	1.3729	0.0000	0.2332	0.0000	0.0000	0.0000	0.0000	0.0000
C13	0.0000	0.0000	0.6429	0.0000	-0.4968	0.0000	0.0000	0.0001	0.0000	0.0000
C14	0.0000	0.0000	0.6429	0.0000	-0.4968	0.0000	0.0000	0.0005	0.0000	0.0000
C15	0.0000	0.0000	0.6429	0.0000	-0.4968	0.0000	0.0000	0.0014	0.0000	0.0000
C16	0.0000	0.0000	5.1229	0.0000	3.9832	0.0000	0.0001	0.0005	0.0000	0.0000
C17	0.0000	0.0000	5.1229	0.0000	3.9832	0.0000	0.0001	0.0002	0.0000	0.0000
C18	0.0000	0.0000	5.1229	0.0000	3.9832	0.0000	0.0001	0.0001	0.0000	0.0000
C19	0.0000	0.0000	5.1229	0.0000	3.9832	0.0000	0.0001	0.0002	0.0000	0.0000
C20	0.0000	0.0000	5.1229	0.0000	3.9832	0.0000	0.0001	0.0006	0.0000	0.0000
	0.5552	1.0		1.14			15.0308	30.0081		1.0000

6.4 Regularity Check

In general, the regularity in the plan can be checked when the structural model is defined. The criteria for regularity in the plan are described in EN 1998-1 (4.2.3.2):

- The slenderness of the building shall be not higher than 4 ($\lambda = \frac{L_{max}}{L_{min}}$)
- The structural eccentricity shall be smaller than 30% of the torsional radius ($e_{ox} \leq 0.30r_x$, $e_{oy} \leq 0.30r_y$).
- The torsional radius shall be larger than the radius of the gyration of the floor mass in plan ($r_x \geq l_s$, $r_y \geq l_s$).

Considering the first point, the check is satisfied since the $\lambda = L_{max}/L_{min} = 22/11 = 2$ is less than 4.

The eccentricity of the plan was computed by determining the center of stiffness with respect to the center of gravity. The resulting eccentricity is computed as:

$$e_x = 1.64\text{m}$$

$$e_y = 1.14\text{m}$$

In order to compute the admissibility of the eccentricity, the torsional radii are computed using the following formula.

$$x_{CR} = \frac{\sum (x E I_y)}{\sum (E I_y)}; \quad y_{CR} = \frac{\sum (y E I_x)}{\sum (E I_x)}$$
$$r_x = \sqrt{\frac{\sum [(x - x_{CR})^2 E I_y + (y - y_{CR})^2 E I_x]}{\sum (E I_y)}}$$
$$r_y = \sqrt{\frac{\sum [(x - x_{CR})^2 E I_y + (y - y_{CR})^2 E I_x]}{\sum (E I_x)}}$$

The torsional radii were determined as:

$$r_x = 7.36\text{m}$$

$$r_y = 7.79\text{m}$$

The check for eccentricity is:

$$e_x = 1.64\text{m} < 0.3r_x = 0.3 \times 7.36 = 2.21\text{m}$$

$$e_y = 1.14\text{m} < 0.3r_y = 0.3 \times 7.79 = 2.34\text{m}$$

The check is satisfied for both directions. Therefore, the second condition is satisfied.

Considering the third condition, the radius of gyration of the floor mass is defined as:

$$A = 22 \text{ m}$$

$$B = 11 \text{ m}$$

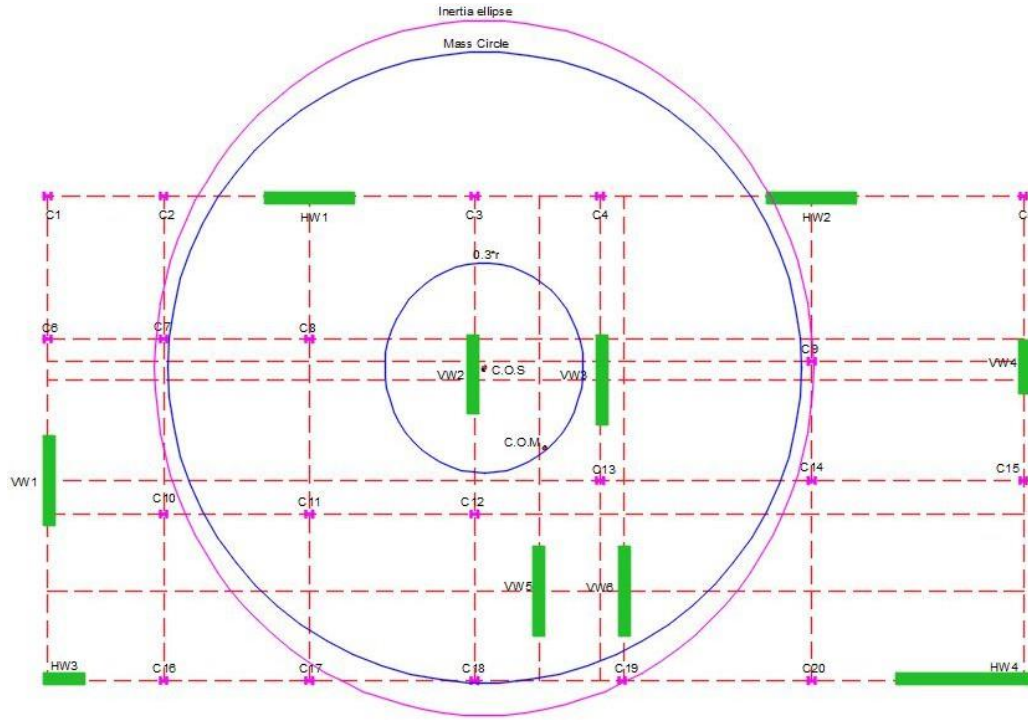


Figure 6.1: Plan Eccentricity

$$I_s = \sqrt{\frac{A^2 + B^2}{12}} = 7.1 \text{ m}$$

Since, both r_x and r_y are greater than I_s , therefore, the third condition for regularity is satisfied.

The above-discussed geometrical parameters are shown for a typical plan in the below figure.

6.5 Seismic Vulnerability

The seismic vulnerability of the building is assessed based on two indices:

- Wall Area Index - I
- Wall Area Index - II

The first index is related to an outline of earthquake provisions from the Japanese Building Code (Aoyama H,1981) and the second one is a study by Hassan and Sozen based on the assessment of seismic vulnerability.

6.5.1 Wall Area Index - I

The seismic vulnerability is assessed by plotting the ratio of the total wall area to floor area (WI%) against the average shear stress computed by dividing weight

by the cumulative cross-sectional area of walls and columns. The parameters are plotted by superimposing them over a literature graph from the Japanese Building Code. The values of the indices are given in the table below for each of the floors and cumulatively. In each of these cases, it is observed that:

WI% is greater than 0.3% and Stress is less than 1.2; therefore, the check is within the safe region of the literature graph (see picture below with superimposed points for both X and Y directions).

6.5.2 Wall Area Index - II

In order to assess the Wall Area Index II, the indices proposed by Hassan and Sozen are to be computed. The formulation of both indices is given below:

$$WI\% = \frac{A_{wt}}{A_{ft}} * 100 \quad (6.2)$$

$$CI\% = \frac{A_{ce}}{A_{ft}} * 100 \quad (6.3)$$

Where:

- WI and CI represent the wall index and column index, respectively;
- $A_{wt} = A_{cw} + A_{mw}/10$ = Effective cross-sectional area of walls in a given horizontal direction;
- A_{cw} is the total cross-sectional area of the reinforced concrete walls in one horizontal direction at the base;
- A_{mw} is the cross-sectional area of the non-reinforced masonry filler walls in one horizontal direction at the base;
- A_{ft} is the total floor area above the base;
- $A_{ce} = A_{col}/2$ = Effective cross-sectional area of columns at the base;
- A_{col} is the total cross-sectional area of columns above the base.

Based on the formulas above, the wall and column indexes were calculated for each floor and are reported in the following tables.

Table 6.10: Wall Area Indices - x direction

Wall Area Index (X-Direction)		Wall Area Index 1						Wall Area Index 2		
Floor	A _w [m ²]	A _c [m ²]	A _f [m ²]	W _l [%]	A _w + A _c [m ²]	W [kN]	W/(A _w + A _c) [MPa]	CI [%]	A _{mw} [m ²]	WI [%]
0	1.975	0.074	240.1	0.82%	2.05	-	-	0.015%	3.24	0.96%
1	1.975	0.074	228.29	0.87%	2.05	-	-	0.016%	3.65	1.03%
2	1.975	0.070	228.29	0.87%	2.05	-	-	0.015%	3.21	1.01%
3	1.975	0.070	228.29	0.87%	2.05	-	-	0.015%	3.21	1.01%
4	1.975	0.070	228.29	0.87%	2.05	-	-	0.015%	3.21	1.01%
5	1.975	0.070	228.29	0.87%	2.05	-	-	0.015%	3.21	1.01%
6	1.000	0.056	226.06	0.44%	1.06	-	-	0.012%	3.13	0.58%
Total	12.85	0.48	1607.61	0.80%	13.33	8560.50	0.64	0.015%	22.86	0.94%

Table 6.11: Wall Area Indices - y direction

Wall Area Index (Y-Direction)	Wall Area Index 1							Wall Area Index 2		
Floor	A _w [m ²]	A _c [m ²]	A _f [m ²]	W _I [%]	A _w + A _c [m ²]	W [kN]	W/(A _w + A _c) [MPa]	CI [%]	A _{mw} [m ²]	WI [%]
0	2.738	0.074	240.1	1.14%	2.81	-	-	0.015%	1.98	1.22%
1	2.738	0.074	228.29	1.20%	2.81	-	-	0.016%	1.9	1.28%
2	2.738	0.070	228.29	1.20%	2.81	-	-	0.015%	1.54	1.27%
3	2.738	0.070	228.29	1.20%	2.81	-	-	0.015%	1.54	1.27%
4	2.738	0.070	228.29	1.20%	2.81	-	-	0.015%	1.54	1.27%
5	2.738	0.070	228.29	1.20%	2.81	-	-	0.015%	1.54	1.27%
6	2.738	0.056	226.06	1.21%	2.79	-	-	0.012%	0.5	1.23%
Total	19.16	0.48	1607.61	1.19%	19.65	8560.50	0.44	0.015%	10.54	1.26%

The values of WI and CI are superimposed on the limit graph by Hassan and Sozen for each of the directions. It can be observed that the indices are out of the vulnerable zone.

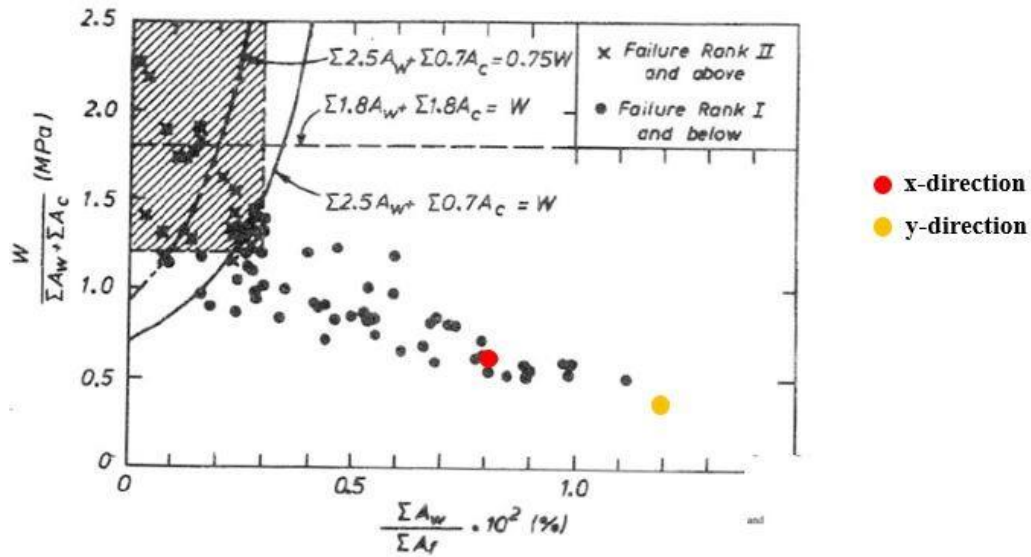


Figure 6.2: Wall Area Index 1 - X and Y Direction

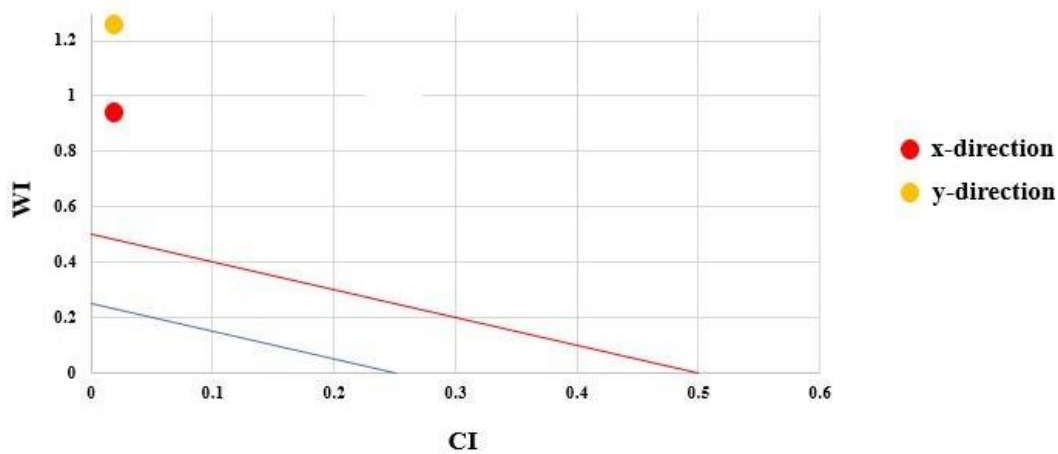


Figure 6.3: Wall Area Index 2 - X and Y Direction

6.6 Deformability Checks

Checks for deformability were performed verifying the inter-story drift according to NTC 7.3.6. To compute it, the reference was to a simple cantilever model with lumped masses at floors. The loading were applied as obtained from analysis of base shear done earlier. The modelling of the deformable cantilever was performed considering the total inertia of columns and walls for each side. From displacement values shown in the model results of the following figure, it was evaluated

the maximum inter story drift of 13.93 mm along Y direction and of 12.41 mm along X direction. Moreover, it is assumed that connections between structural and non-structural elements do not interfere with structural deformations. So, it results that:

$$d_{ry} = 13.93 \text{ mm} < 0.01 \cdot h_i = 0.01 \cdot 3000 = 30 \text{ mm}$$

$$d_{rx} = 12.41 \text{ mm} < 0.01 \cdot h_i = 0.01 \cdot 3000 = 30 \text{ mm}$$

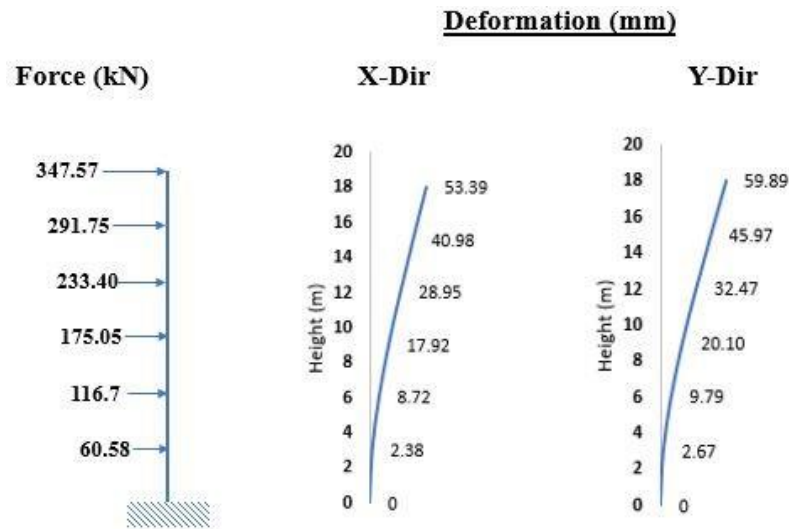
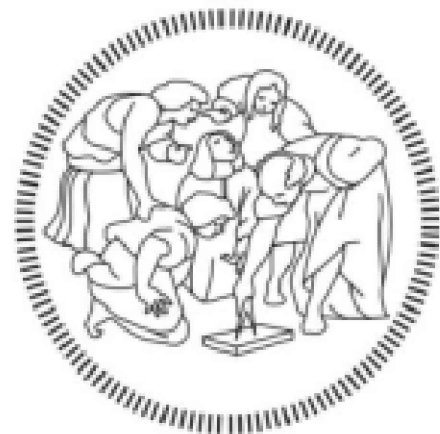


Figure 6.4: Deformability Checks

The checks are satisfied.

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- [2] Lecture Notes - Prof. Angelo MARCHISELLA
- [3] Lecture Notes - Prof. Marco CERVIO
- [4] NTC 2018
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- [11] EC:2-4: Design of Fasteners for use in Concrete.



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ARCHITECTURAL DRAWINGS
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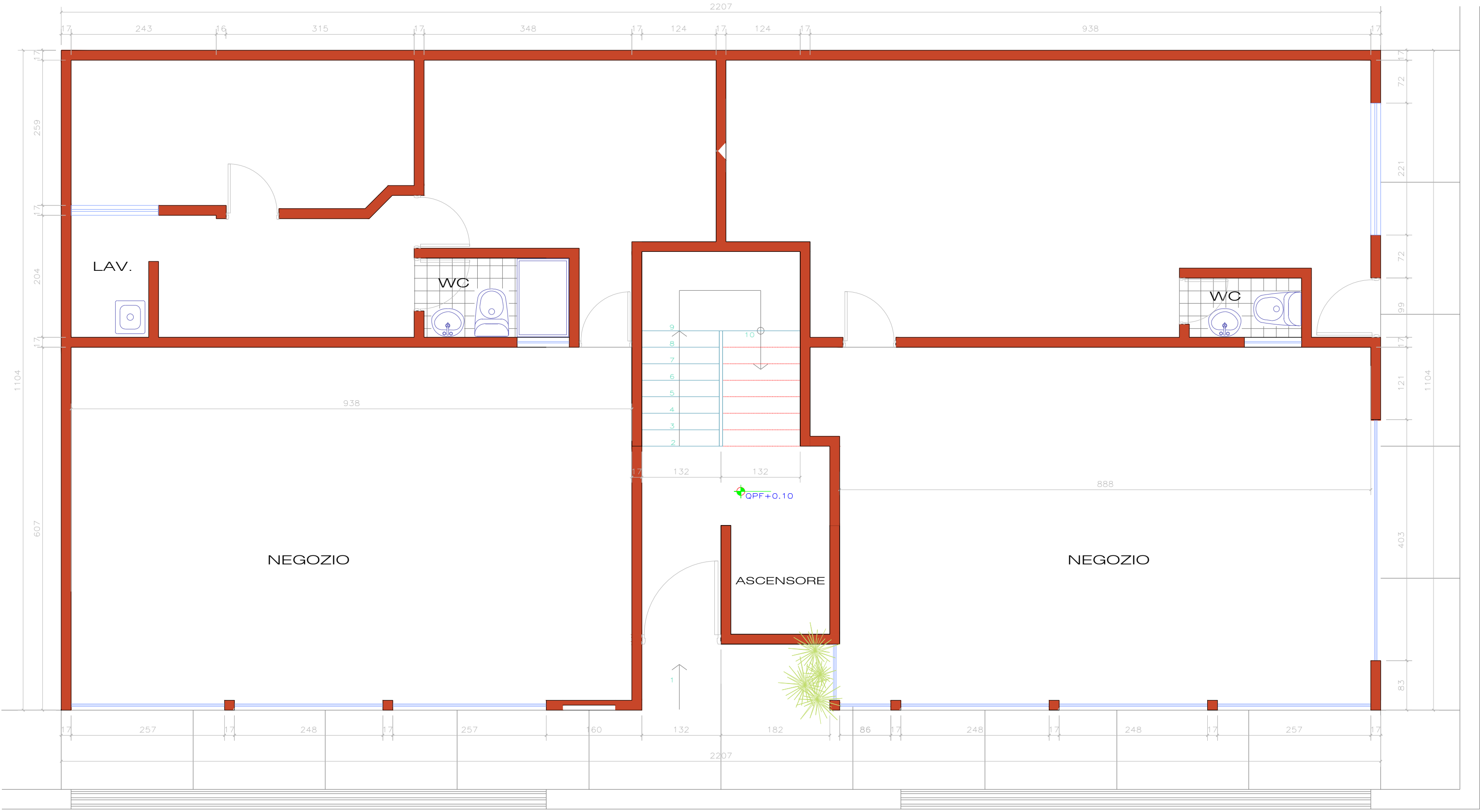
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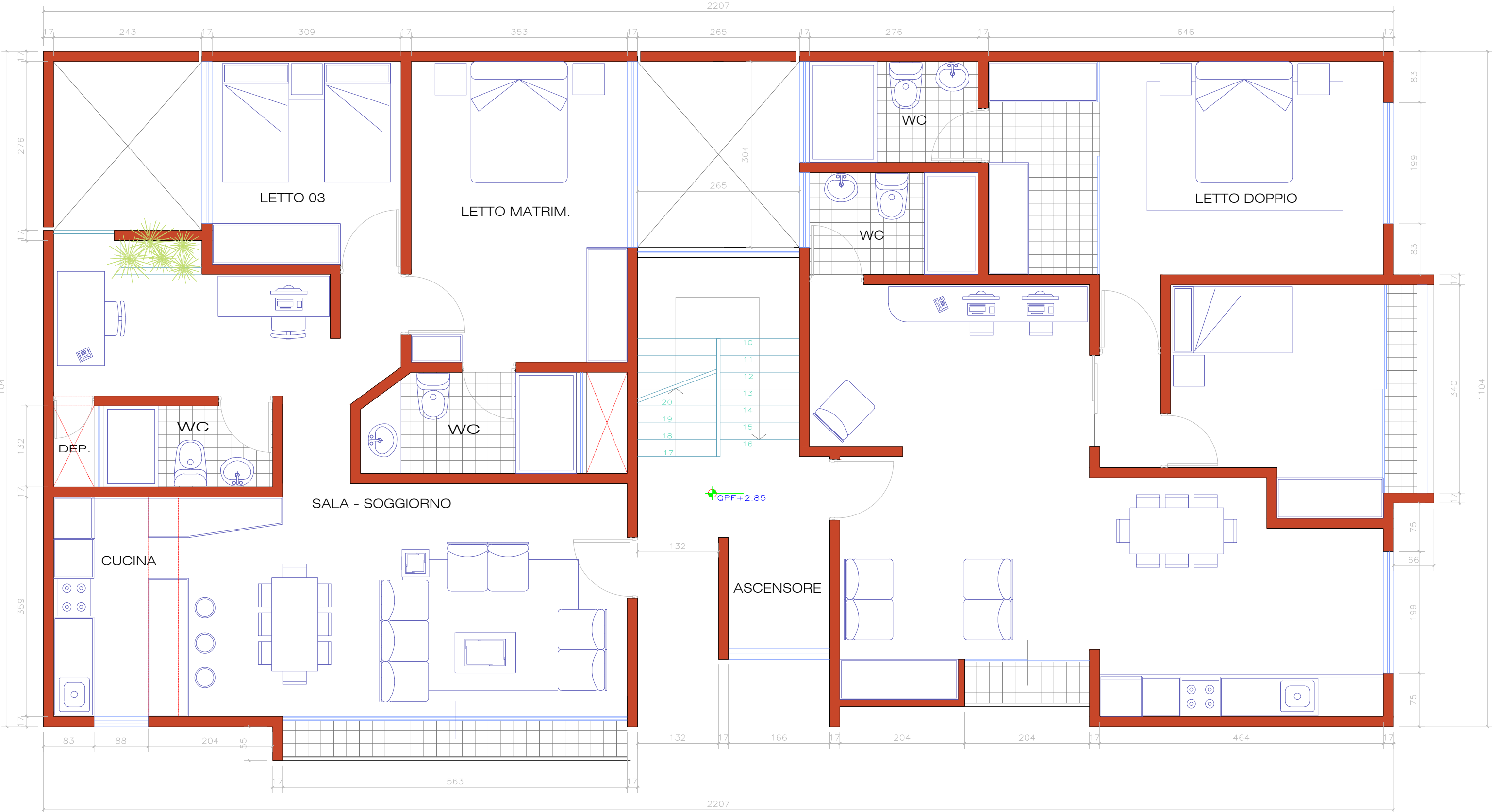
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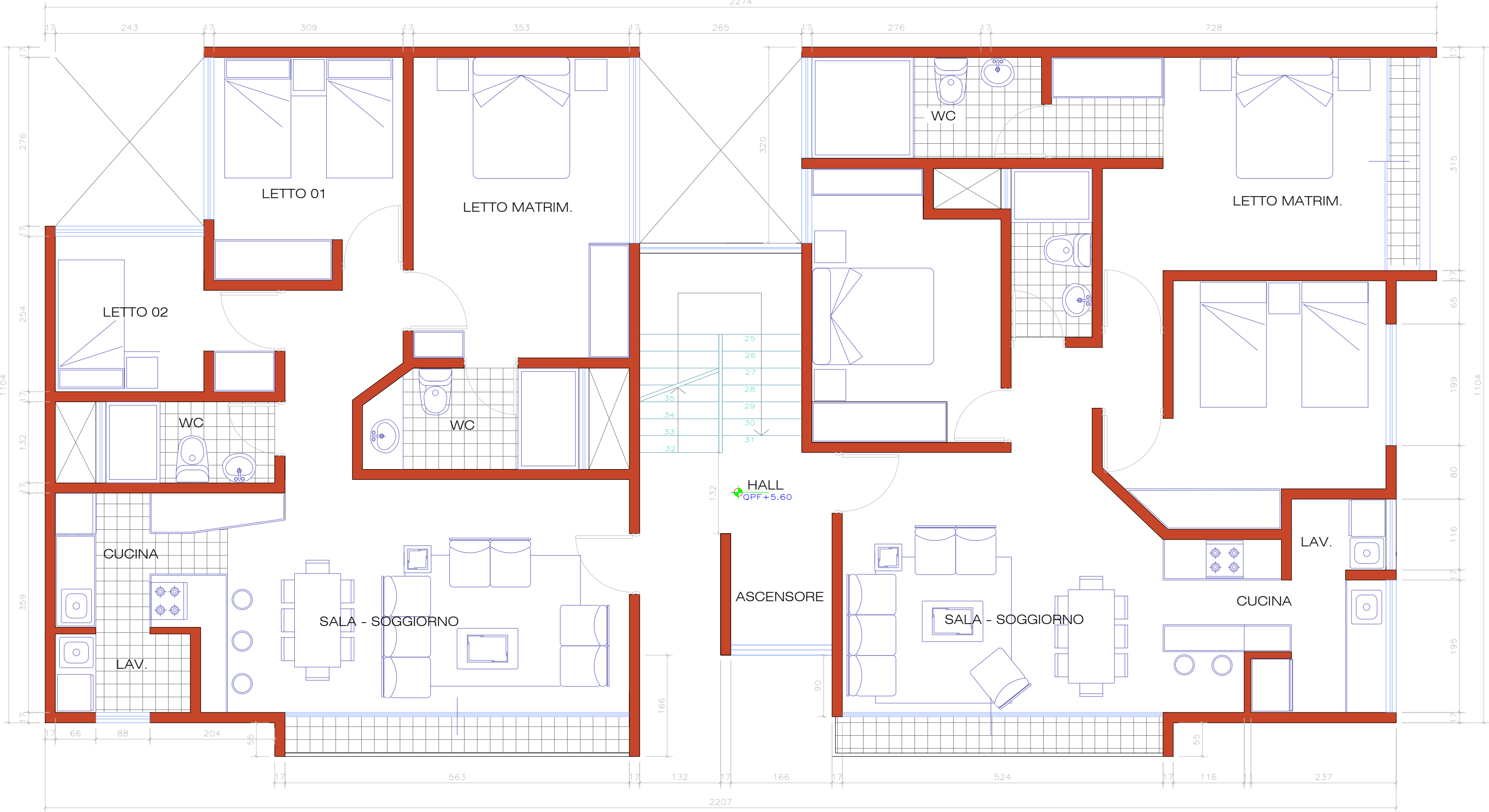
GROUND FLOOR PLAN



FIRST FLOOR PLAN



SECOND, THIRD, FOURTH AND FIFTH FLOOR PLAN



TERRACE PLAN



SCALE 1:60



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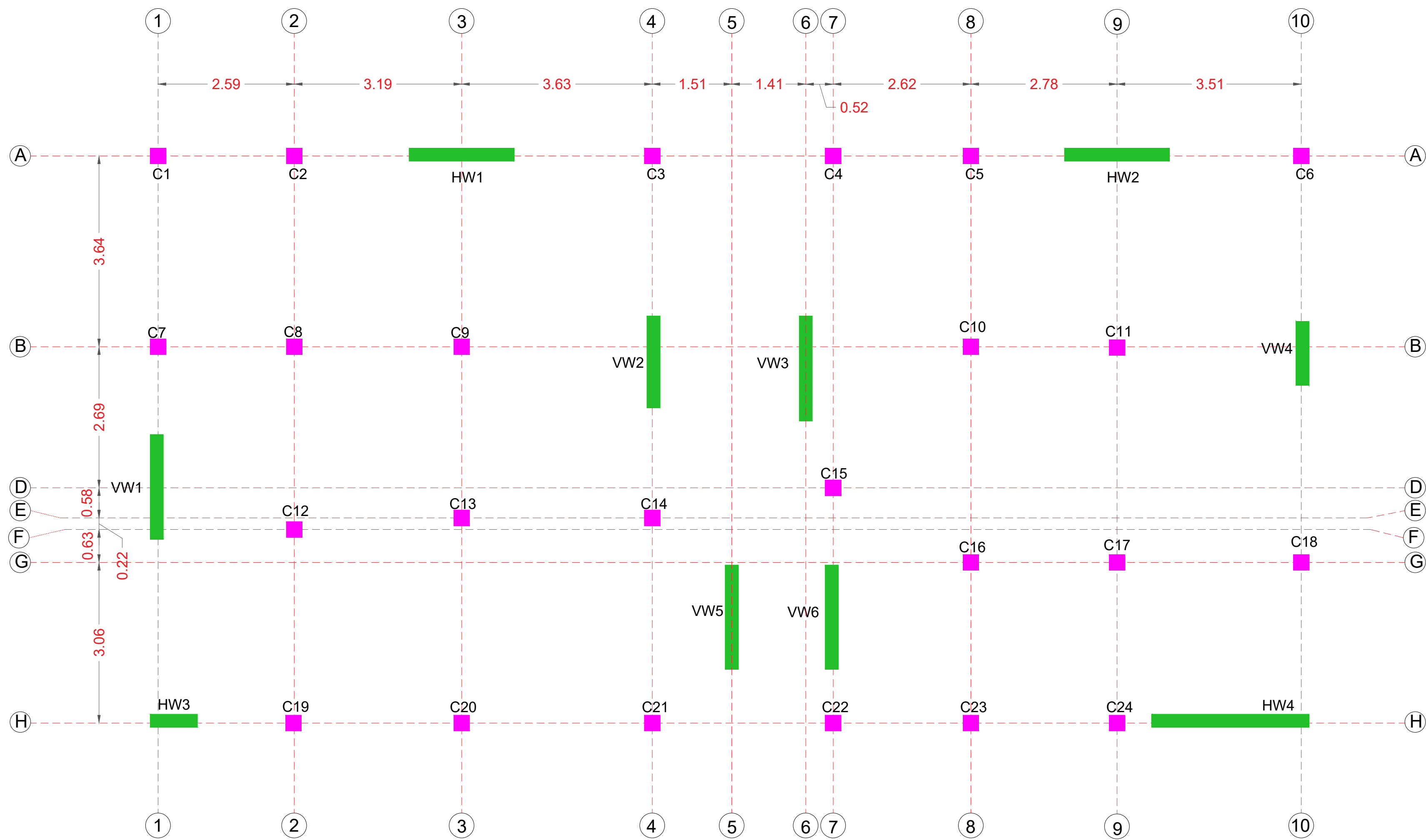
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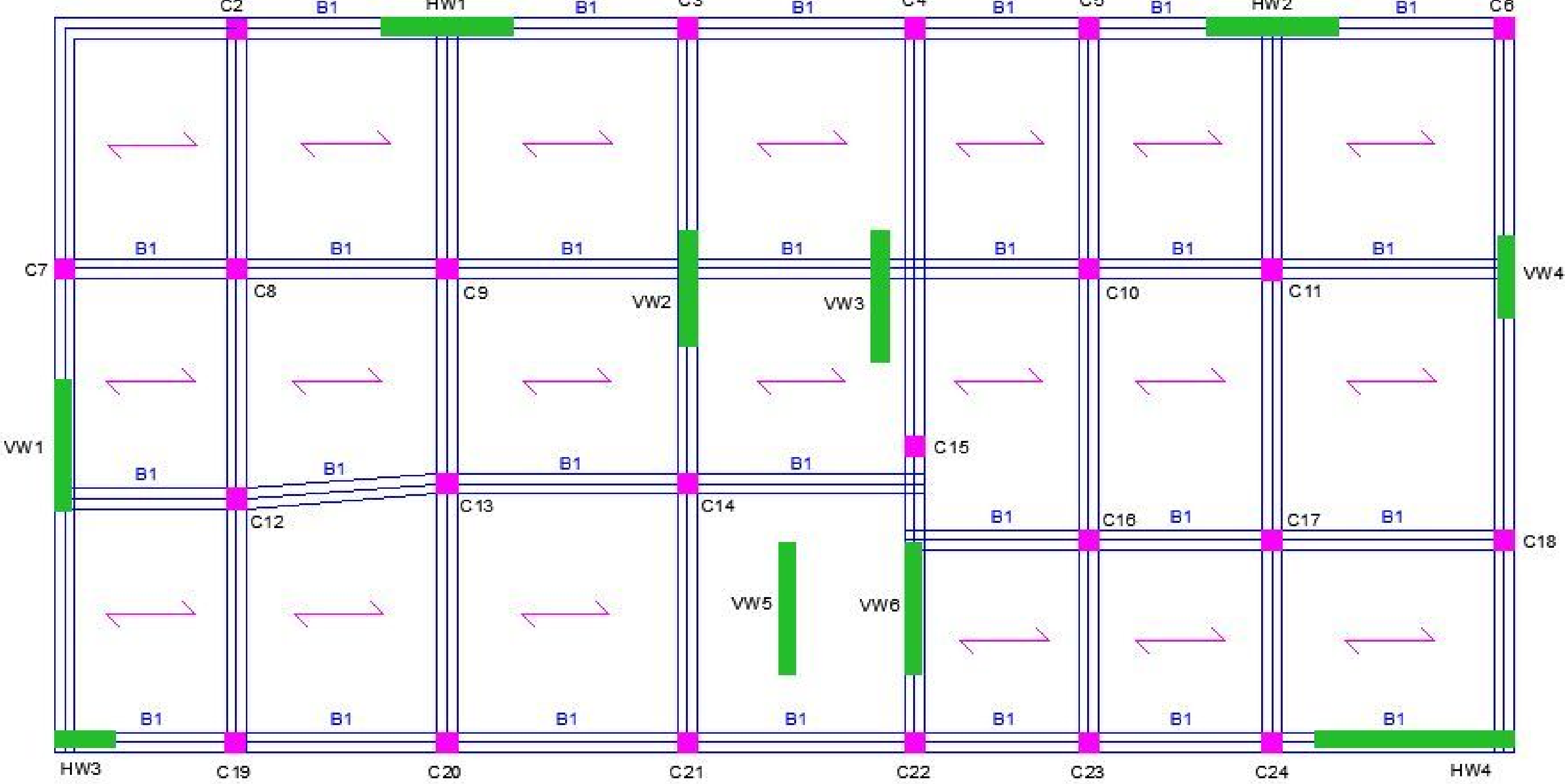
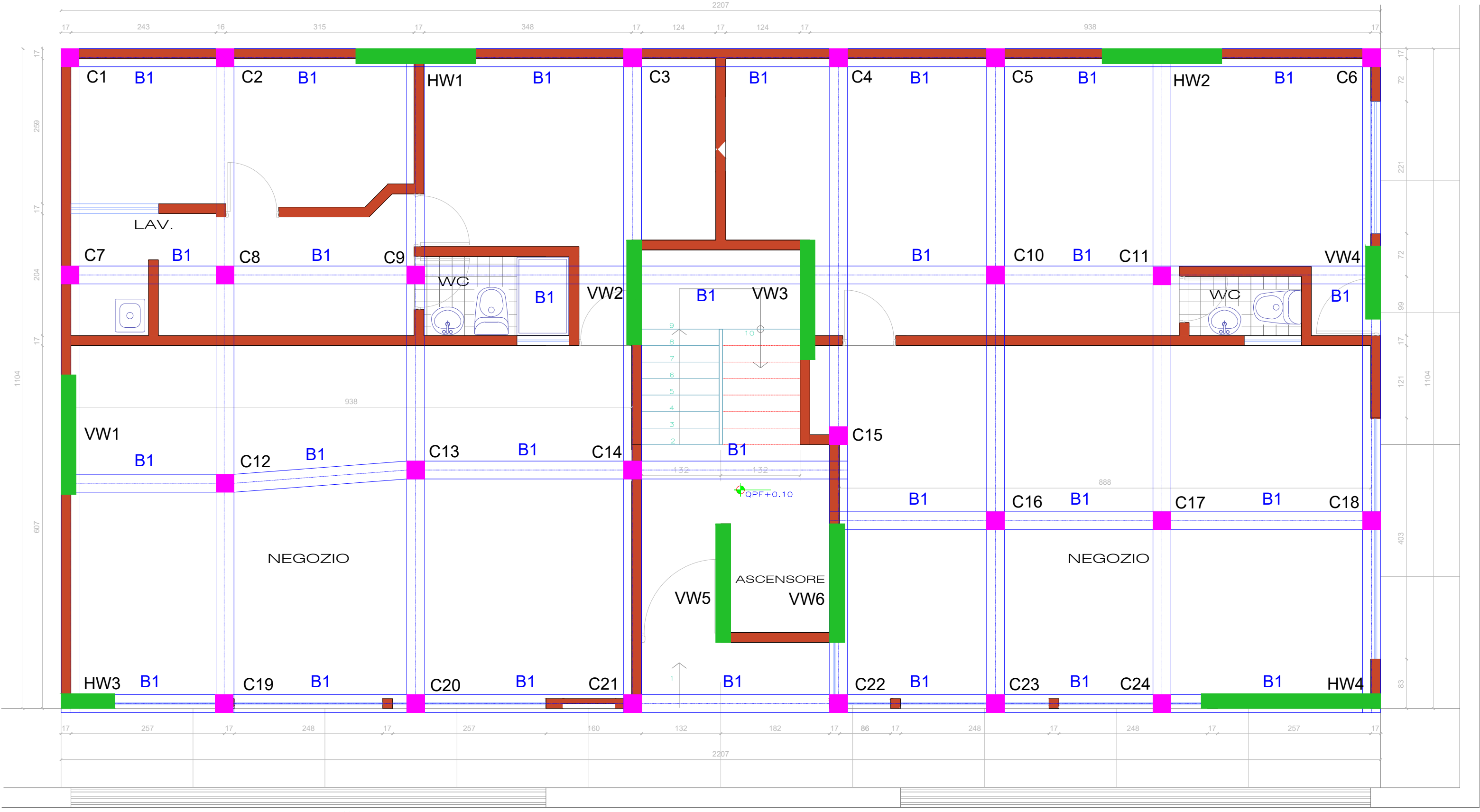
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CENTER LINE PLAN



STRUCTURAL LAYOUT - GROUND AND FIRST FLOOR



SCALE 1:60



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STRUCTURAL LAYOUTS
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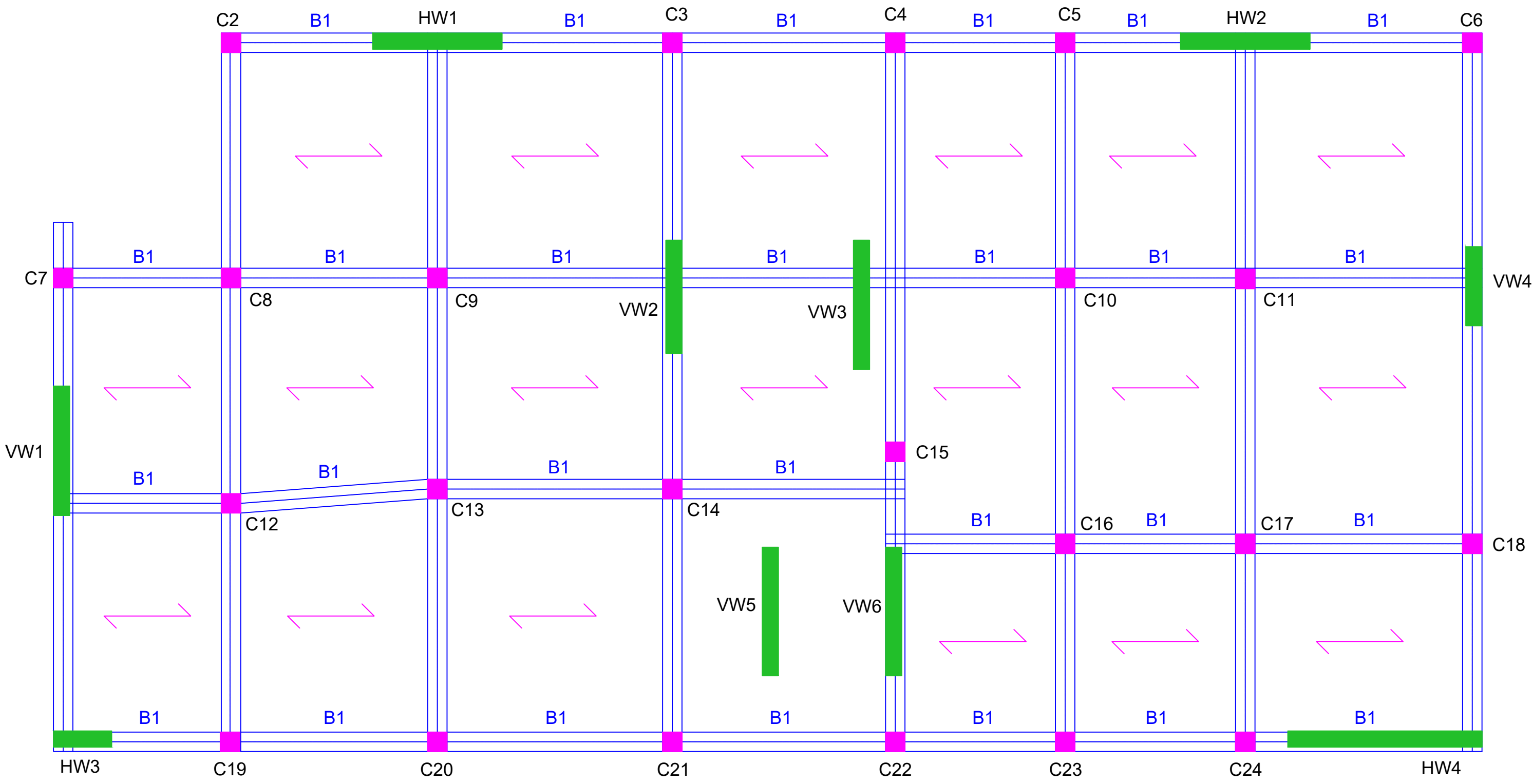
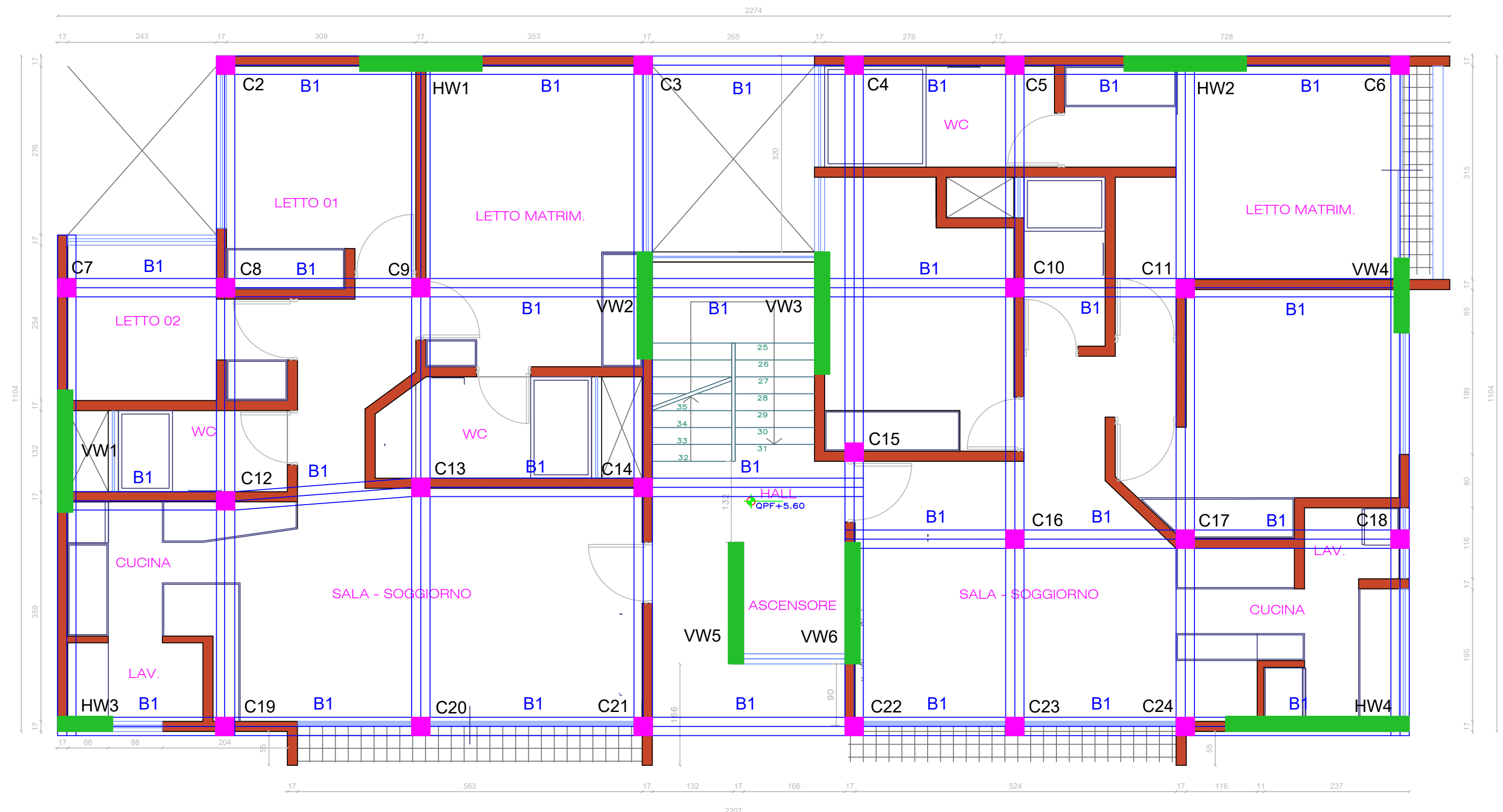
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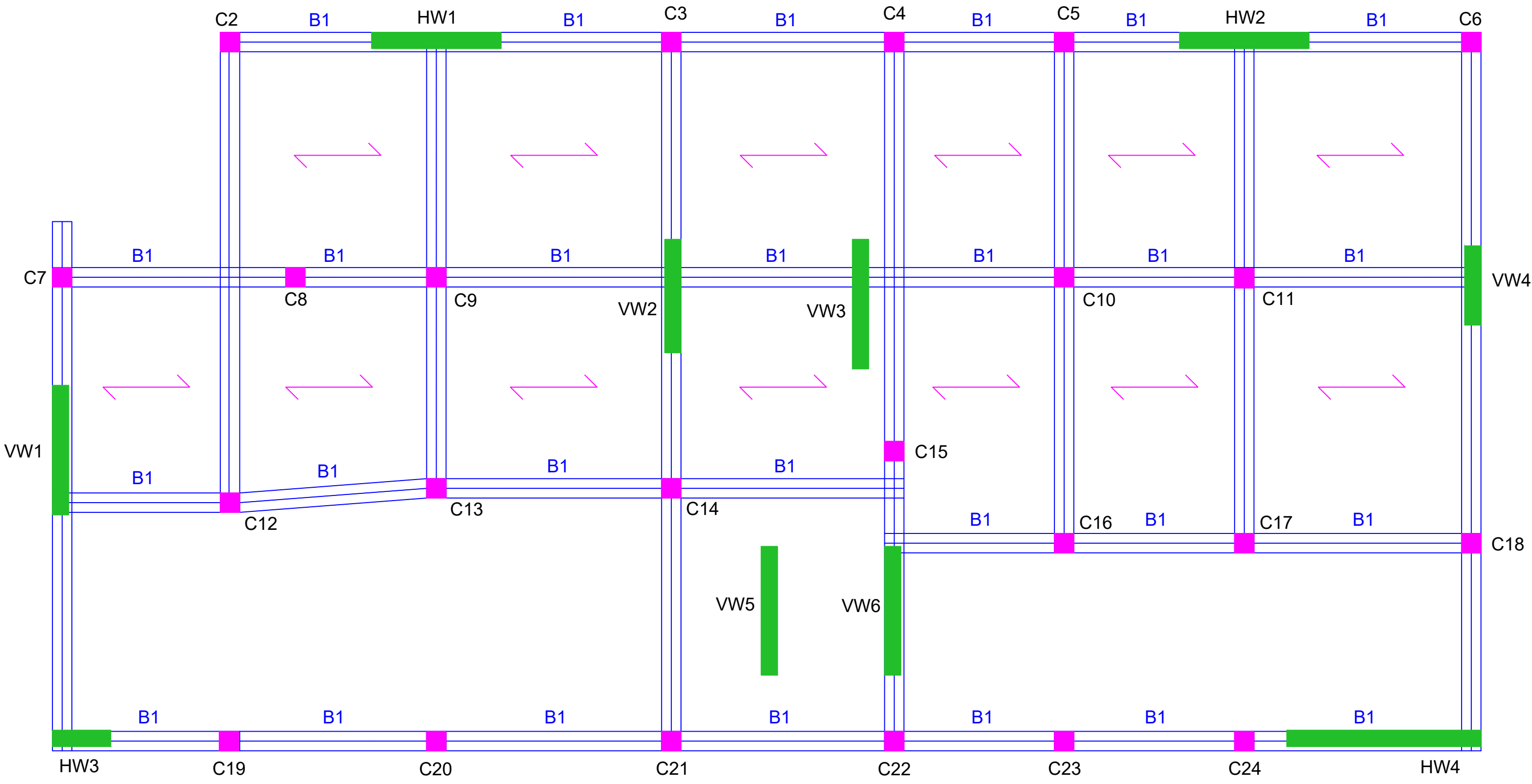
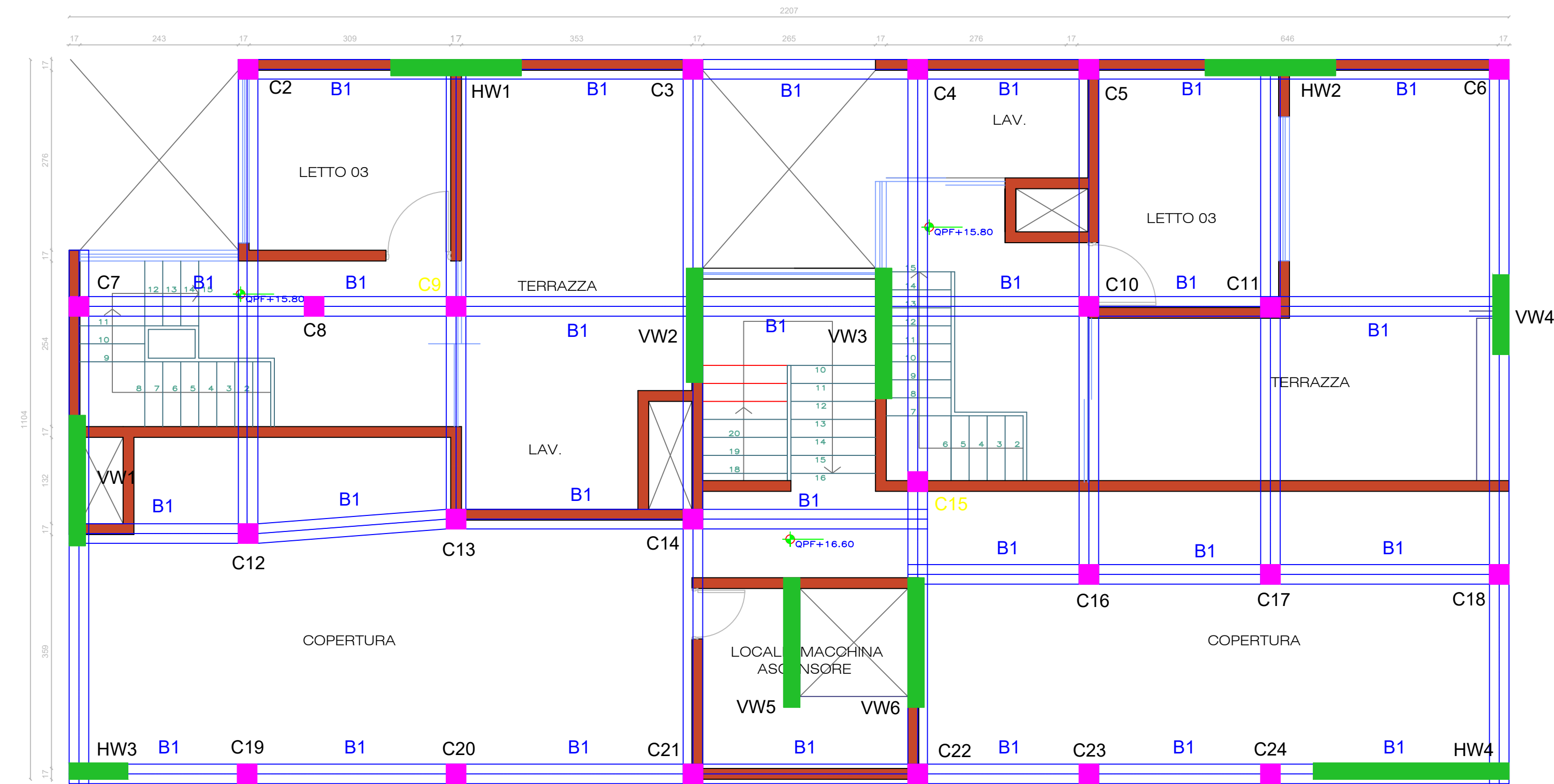
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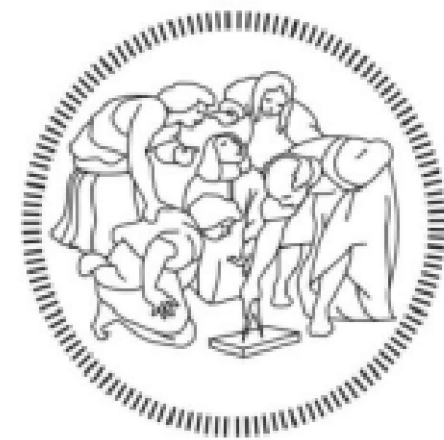
STRUCTURAL LAYOUT - SECOND, THRID, FOURTH AND FIFTH FLOOR



STRUCTURAL LAYOUT - TERRACE



SCALE 1:60



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(2023-24)

STRUCTURAL ELEMENTS
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(PRIMARY BEAM B1)

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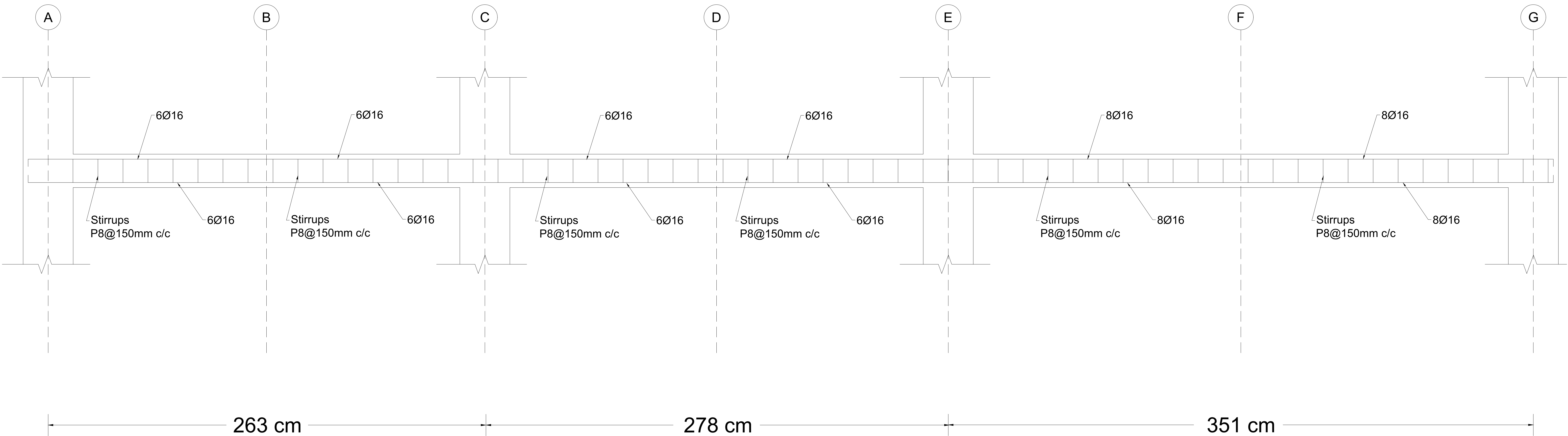
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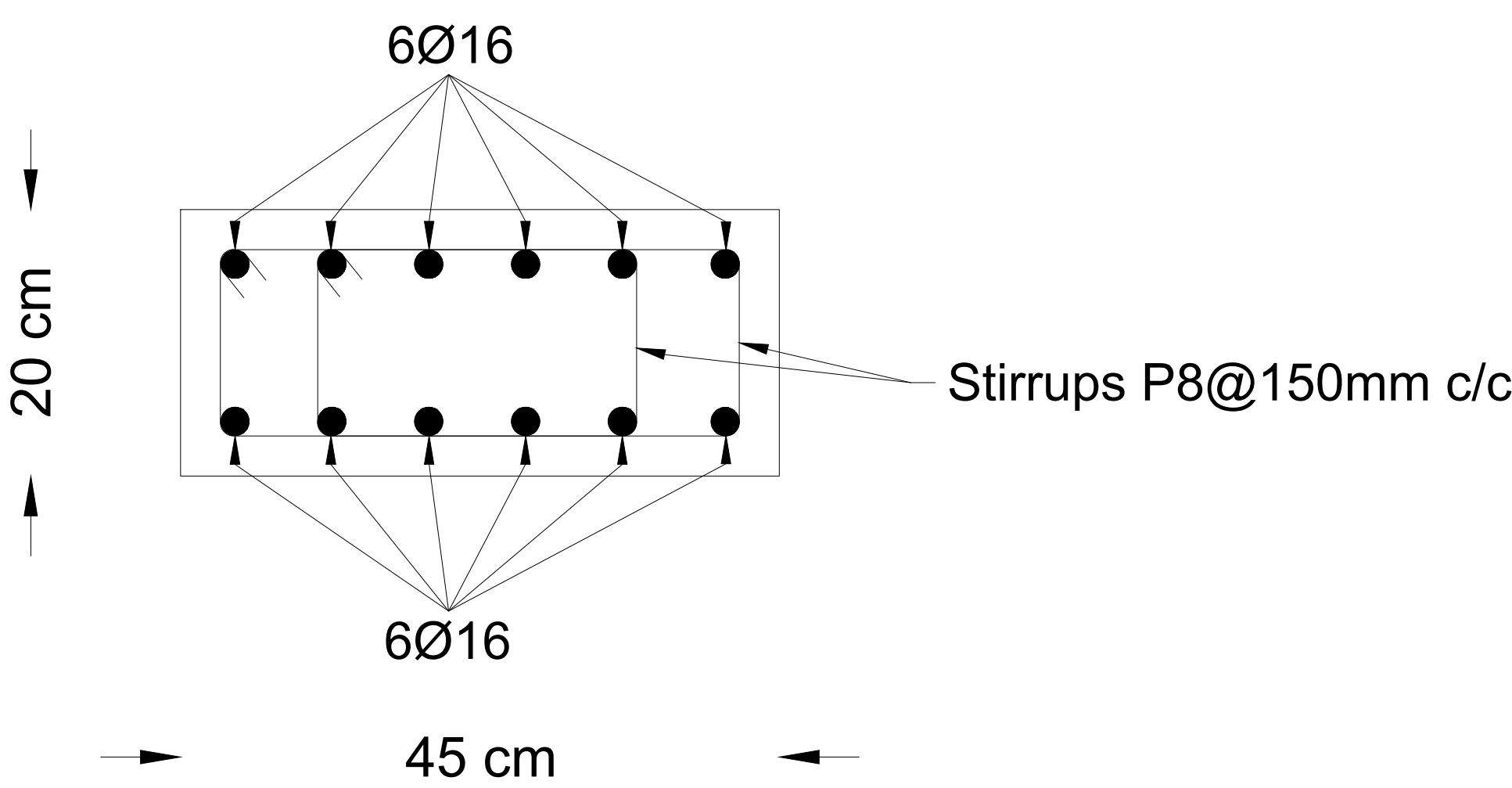
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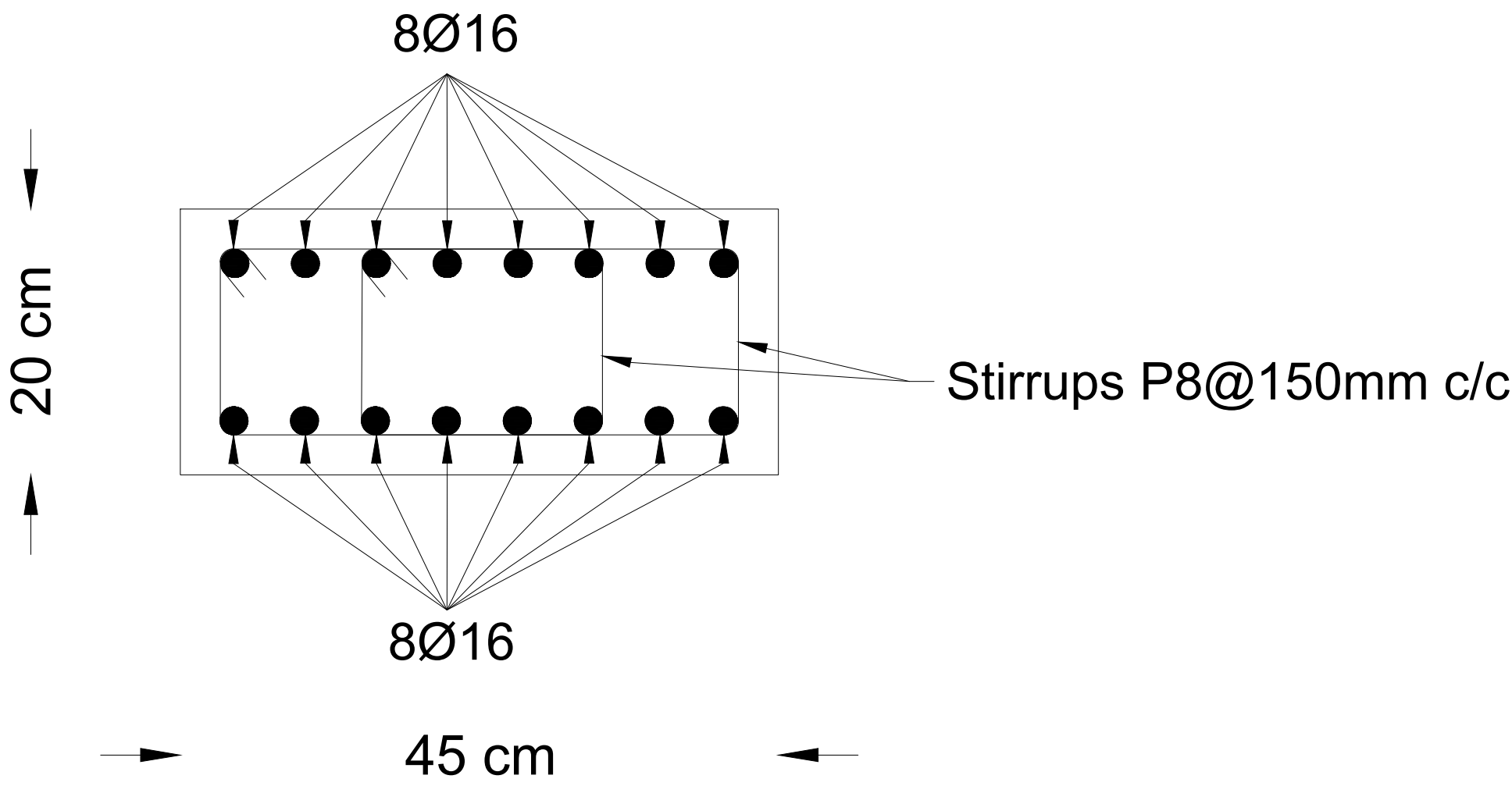
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MULTI BAY BEAM (B1) - CROSS SECTION



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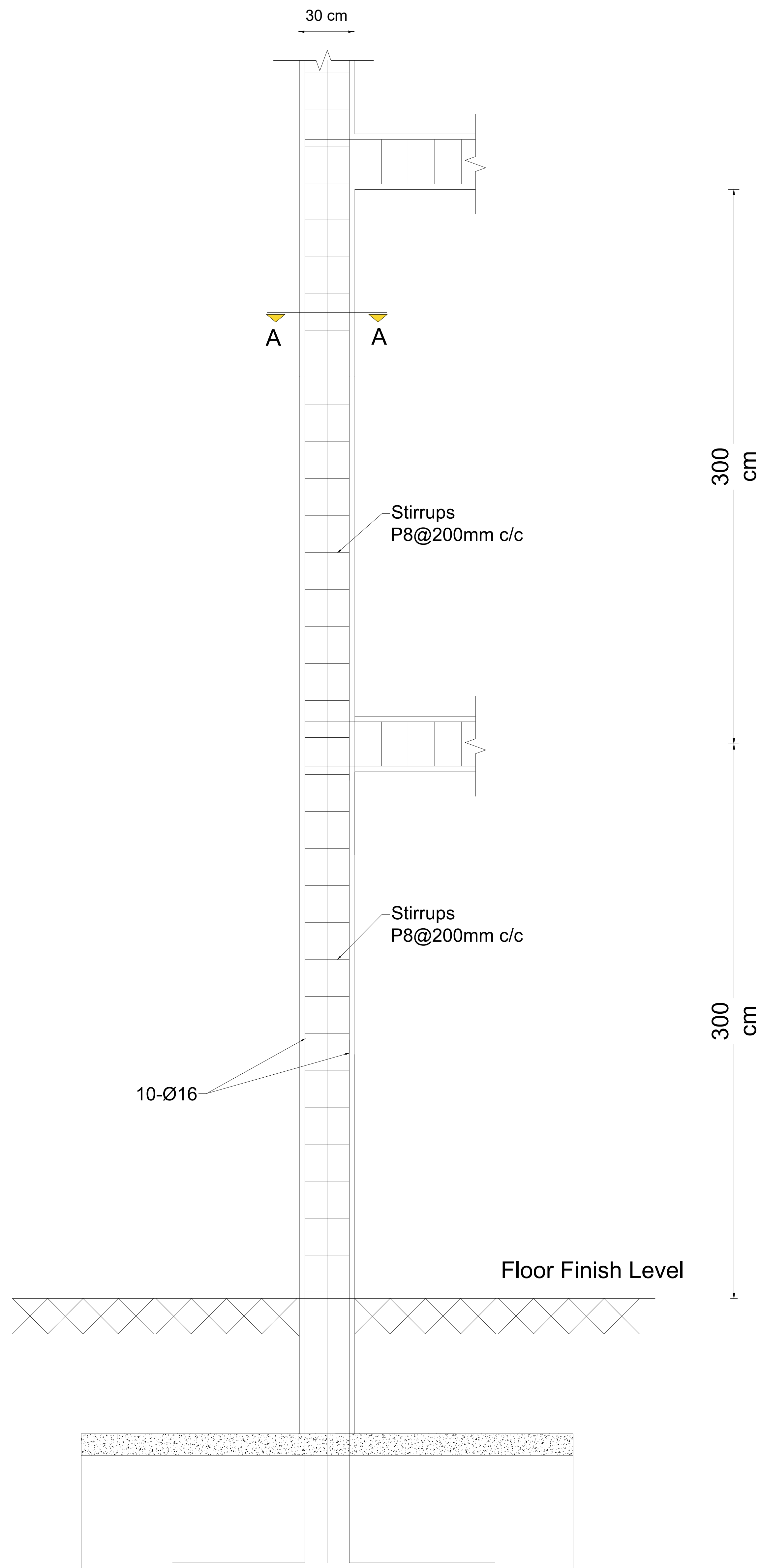


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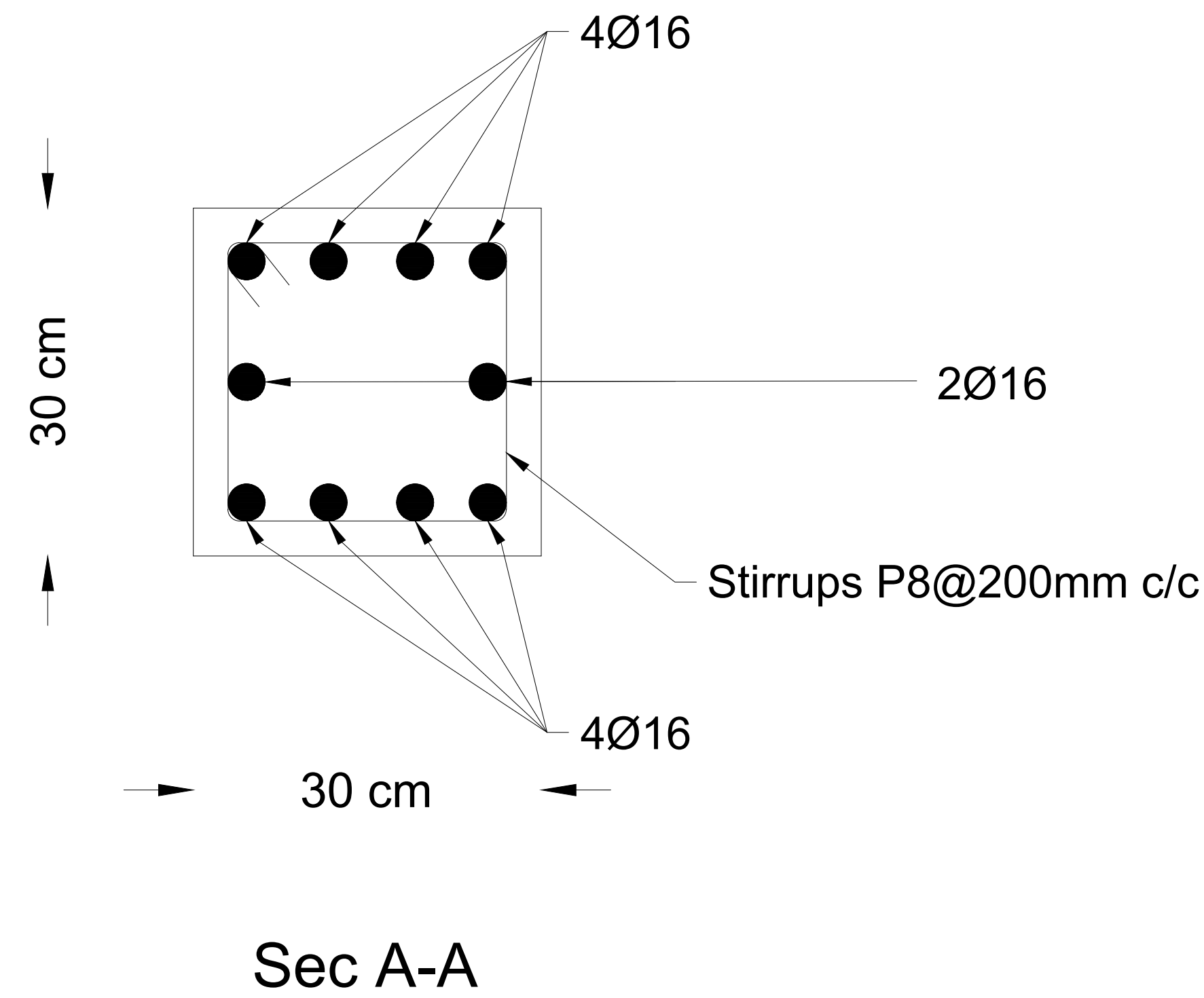
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BUILDING CATEGORY OF USE: A
SOIL TYPE : 'C'
CONCRETE COVER: 30mm

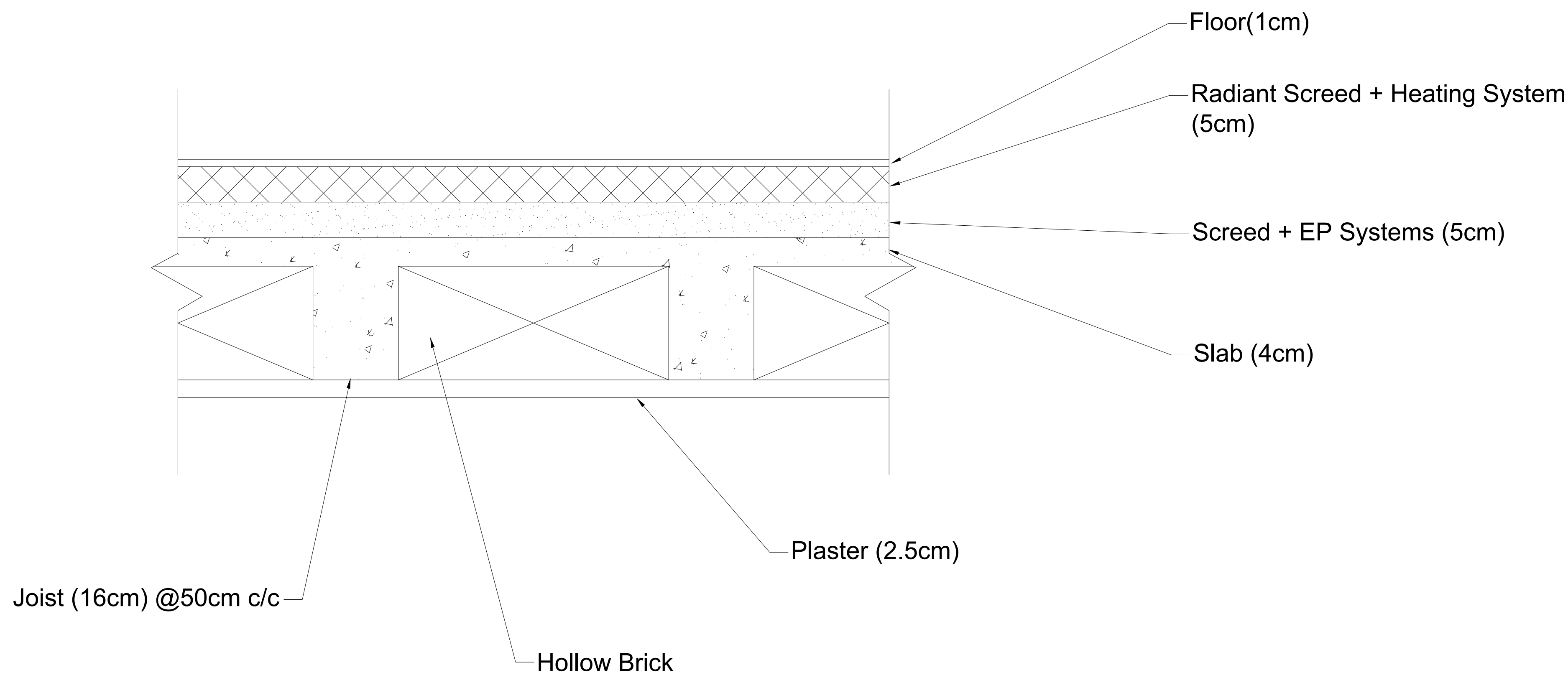
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COLUMN - CROSS SECTION



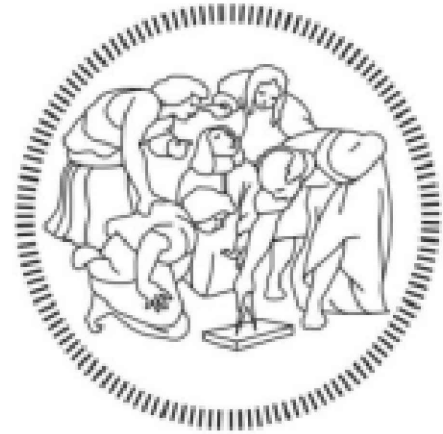
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EXPOSURE CLASS: XC-1
BUILDING CATEGORY OF USE: A
SOIL TYPE : 'C'
CONCRETE COVER: 30mm

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STRUCTURAL ELEMENTS
SECTIONS
(COLUMN AND SLAB)

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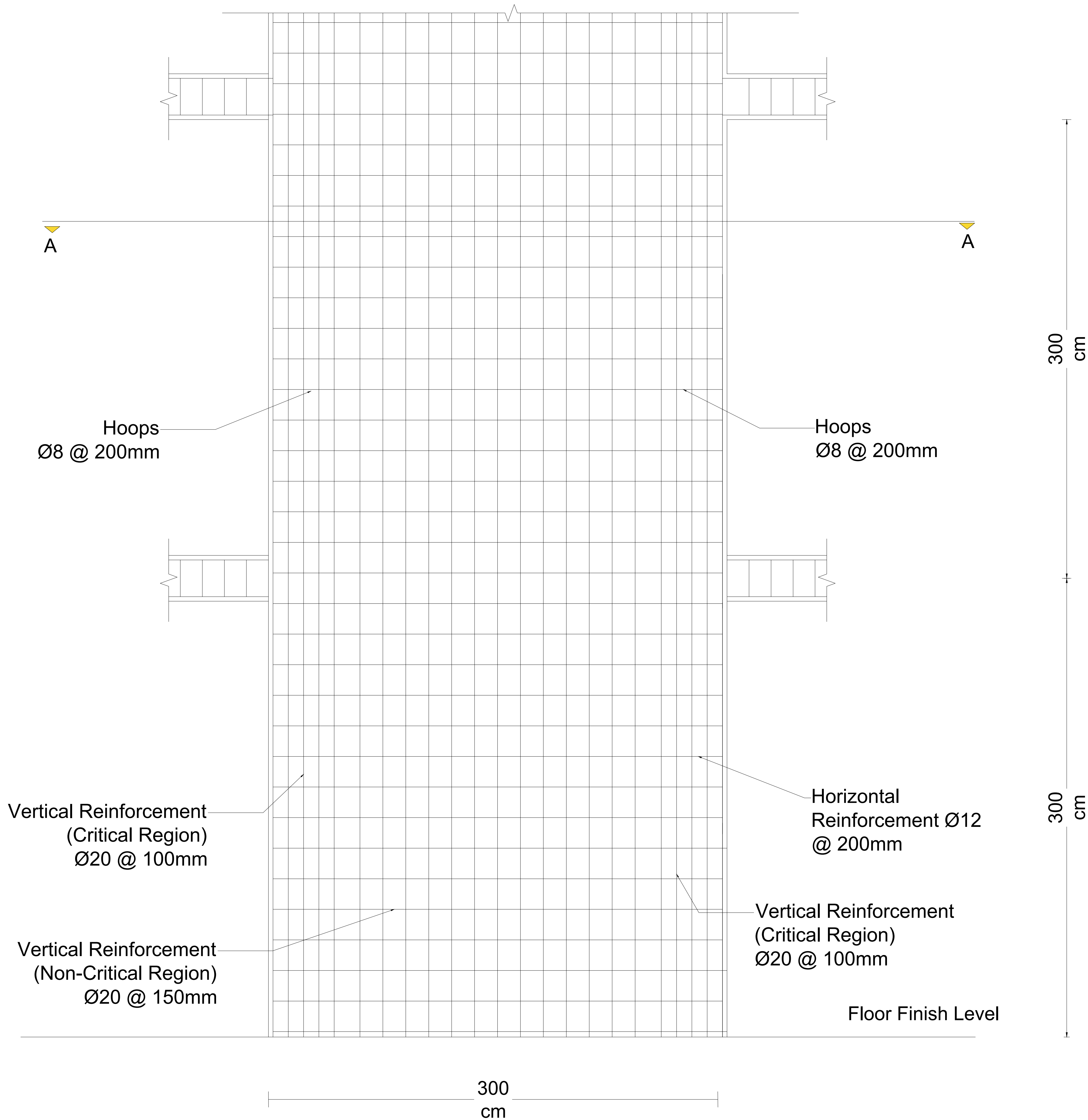
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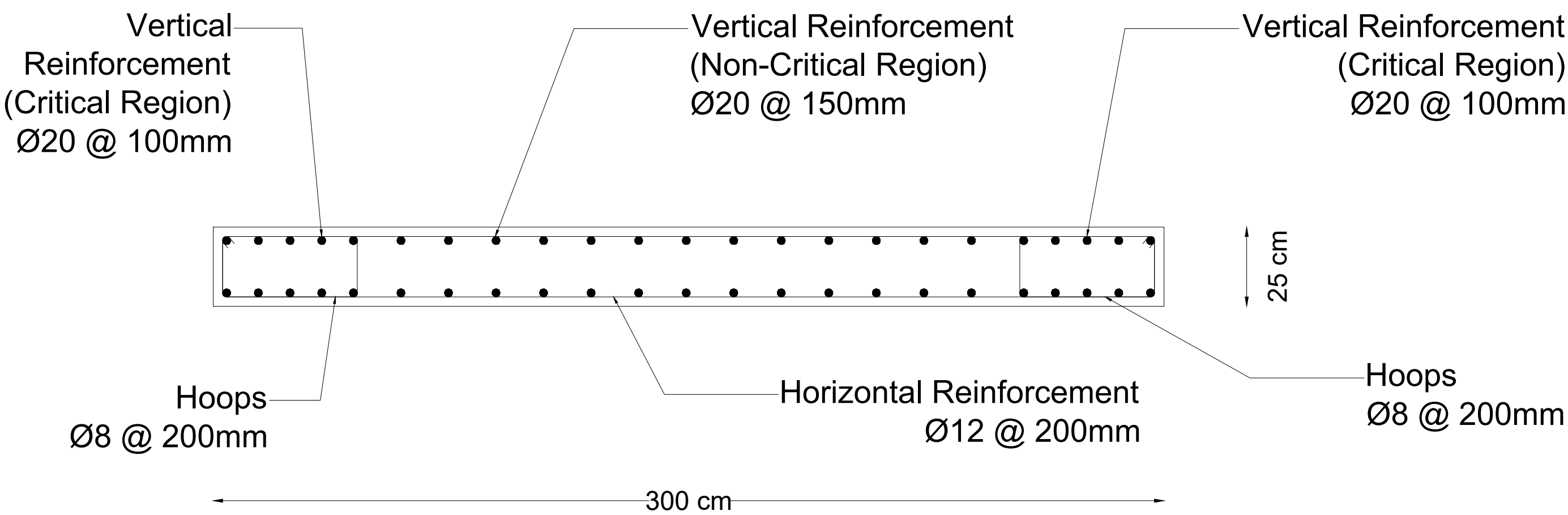
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Area (m2)	0.5	0.225	0.75	0.5	0.44	0.3
Length (m)	2	0.9	3	2	1.75	1.2
Width (m)	0.25	0.25	0.25	0.25	0.25	0.25

TYPICAL SHEAR WALL - CROSS SECTION

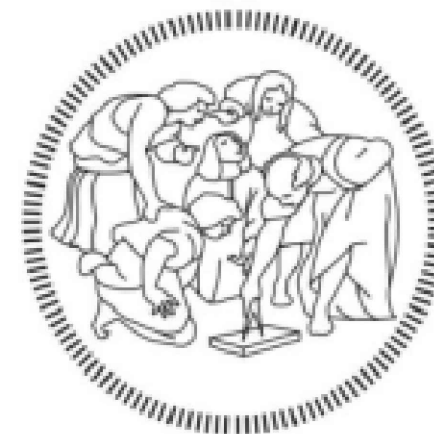


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EXPOSURE CLASS: XC-1
BUILDING CATEGORY OF USE: A
SOIL TYPE : 'C'
CONCRETE COVER: 30mm

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(WALL SCHEDULE AND TYPICAL
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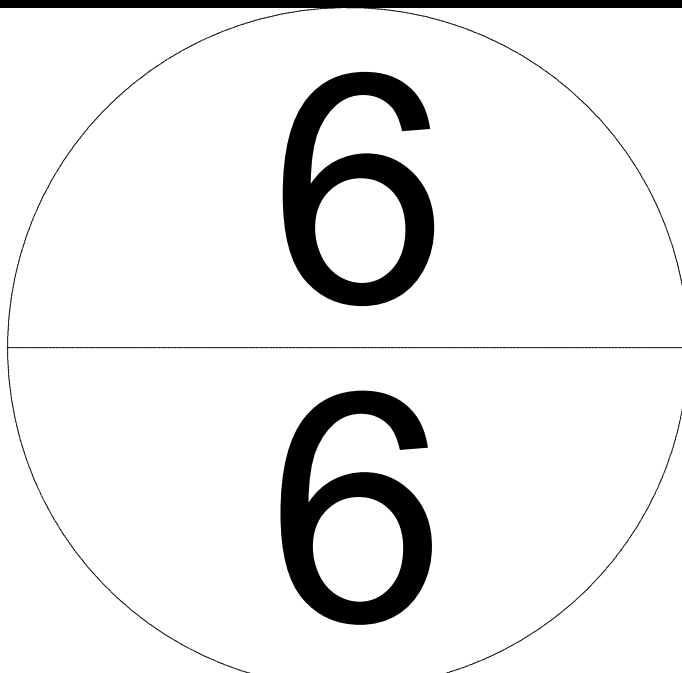
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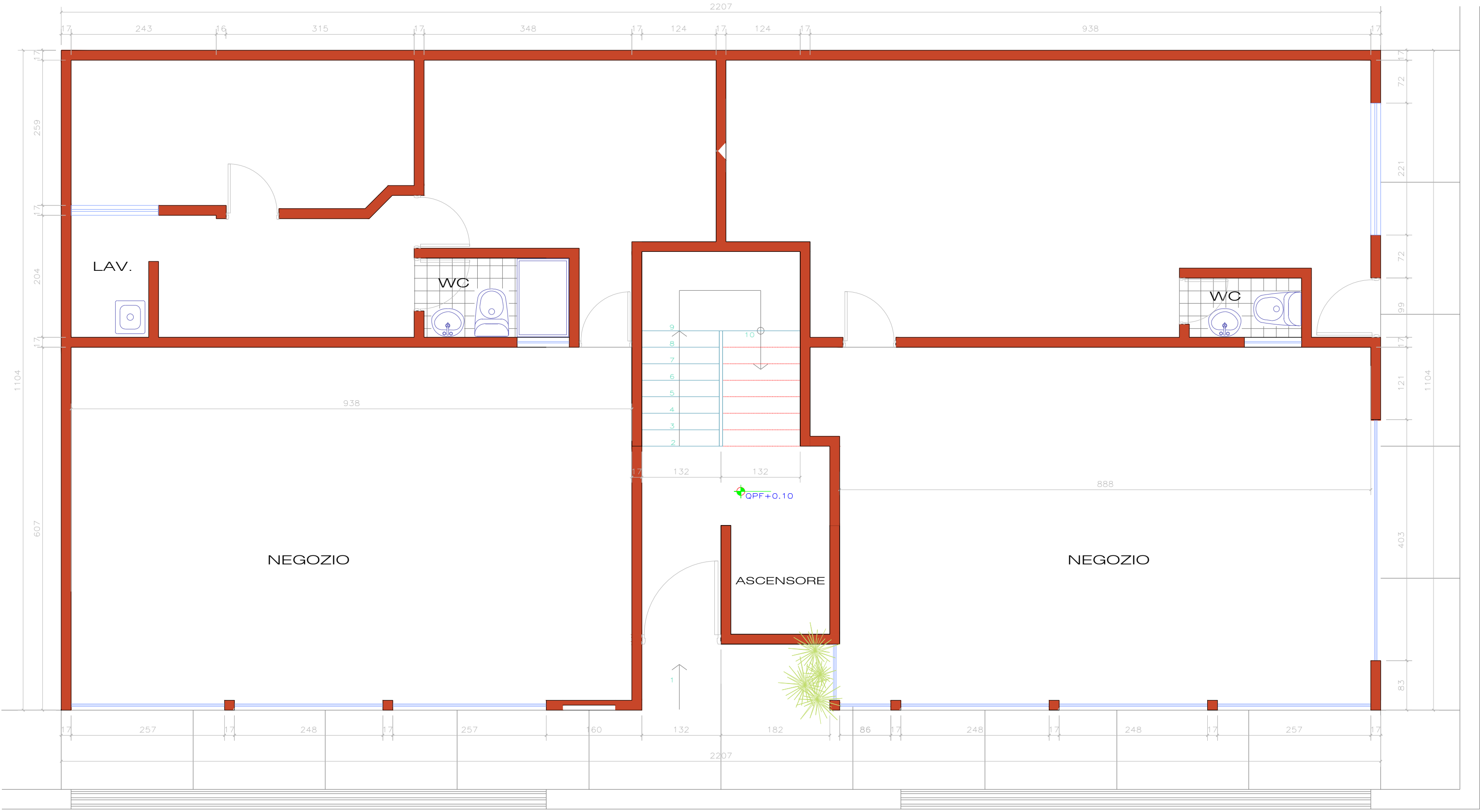
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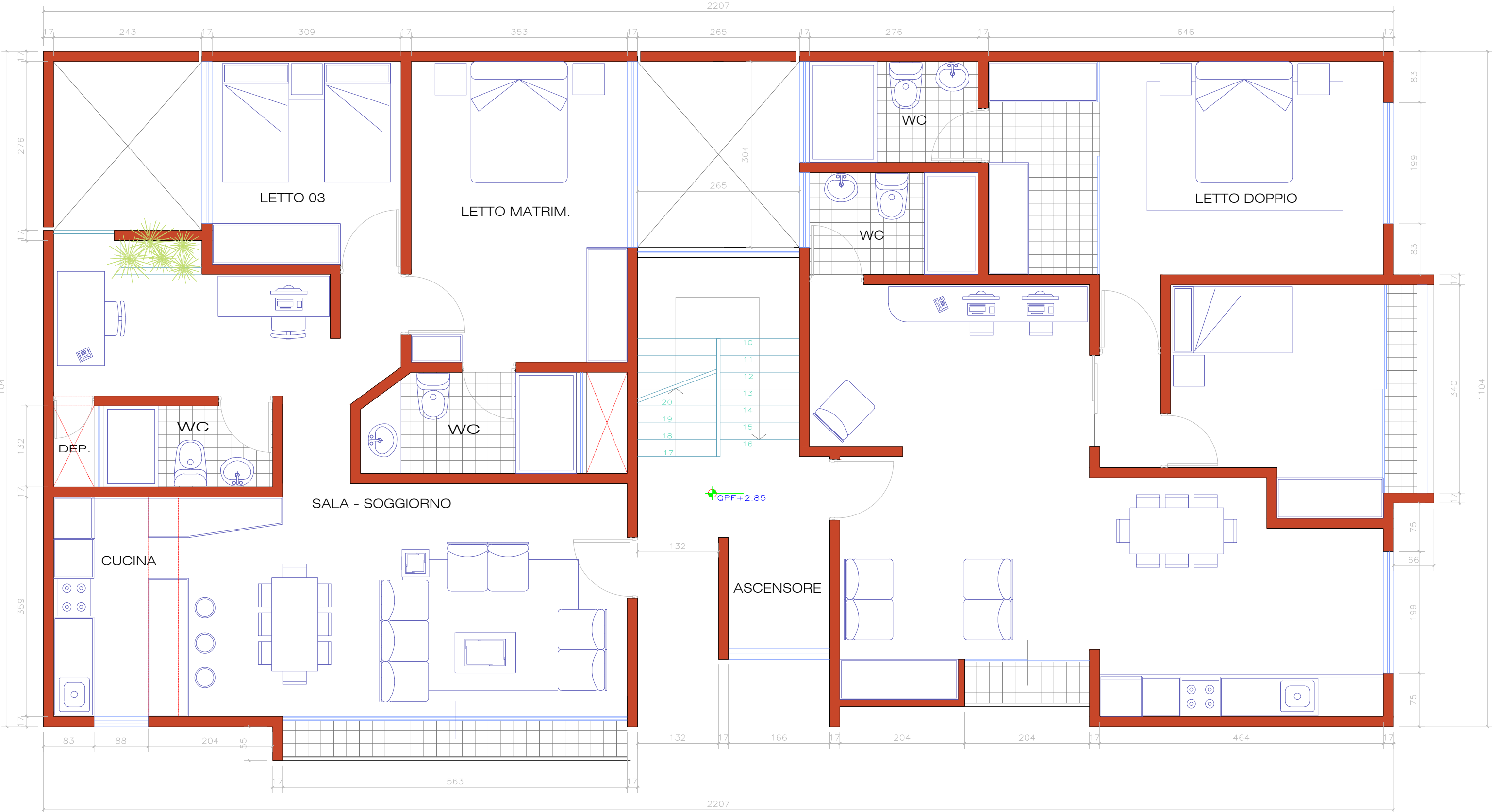
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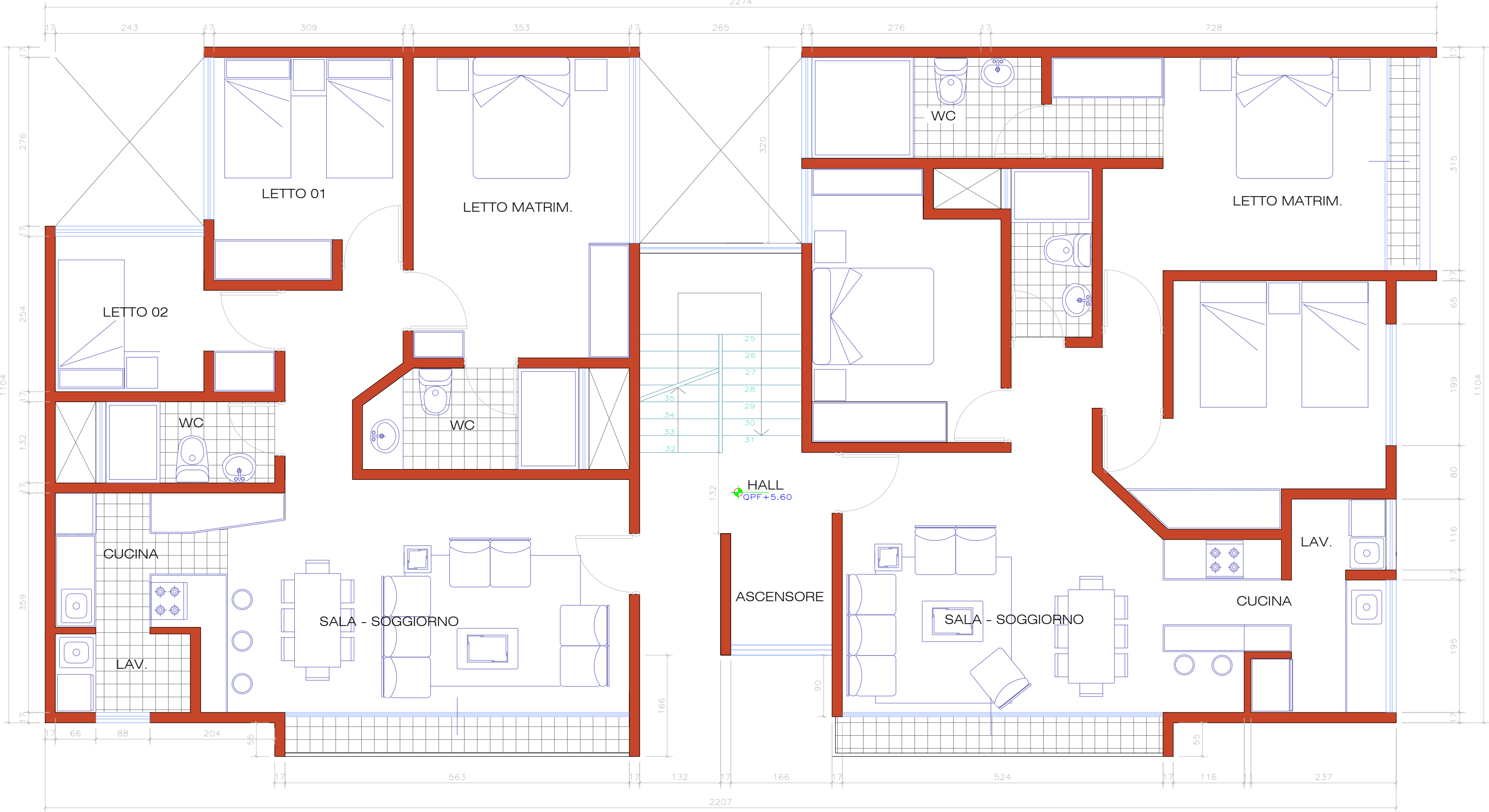
GROUND FLOOR PLAN



FIRST FLOOR PLAN



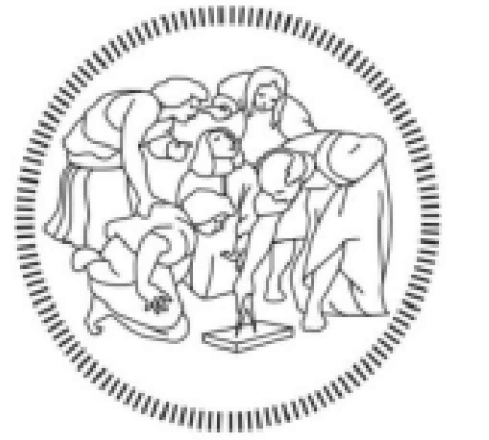
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TERRACE PLAN



SCALE 1:60



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CONCEPTUAL DESIGN OF
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STRUCTURAL LAYOUTS
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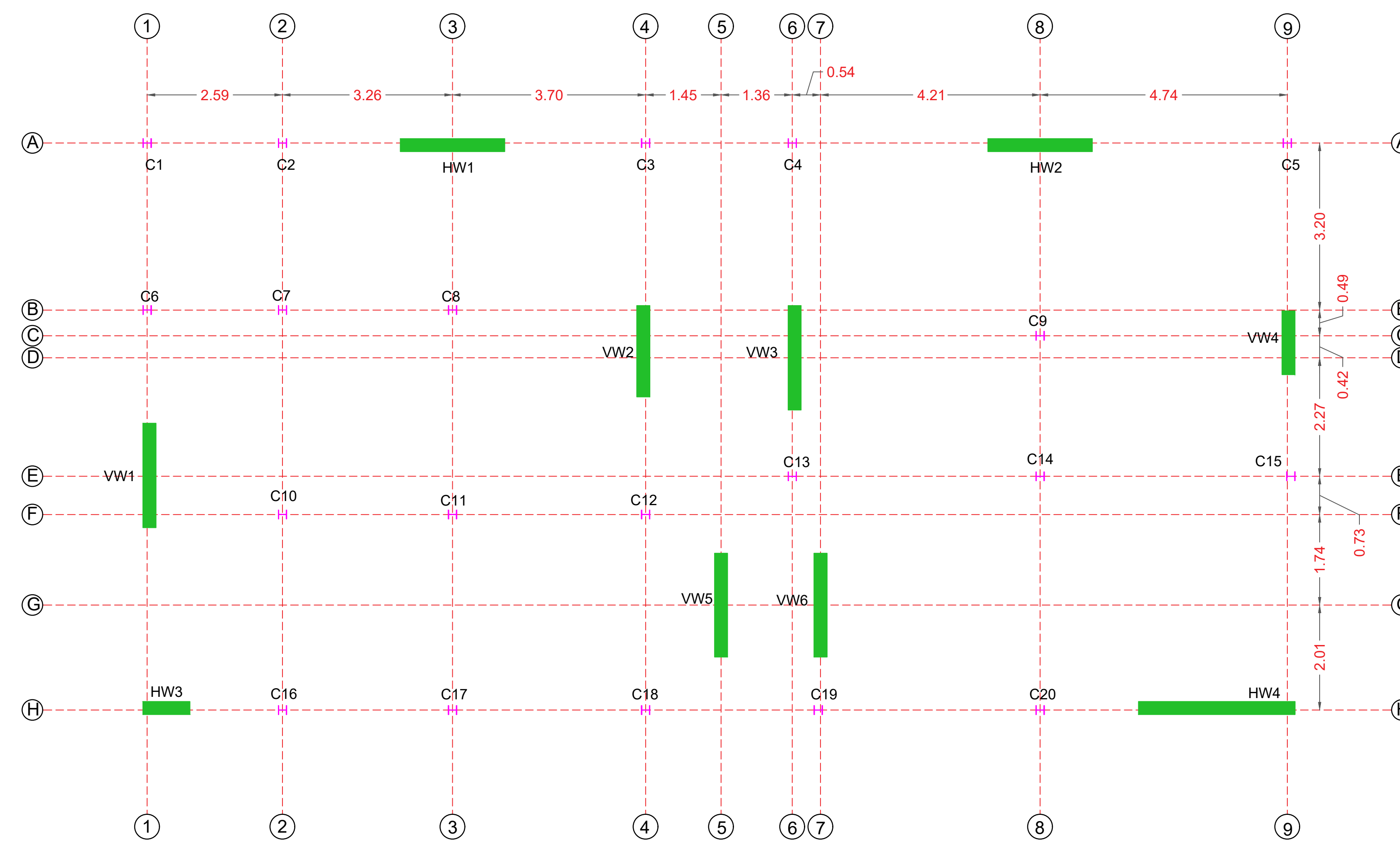
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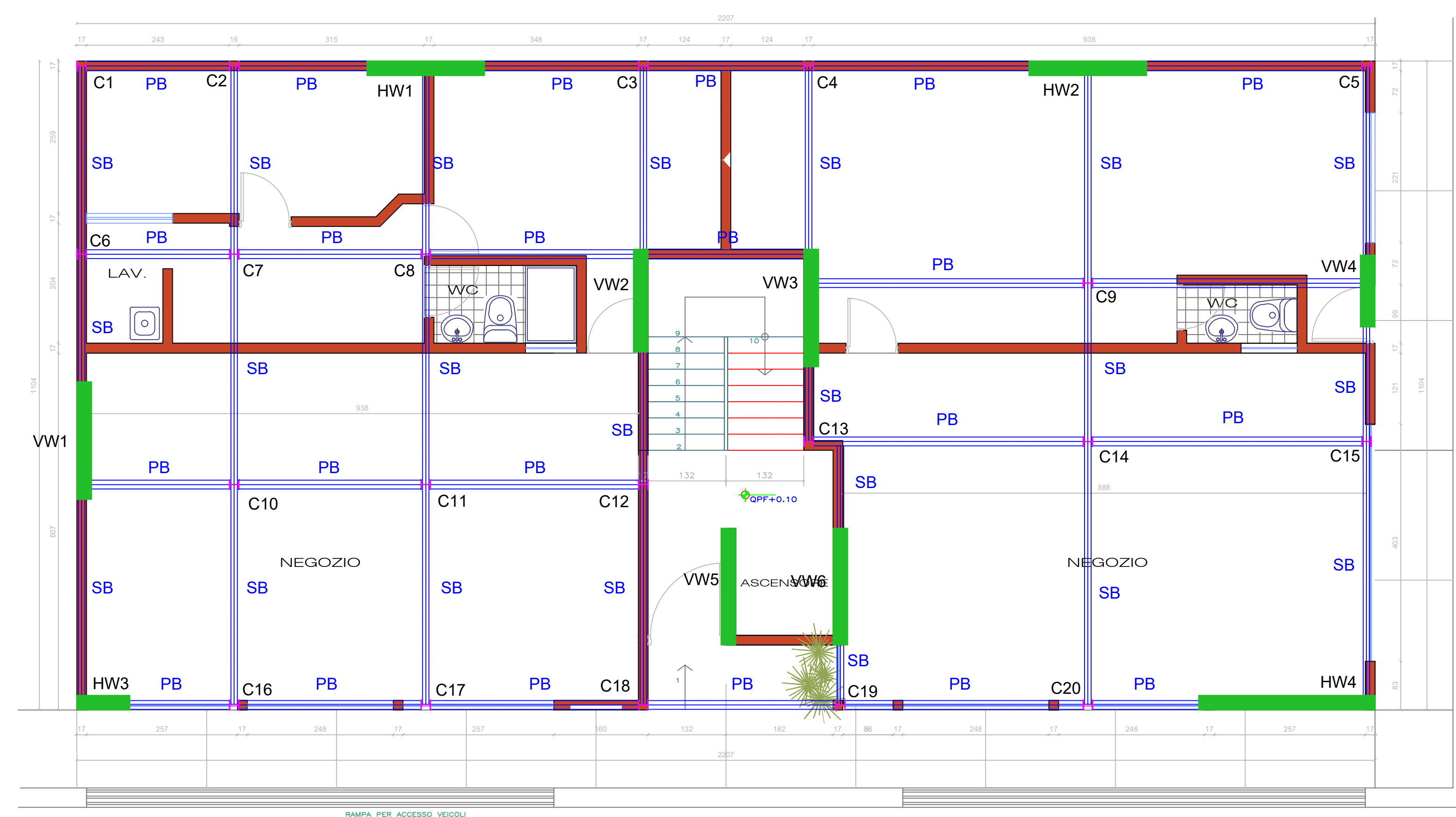
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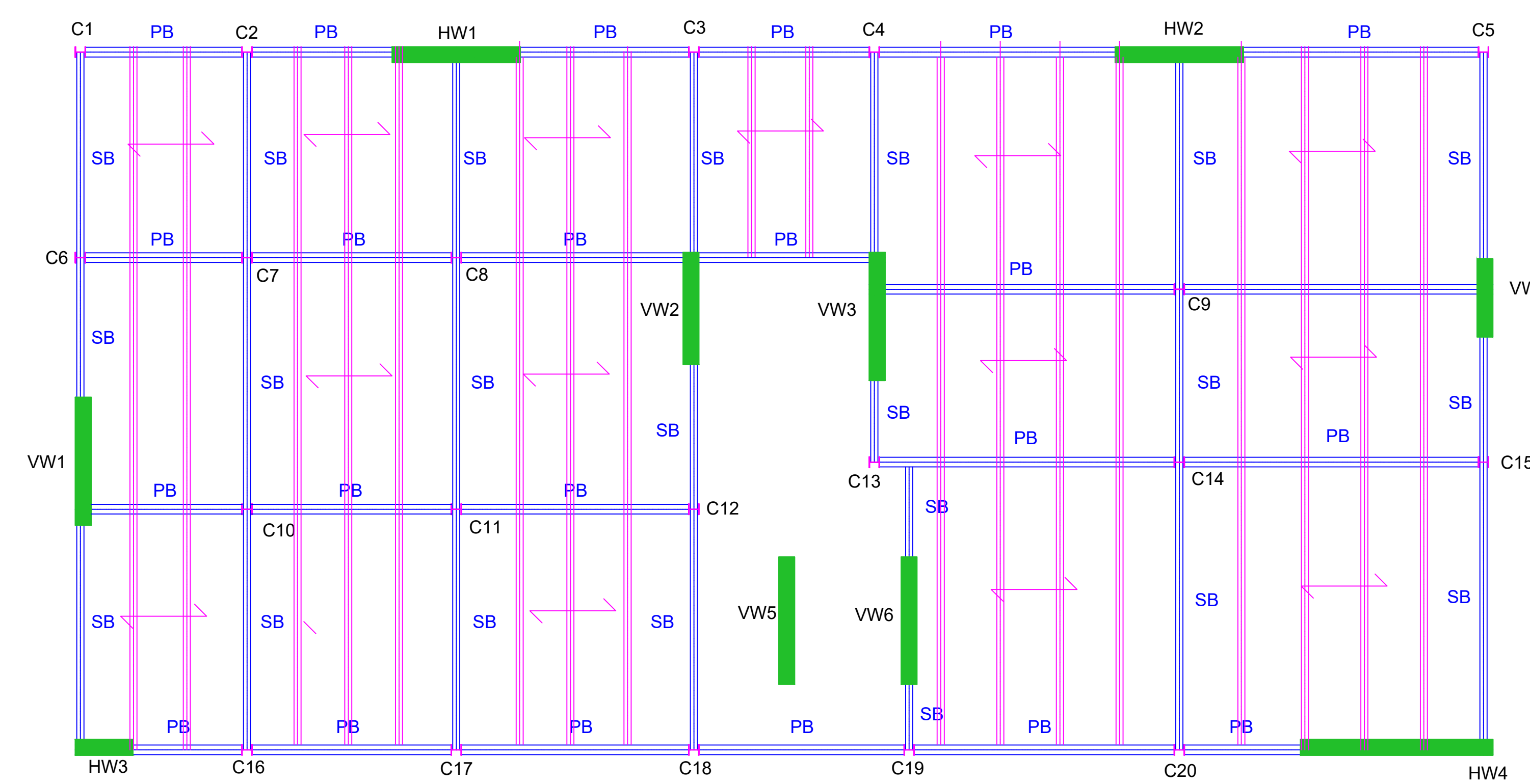
CENTER LINE PLAN



STRUCTURAL LAYOUT - GROUND AND FIRST FLOOR



SCALE 1:60





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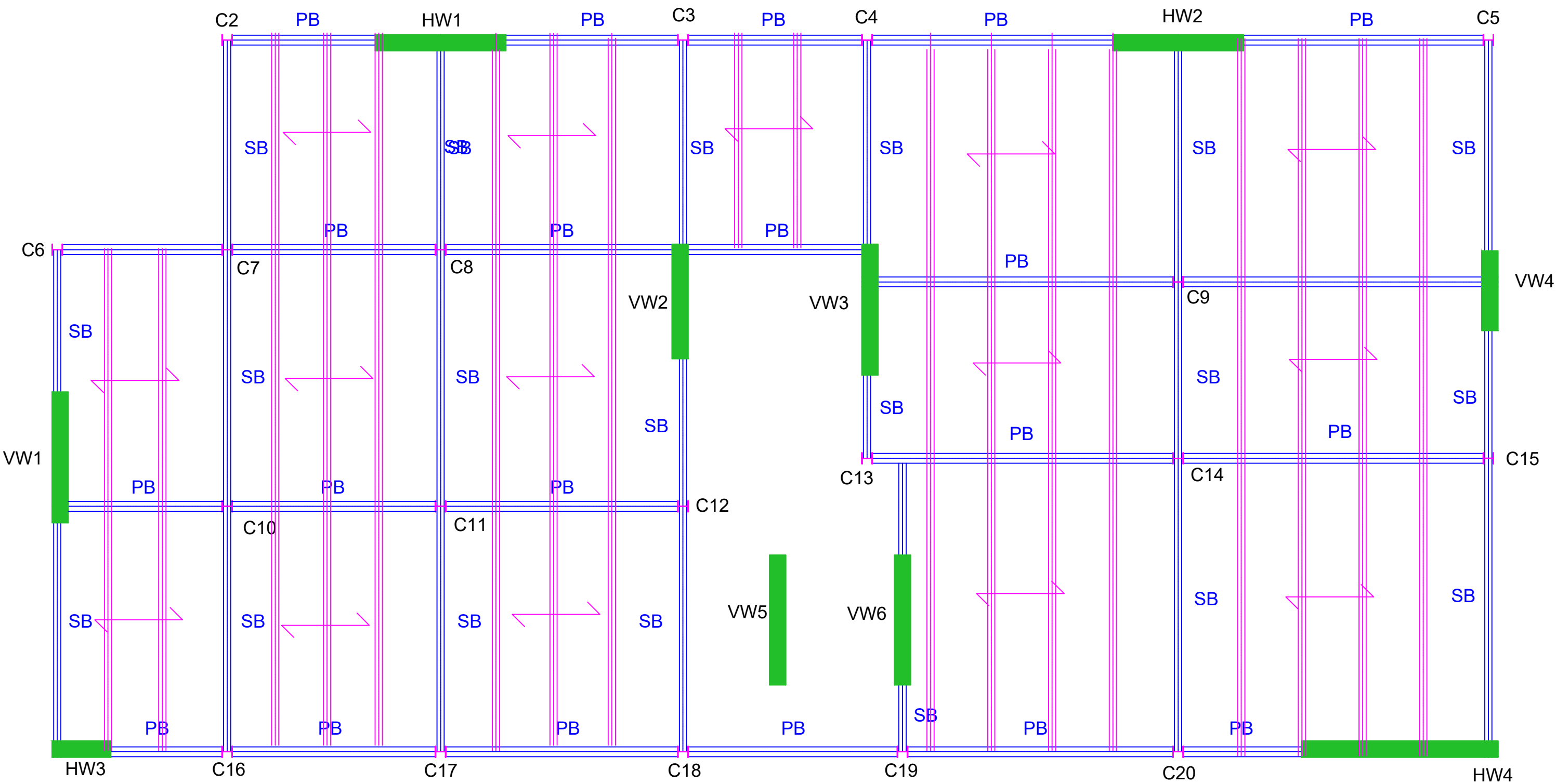
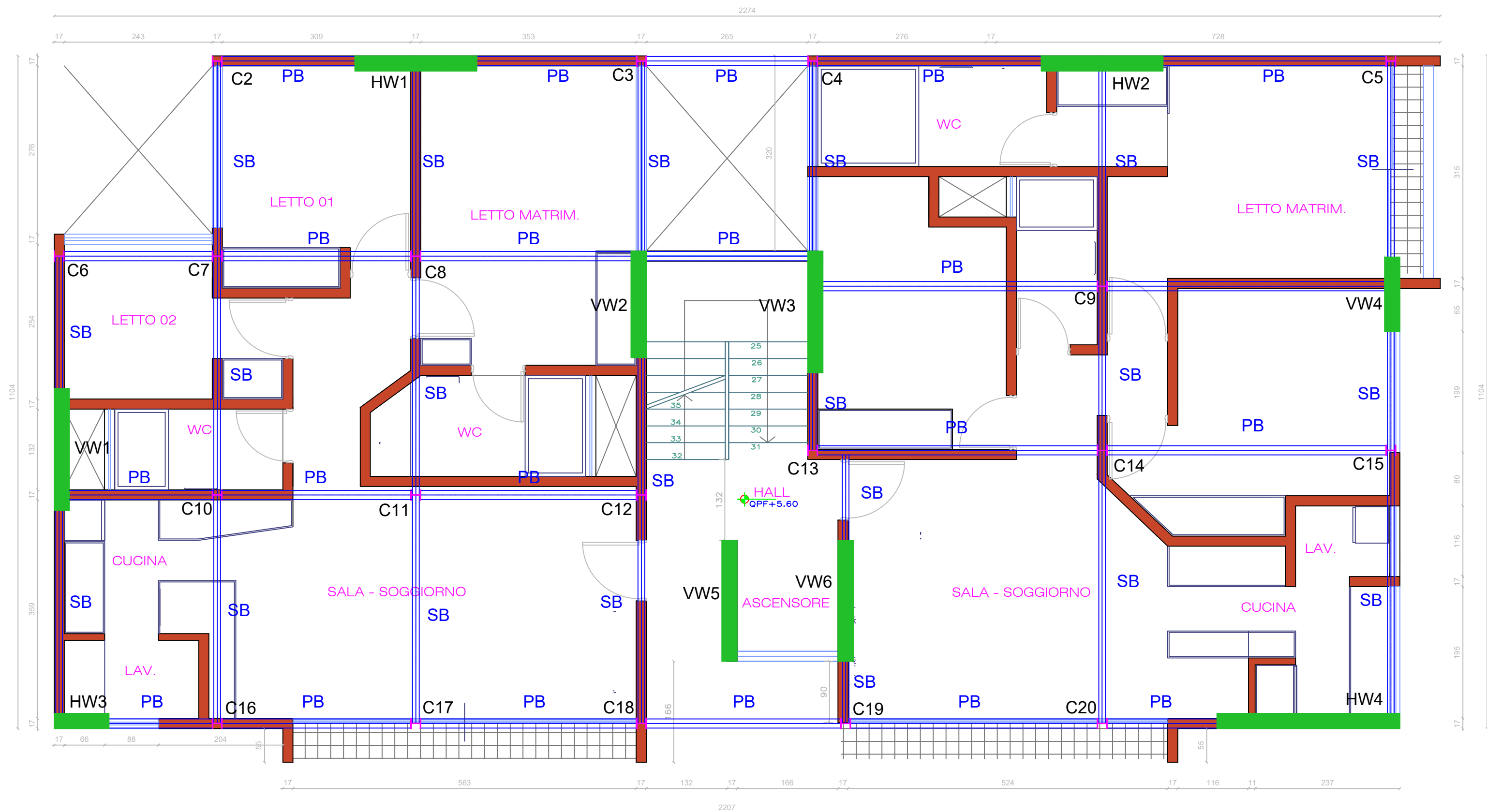
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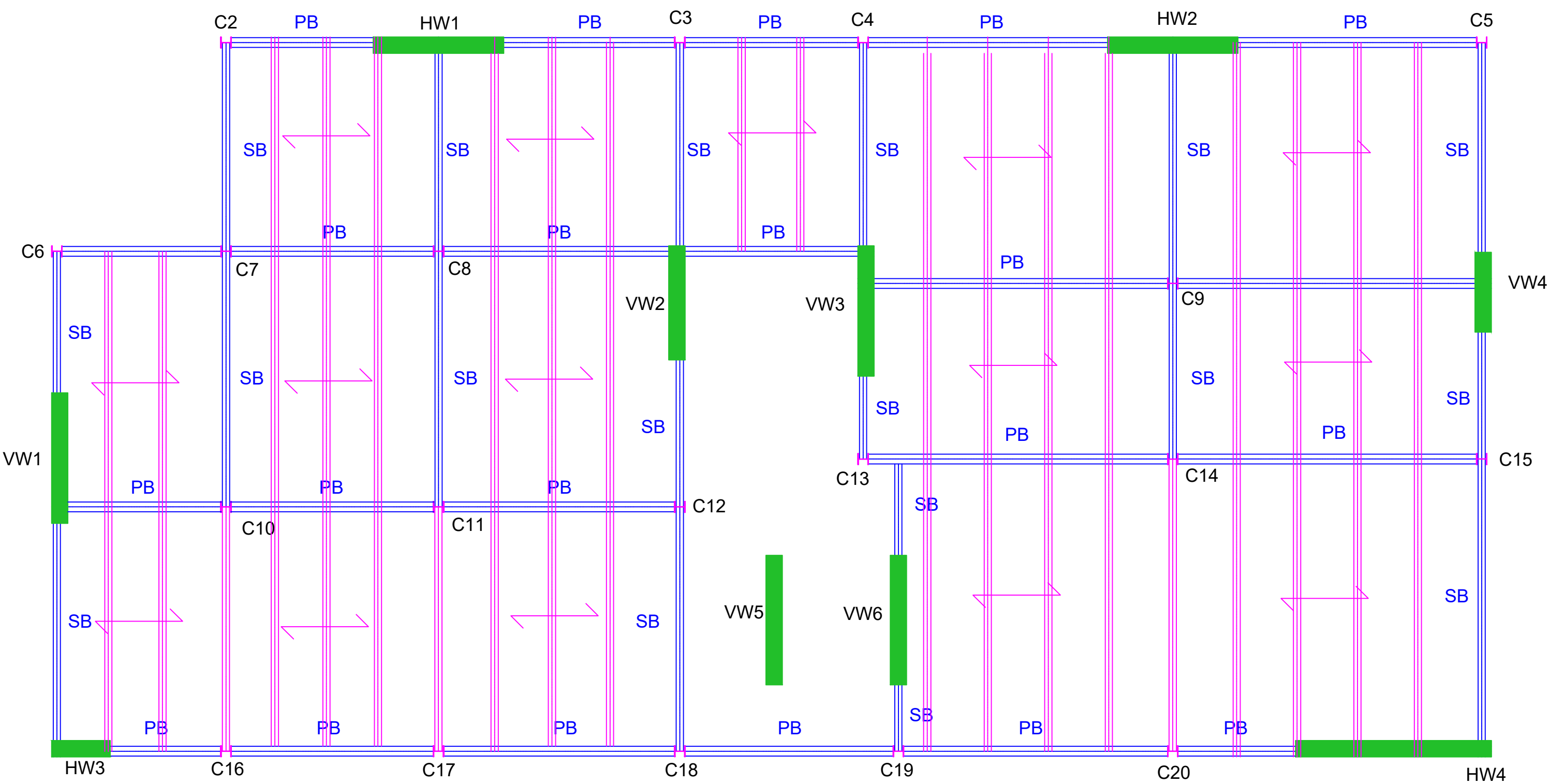
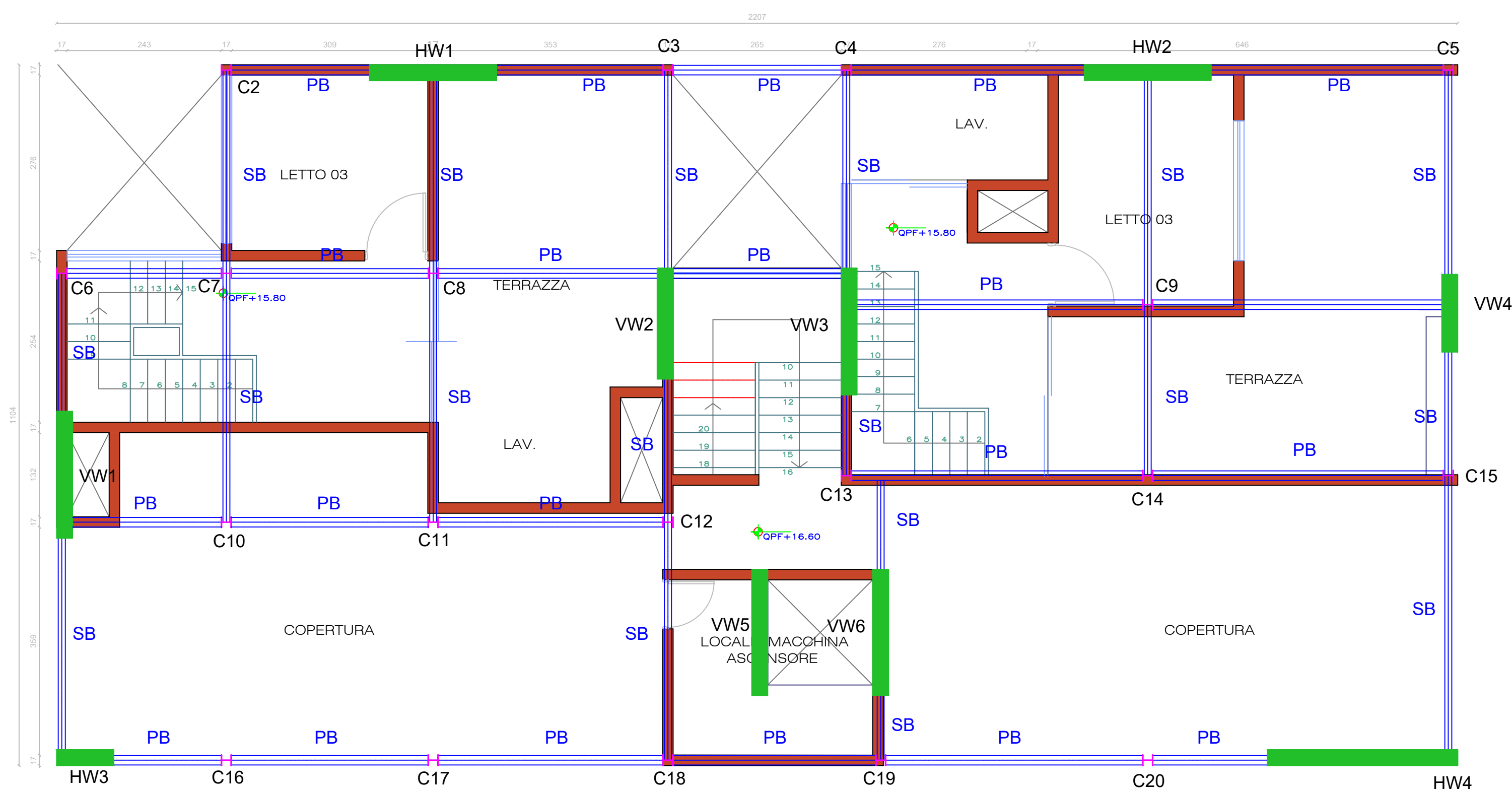
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ENG. MARCO CERVIO

STRUCTURAL LAYOUT - SECOND, THRID, FOURTH AND FIFTH FLOOR

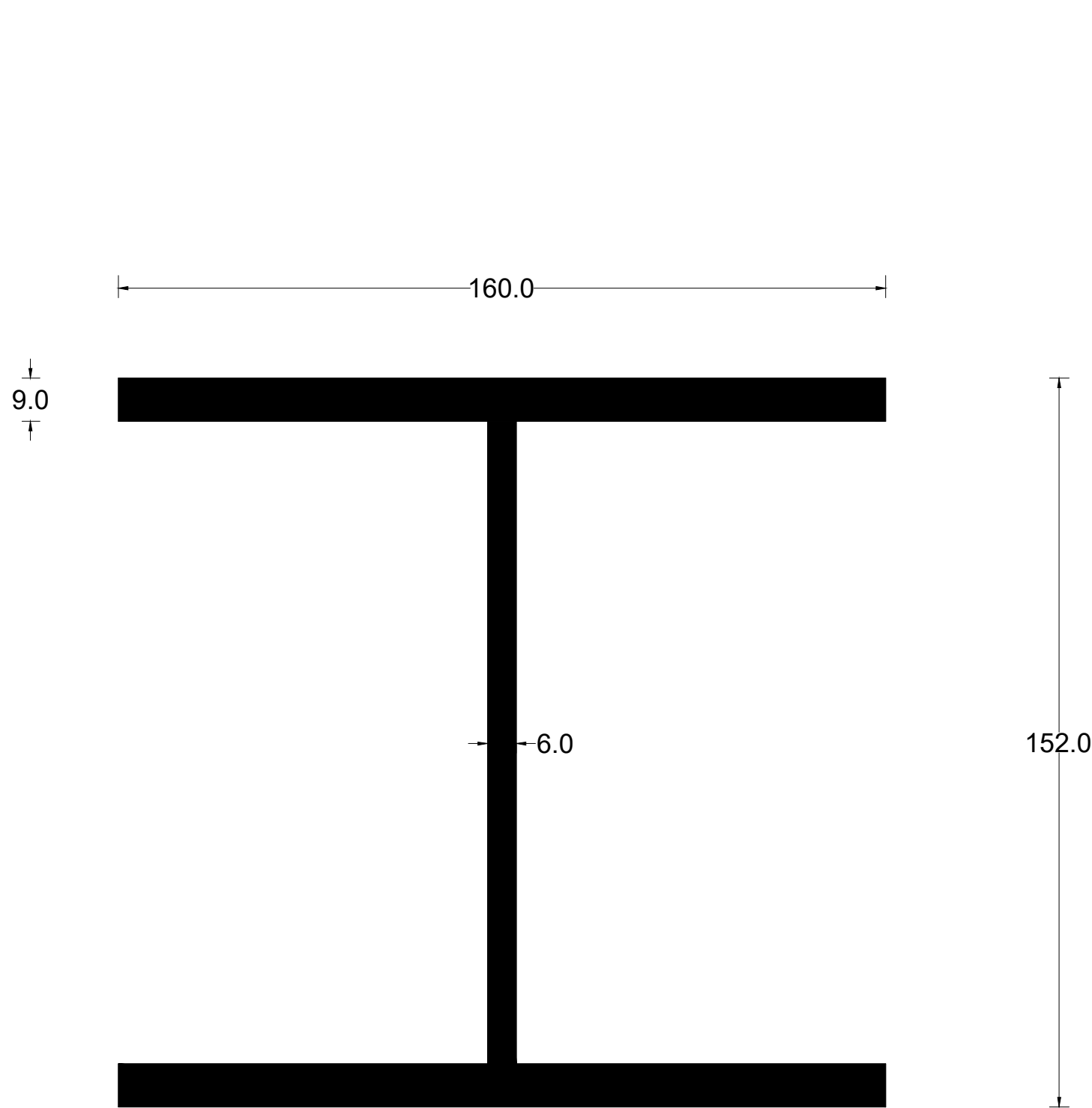


STRUCTURAL LAYOUT - TERRACE

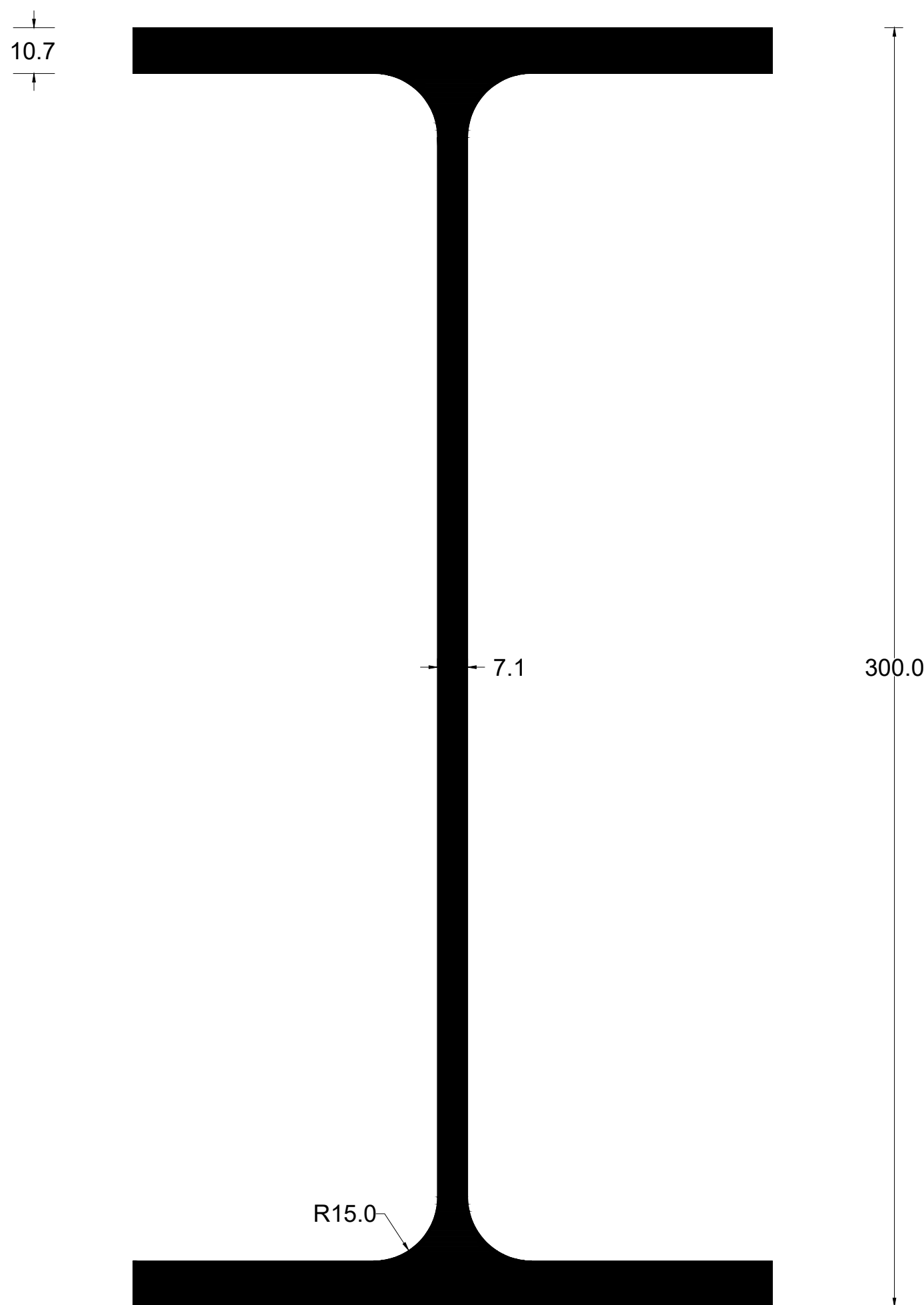


SCALE 1:60

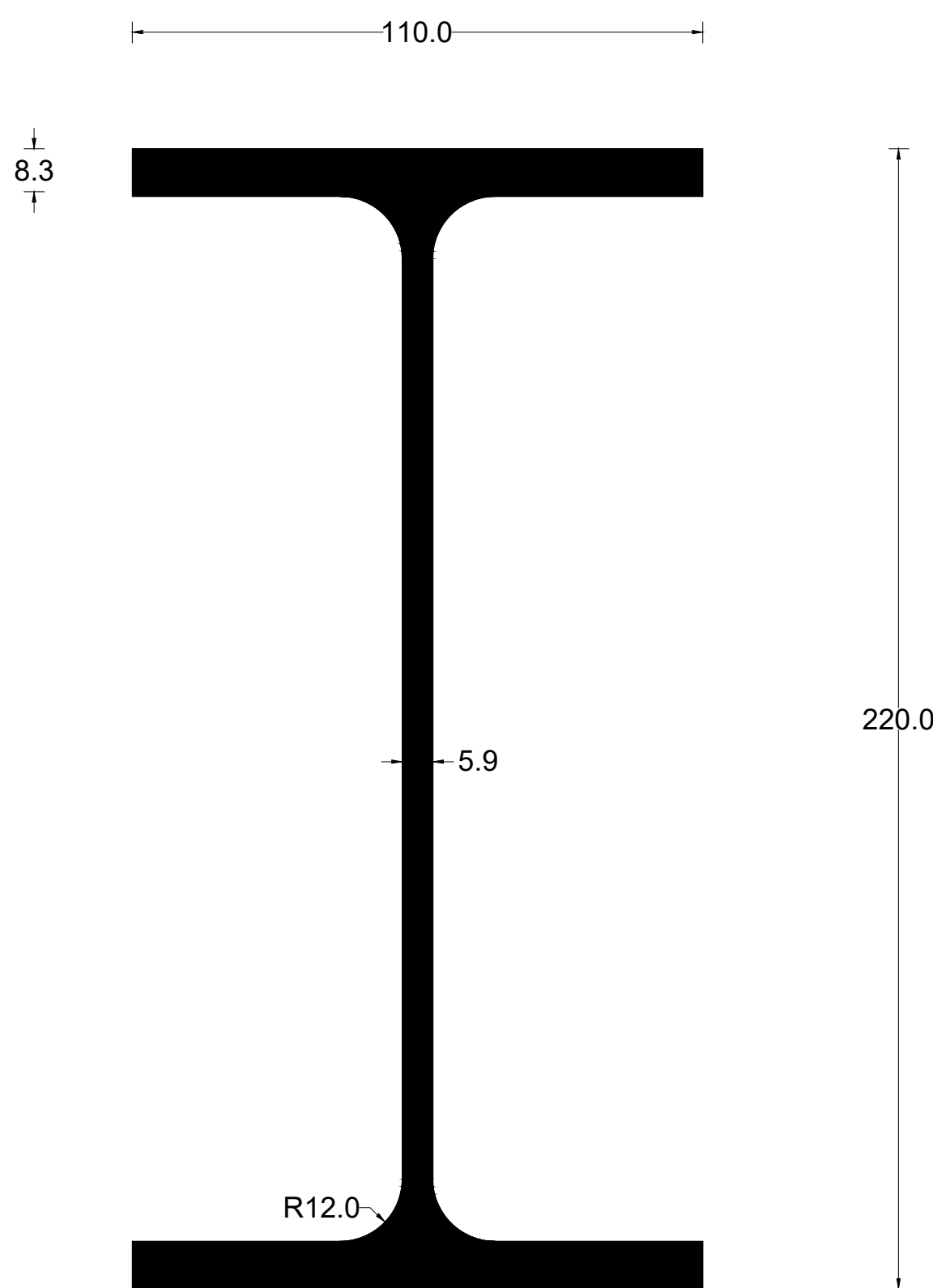
COLUMN SECTION (HE160A)



PRIMARY BEAM SECTION (IPE300)

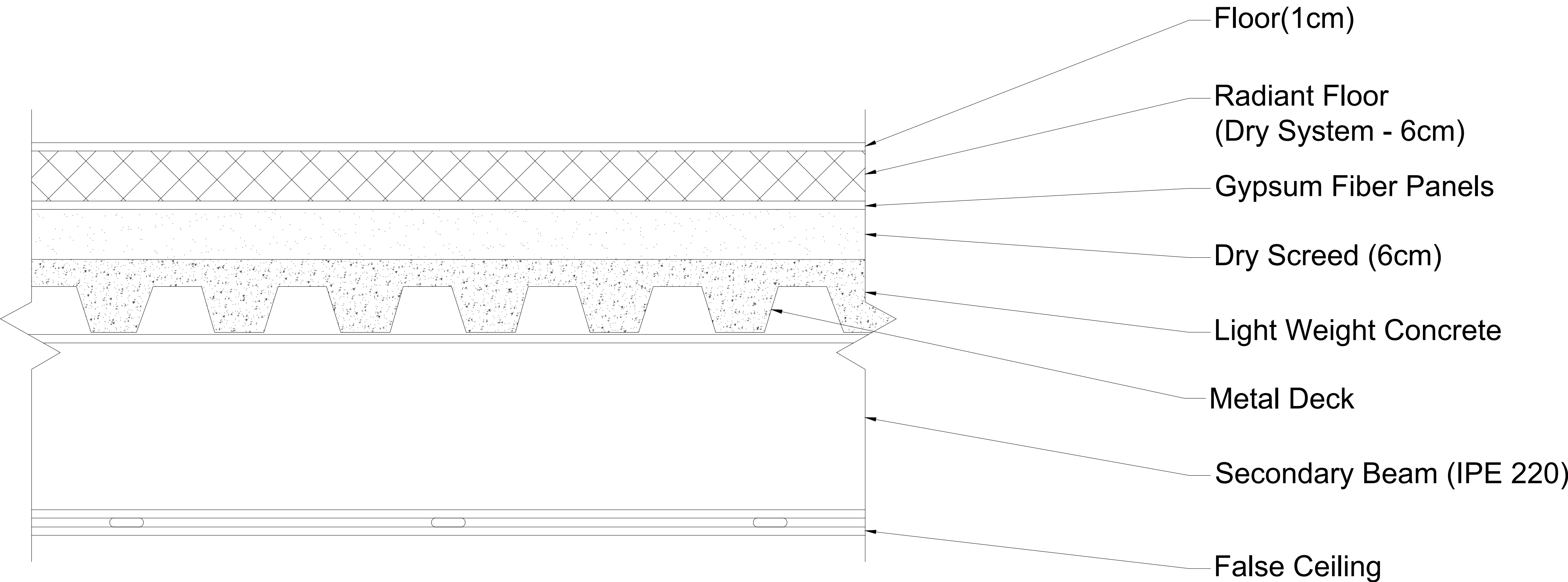


SECONDARY BEAM SECTION (IPE220)



Dimensions of above sections are in millimeters

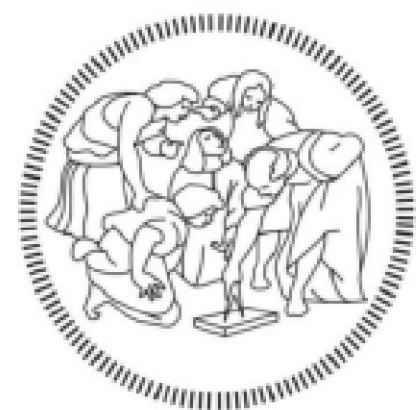
SLAB CROSS SECTION



GENERAL CONDITIONS

- CONCRETE CLASS: C30/37
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: A
- SOIL TYPE : 'C'
- CONCRETE COVER: 30mm
- STRUCTURAL STEEL: S355
- YIELD STRENGTH OF STR. STEEL: 355 MPa
- UNIT WEIGHT OF STR. STEEL: 78.5 kN/m³

M.S. CIVIL ENGINEERING
DESIGN OF NEW STRUCTURES



POLITECNICO
MILANO 1863

CONCEPTUAL DESIGN OF
STRUCTURES
ACADEMIC YEAR
(2023-24)

STRUCTURAL ELEMENTS
SECTIONS

SUBMITTED BY:

TEAM
BRAVO

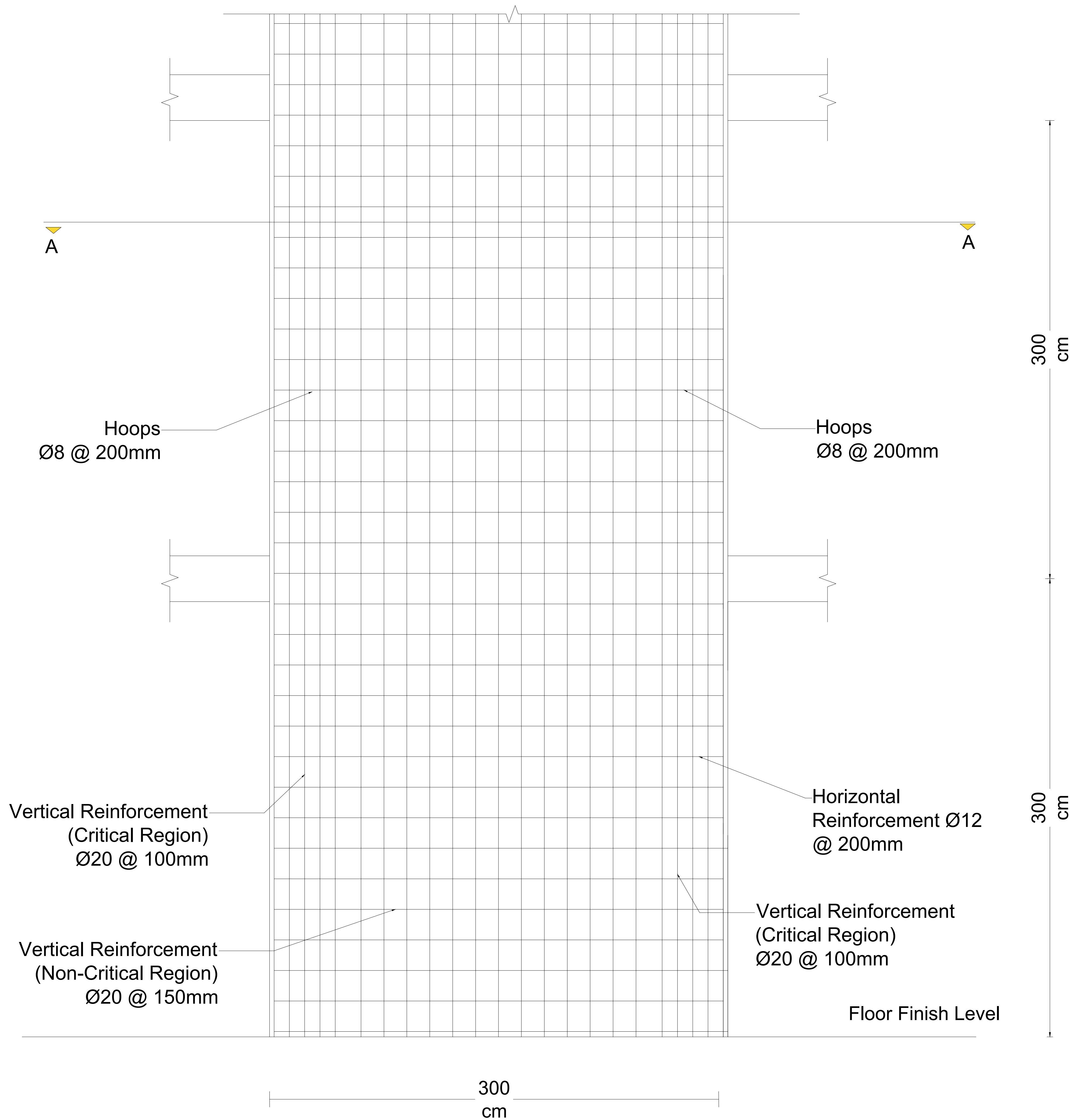
SHEET: A0

SUBMITTED TO:

PROF. GIANPAOLO ROSATI
ENG. ANGELO MARCHISELLA
ENG. MARCO CERVIO

4
6

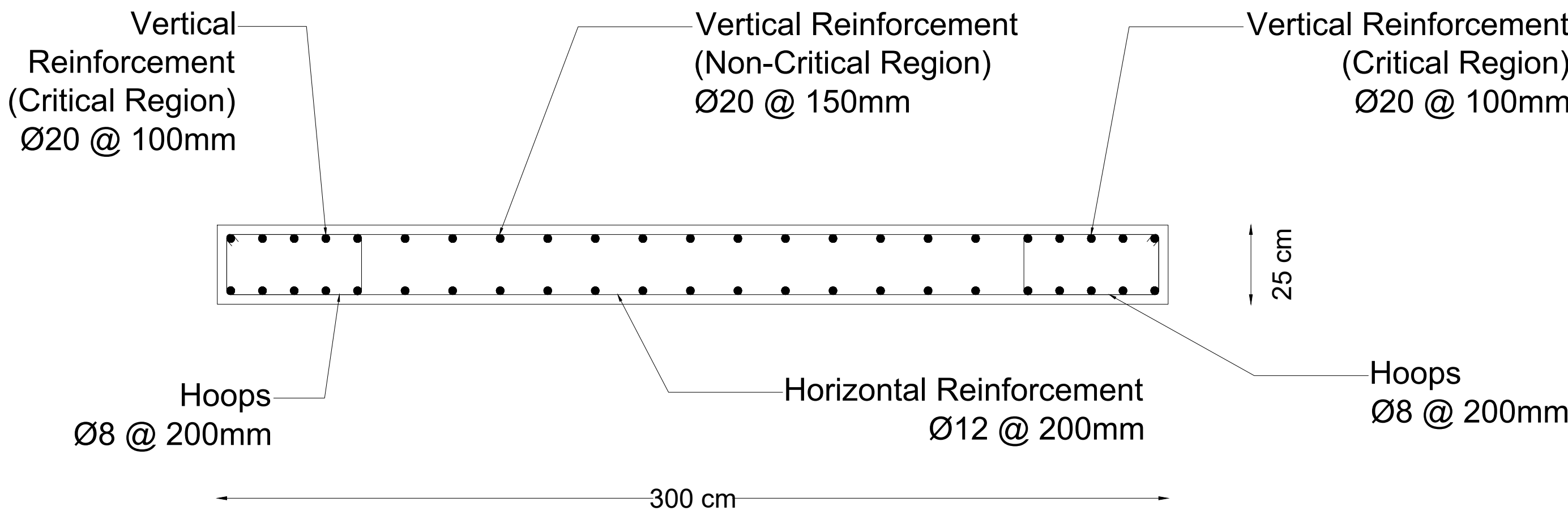
TYPICAL SHEAR WALL - LONGITUDINAL SECTION



SHEAR WALL SCHEDULE

Type	HW1,HW2	HW3	HW4	VW1, VW3 VW5, VW6	VW2	VW4
Area (m2)	0.5	0.225	0.75	0.5	0.44	0.3
Length (m)	2	0.9	3	2	1.75	1.2
Width (m)	0.25	0.25	0.25	0.25	0.25	0.25

TYPICAL SHEAR WALL - CROSS SECTION

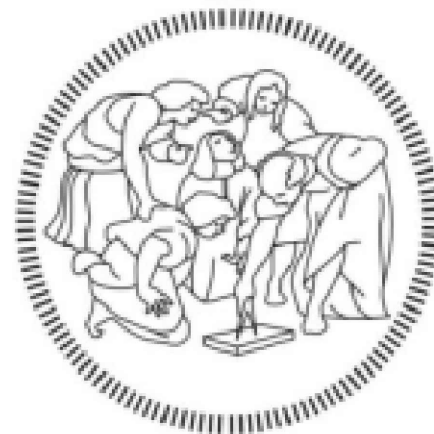


Sec A-A

GENERAL CONDITIONS

CONCRETE CLASS: C30/37
STEEL TYPE: B450C
EXPOSURE CLASS: XC-1
BUILDING CATEGORY OF USE: A
SOIL TYPE : 'C'
CONCRETE COVER: 30mm

M.S. CIVIL ENGINEERING
DESIGN OF NEW STRUCTURES



POLITECNICO
MILANO 1863

CONCEPTUAL DESIGN OF
STRUCTURES
ACADEMIC YEAR
(2023-24)

STRUCTURAL ELEMENTS
SECTIONS
(WALL SCHEDULE AND TYPICAL
WALL)

SUBMITTED BY:

TEAM
BRAVO

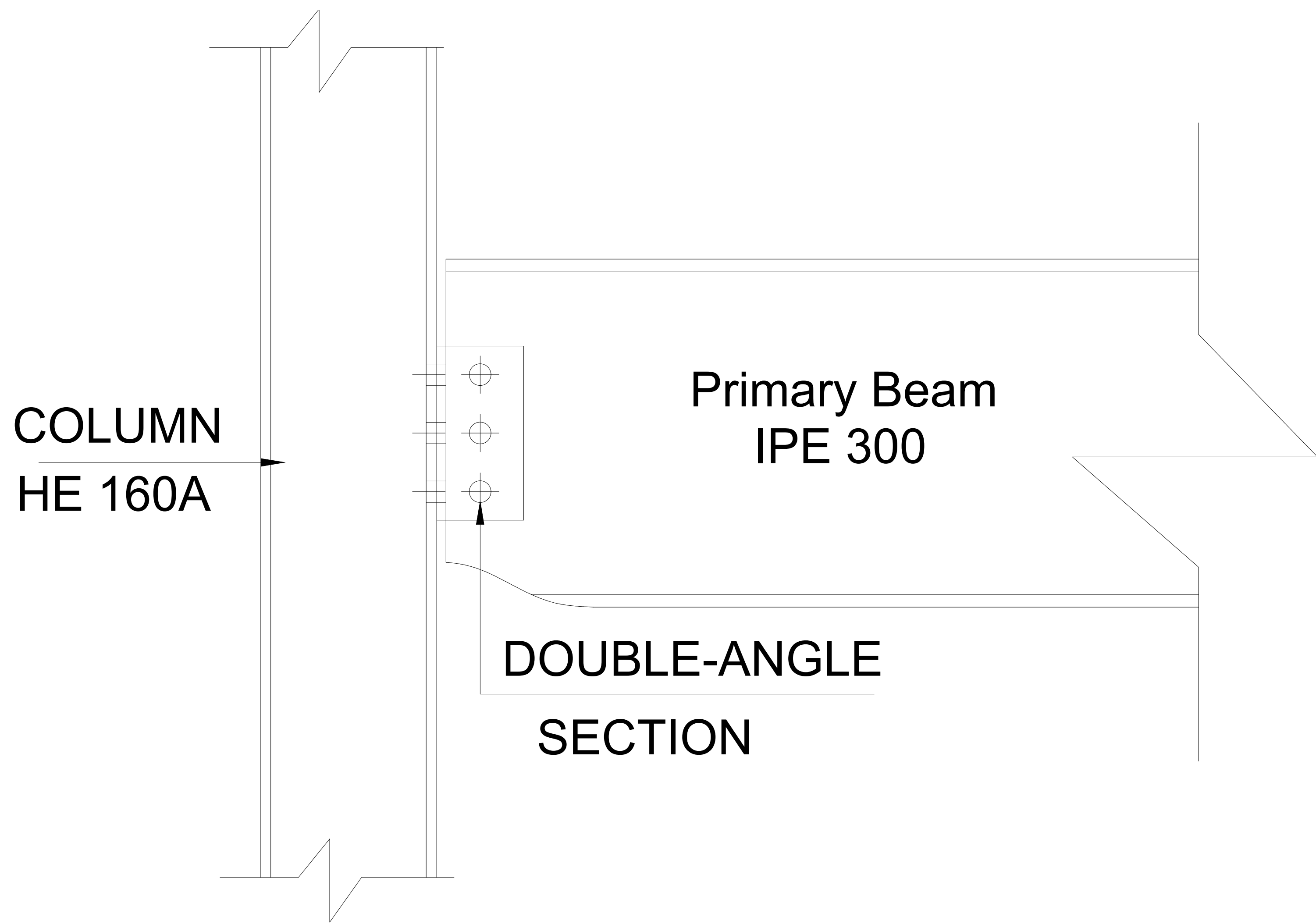
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SUBMITTED TO:

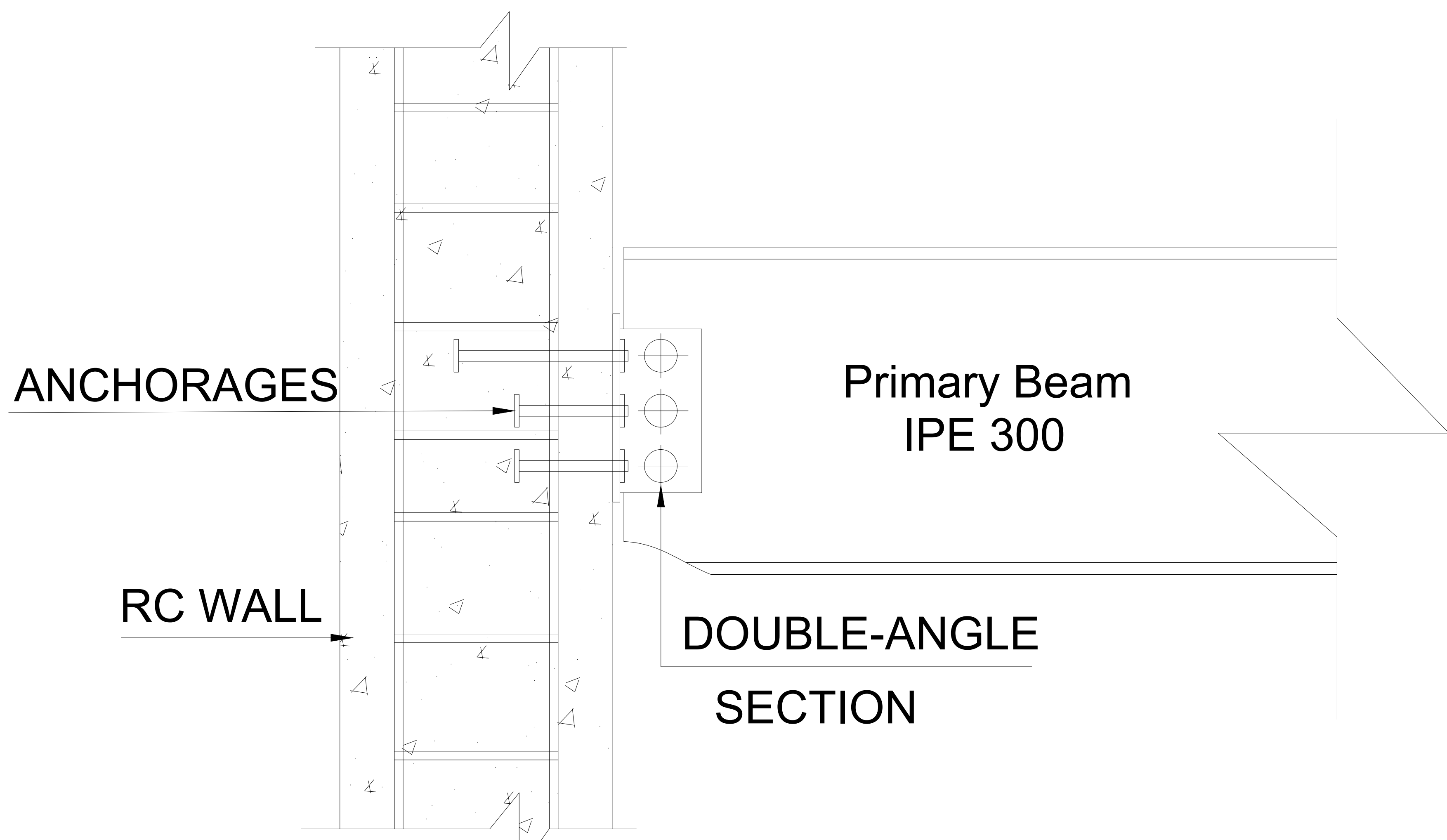
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5
6

TYPICAL COLUMN-PRIMARY BEAM CONNECTION



TYPICAL WALL-PRIMARY BEAM CONNECTION



GENERAL CONDITIONS

- CONCRETE CLASS: C30/37
- STEEL TYPE: B450C
- EXPOSURE CLASS: XC-1
- BUILDING CATEGORY OF USE: A
- SOIL TYPE : 'C'
- CONCRETE COVER: 30mm
- STRUCTURAL STEEL: S355
- YIELD STRENGTH OF STR. STEEL: 355 MPa
- UNIT WEIGHT OF STR. STEEL: 78.5 kN/m³

M.S. CIVIL ENGINEERING
DESIGN OF NEW STRUCTURES



POLITECNICO
MILANO 1863

CONCEPTUAL DESIGN OF
STRUCTURES
ACADEMIC YEAR
(2023-24)

TYPICAL CONNECTIONS

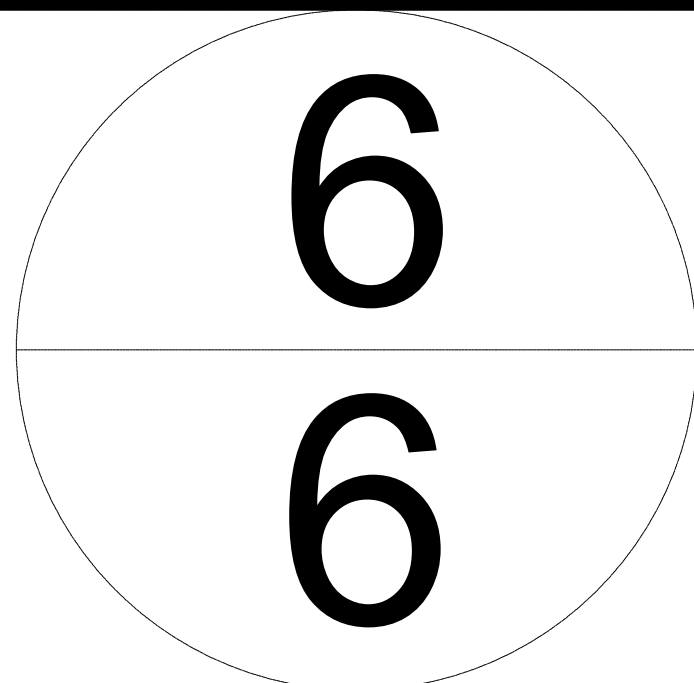
SUBMITTED BY:

TEAM
BRAVO

SHEET: A0

SUBMITTED TO:

PROF. GIANPAOLO ROSATI
ENG. ANGELO MARCHISELLA
ENG. MARCO CERVIO



SUMMARY TABLE - PROJECT 1

Table 1: Volume of RC Elements

Element	Volume (m³)
Columns	44.06
Beams	54.84
Shear Walls	83.25
Slab	63.83
Pre-Cast Joists	63.59
Volume of all RCC Elements (m³)	309.57

Table 2: Seismic Mass

Seismic Mass (Tons)	
Ground	211.55
1	203.69
2	203.69
3	203.69
4	203.69
Roof	202.21
Total Seismic Mass (Tons)	1228.52

Table 3: Seismic Base Shear

Seismic Shear at base (kN) - in X and Y direction	1864.41
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Table 4: Area Indices - X Direction

Floor	WI% (Japanese Code)	CI% (Suzon)	WI% (Suzon)
0	0.82%	0.45%	0.96%
1	0.87%	0.47%	1.03%
2 to 5	0.87%	0.45%	1.01%
6	0.44%	0.37%	0.58%

Table 5: Wall Area Index 1 - X-direction

W/(Aw + Ac) - MPa (<i>Cummulative</i>)	0.45
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Table 6: Area Indices - Y Direction

Floor	WI% (Japanese Code)	CI% (Suzon)	WI% (Suzon)
0	1.14%	0.45%	1.22%
1	1.20%	0.47%	1.28%
2 to 5	1.20%	0.45%	1.27%
6	1.21%	0.37%	1.23%

Table 7: Wall Area Index 1 - Y-direction

W/(Aw + Ac) - MPa (<i>Cummulative</i>)	0.37
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Table 8: Natural Period of Building

Natural Period - in X and Y Direction (sec)	0.66
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Table 9: Eccentricity of Mass and Stiffness

Eccentricities (m)	
X	1.62
Y	1.12

Table 10: Cross-Sectional Dimension (b x h)

Column	30 cm x 30 cm
Beam	45 cm x 20 cm

***** THE END *****

SUMMARY TABLE - PROJECT 2

Table 1: Volume of Elements

Element	Volume (m ³)
Columns	1.50
Primary Beams	3.23
Shear Walls	83.25
Secondary Beams	5.53
All Elements	93.52
<i>*Slab volume not included</i>	

Table 2: Seismic Mass

Seismic Mass (Tons)	
Ground	147.9
1	142.5
2	142.5
3	142.5
4	142.5
Roof	141.4
Total Seismic Mass (Tons)	859.14

Table 3: Seismic Base Shear

Seismic Shear at base (kN) - in X and Y direction	1225.04
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Table 4: Area Indices in x-direction

Floor	WI% (Japanese Code)	CI% (Suzon)	WI% (Suzon)
0	0.82%	0.015%	0.96%
1	0.87%	0.016%	1.03%
2 to 5	0.87%	0.015%	1.01%
6	0.44%	0.012%	0.58%

Table 5: Wall Area Index 1 - X-direction

W/(Aw + Ac) - MPa (<i>Cummulative</i>)	0.64
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Table 6: Area Indices in y-direction

Floor	WI% (Japanese Code)	CI% (Suzon)	WI% (Suzon)
0	1.14%	0.02%	1.22%
1	1.20%	0.02%	1.28%
2 to 5	1.20%	0.02%	1.27%
6	1.21%	0.01%	1.23%

Table 7: Wall Area Index 1 - Y-direction

W/(Aw + Ac) - MPa (<i>Cummulative</i>)	0.44
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Table 8: Natural Period of the building

Natural Period - in X and Y Direction (sec)	0.743
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Table 9: Plan Eccentricities

Eccentricities (m)		
X	1.64	m
Y	1.14	m

Table 10: Sections of basic elements

Column Section	HE 160A
Primary Beam Section	IPE300
Secondary Beam Section	IPE220

***** THE END *****